Beyond Frozen Encoders: Adapting 3D CT Foundation Models to Multi-Label Classification and Segmentation

George Yiasemis G.YIASEMIS@NKI.NL
Razvan-Andrei Matisan R.MATISAN@NKI.NL
??? ???@NKI.NL
Jonas Teuwen J.TEUWEN@NKI.NL
AI for Oncology, Netherlands Cancer Institute, Amsterdam, The Netherlands

Editors: Under Review for MIDL 2027

Abstract

Foundation models for 3D computed tomography (CT) are pretrained to be adapted to clinical datasets that carry several labels per scan, such as multiple findings or multiple structures. However, their published evaluations mostly train a lightweight head on the frozen encoder, and do not inform two decisions: which parameters to train, and whether to train one model per label or one for the whole dataset. We compare 11 classification and 7 segmentation adaptation methods, from frozen probes to low-rank adapters and full fine-tuning, each trained per label and jointly with one default setting and a fixed budget. We evaluate three public 3D CT encoders, one of which is restricted to head CT, on 39 binary labels from five public classification cohorts and 37 structures from five public segmentation datasets, totaling ∼2k training runs. In classification, adapting the encoder with low-rank adaptation (LoRA) improves the held-out area under the receiver operating characteristic curve (AUROC) by 0.034 over the linear probe and by 0.027 over the frozen attentive probe, and fine-tuning all attention weights performs comparably at similar memory and time per training step, although LoRA updates 19 to 54 times fewer encoder weights. A single joint model per cohort is not detectably different from per-label models on average (AUROC difference +0.004, 95% interval −0.021 to +0.030). In segmentation, a multi-depth decoder on the frozen encoder improves Dice over a linear probe by 0.085 (TAP-CT) and 0.061 (SPECTRE), most of the gain of adapting the encoder, whereas per-structure models remain more accurate than joint models at our training budgets. We condense these findings into practical guidelines.

Keywords: Foundation models, computed tomography, parameter-efficient fine-tuning, low-rank adaptation, multi-task learning, segmentation

1. Introduction

A cohort collected for trauma or hemorrhage carries several findings per scan, and a segmentation dataset delineates several organs or lesions. Several pretrained 3D CT encoders are now public, among them TAP-CT (Veenboer et al., 2026), SPECTRE (Claessens et al., 2026), Pillar-0 (Agrawal et al., 2025), CT-FM (Pai et al., 2025), Merlin (Blankemeier et al., 2026), CT-CLIP (Hamamci et al., 2026) and FM-HCT (Zhu et al., 2026). Their evaluations mostly rank frozen representations with a linear, attentive or nearest-neighbor head (Veenboer et al., 2026; Claessens et al., 2026), and only some report end-to-end fine-tuning alongside (Pai et al., 2025; Zhu et al., 2026). For natural images, comparisons on the same pretrained models showed that a frozen evaluation can be misleading. The linear-probe

© 2027 CC-BY 4.0, G. Yiasemis, R.-A. Matisan, R.-A. ??? & J. Teuwen.

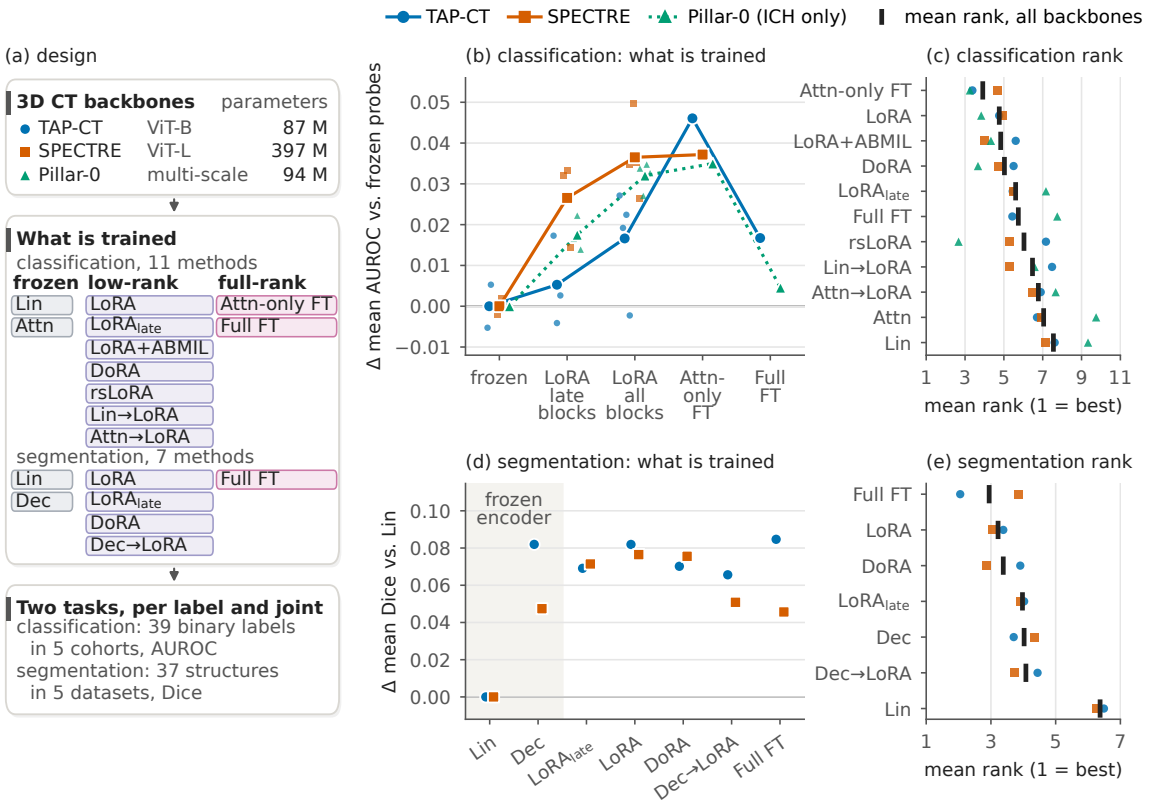


Figure 1: Overview of the study. (a) Backbones, methods grouped by what they train (Section 2), and tasks. (b) Change in pooled held-out AUROC relative to the mean of each backbone's two frozen probes; small markers are methods (late blocks: $\text{LoRA}_{\text{late}}$ and the two-stage methods; all blocks: LoRA, LoRA+ABMIL, DoRA, rsLoRA), lines group means. (c) Mean classification rank per backbone (markers) and pooled (bars). (d) Change in mean held-out Dice relative to Lin of the same backbone, over the 118 structure–mode pairs with every method (TotalSegmentator per-structure models excluded), hence different from Tab. 1; shading marks a frozen encoder. (e) Mean segmentation rank. SPECTRE Full FT, which did not train, is excluded from (b) and (c); ranks are defined in Appendix B.

and fine-tuning accuracies of masked autoencoders were largely uncorrelated, and tuning a single transformer block raised ImageNet accuracy from 73.5% to 81.0% (He et al., 2022). Similarly, prompt tuning with under 1% of the parameters outperformed full fine-tuning on 20 of 24 tasks (Jia et al., 2022), and full fine-tuning outperformed linear probing in distribution but not under distribution shift (Kumar et al., 2022). For 3D medical images such as CT, comparisons of this kind are few. Still, a group that adapts an encoder to a multi-label cohort or a multi-structure dataset, often with a few hundred to a few thousand labeled scans, has to decide whether to keep the encoder frozen, adapt it with LoRA (Hu et al., 2022) or fine-tune part or all of it, whether to train one model per label or one model for all labels of the dataset, and, for segmentation, which head turns the encoder tokens into a voxel map. A frozen probe answers none of these questions.

In medical imaging, the evidence for these choices comes mostly from 2D backbones and single tasks. On 2D medical classification, LoRA matches or surpasses full fine-tuning (Dutt et al., 2024; Zhu et al., 2024), and probing and fine-tuning of DINOv2 (Oquab et al., 2024) have been compared on radiology tasks (Baharoon et al., 2023). For 3D classification, Liu et al. (2026) found that frozen probing understates 2D and 3D foundation models, and that on the spleen-injury label of RSNA23 trauma, one of our labels, full fine-tuning of a convolutional and a Swin 3D model outperformed LoRA. Otherwise, parameter-efficient adaptation of 3D medical encoders has mostly been studied for segmentation, with adapters that lift the 2D Segment Anything model (Kirillov et al., 2023) to volumes (Gong et al., 2024; Chen et al., 2024a; Wu et al., 2025), prompt tuning of a 3D U-Net (Fischer et al., 2024), and LoRA, adapters and full fine-tuning in few-shot transfer (Silva-Rodríguez et al., 2025; Ly and Nguyen, 2026). For a 3D CT vision–language model, full and parameter-efficient fine-tuning performed similarly on one classification task (Kartika et al., 2026). However, none of these studies compares LoRA variants (Liu et al., 2024; Kalajdzievski, 2023), probing before fine-tuning (Kumar et al., 2022) or attention-only fine-tuning (Touvron et al., 2022) on 3D CT encoders. On 75 head-CT findings, a mixture of low-rank experts applied to the pooled embedding of a fully fine-tuned Pillar-0 raised its AUROC by 0.002 (Yoo et al., 2026). Joint training on many labels can help (Kim et al., 2023) or hurt through conflicting gradients (Yu et al., 2020; Zhao et al., 2023). Yet joint and per-task adaptation of a 3D encoder have not been compared on the same labels (Veenboer et al., 2026; Liu et al., 2026).

We compared adaptation methods on five multi-label classification cohorts and five multi-structure segmentation datasets, under one protocol and on three encoders: two plain vision transformers (TAP-CT, SPECTRE) and the multi-scale transformer of Pillar-0 with its convolutional stem (Figs. 1 and 2). Classification was evaluated on all three encoders, with Pillar-0 on the head-CT cohort only and SPECTRE on the other four cohorts, and segmentation on TAP-CT and SPECTRE; every method was trained both per task and jointly. Our contributions can be summarized as follows:

- We present a controlled comparison of 11 classification and 7 segmentation adaptation methods, from frozen probes to full fine-tuning, on three 3D CT encoders from two architectural families, each trained per task and jointly (∼2k runs).
- We find that, in classification at a fixed training budget per method, adapting the attention weights with LoRA outperforms both frozen probes on every encoder, at similar memory and time per training step as fine-tuning all attention weights.
- We find that, in segmentation at the same budget, a multi-depth decoder on the frozen encoder accounts for most of the gain of adapting the encoder, and LoRA beneath it changes Dice little on average.
- We find that one joint model per dataset is not detectably different from per-label models on average in classification but loses Dice in segmentation at our budgets.
- We condense these findings into practical guidelines (Section 5) and release our code and data splits (`https://github.com/NKI-AI/beyond_frozen_encoders`).

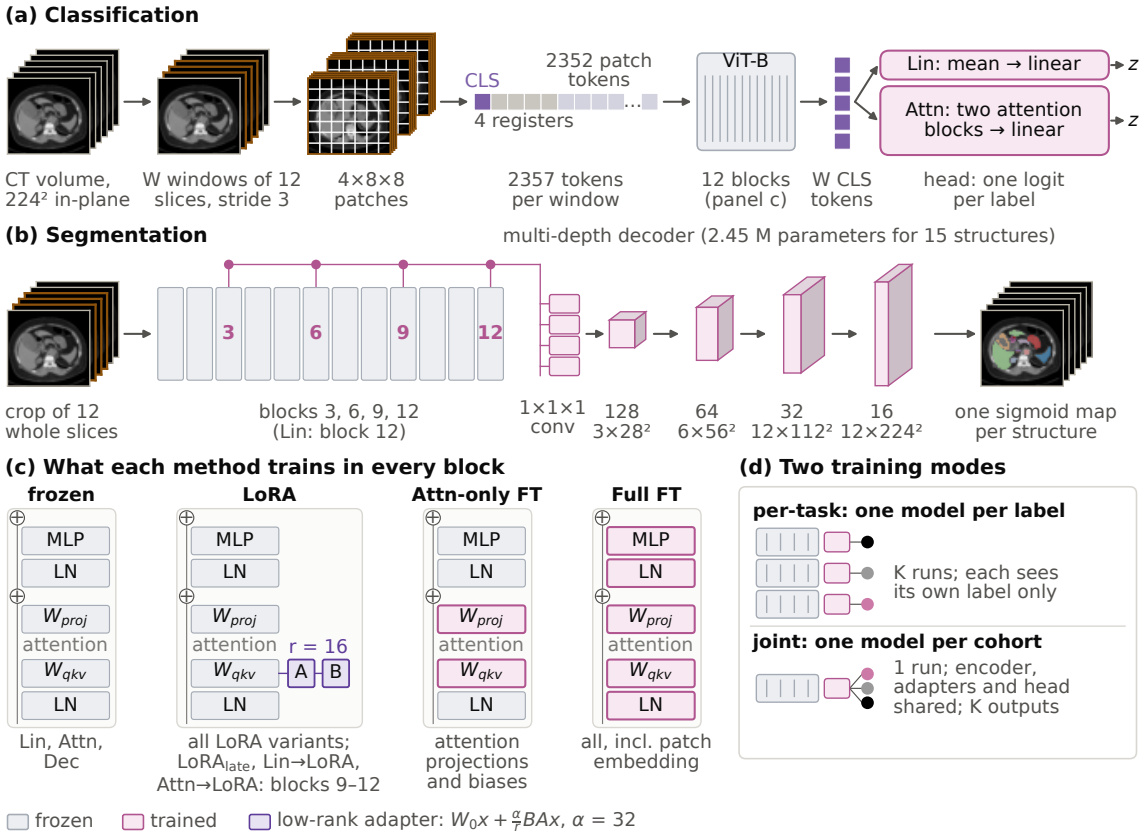


Figure 2: How each task is run, shown for TAP-CT (other backbones: Figs. 6 and 7, Appendix A). (a) Classification: the volume is split into W 12-slice windows starting every 3 slices ($4 \times 8 \times 8$ patches), and the CLS token of every window enters the head, a linear layer on their mean (Lin) or an attentive probe (Attn), with one logit z per label. (b) Segmentation: the tokens of blocks 3, 6, 9 and 12 feed a multi-depth decoder (2.45 M parameters for 15 structures) with one sigmoid map per structure; Lin uses block 12 only. (c) Parameters each method trains in every encoder block; the head is always trained. (d) Per-task and joint training; in segmentation, a label is a structure and a cohort a dataset, and the 15-structure joint models train for three times the steps.

2. Methods

Backbones **TAP-CT-B-3D** (Veenboer et al., 2026), pretrained by DINOv2-style self-supervision on in-house CT, is a plain 12-block vision transformer (ViT-B; Dosovitskiy et al., 2021) with $4 \times 8 \times 8$ voxel patches, a class (CLS) token and four register tokens (Darcet et al., 2024), applied to windows of $12 \times 224 \times 224$ voxels at native slice spacing (Fig. 2). **SPECTRE-Large** (Claessens et al., 2026), pretrained by self-distillation and vision–language alignment with radiology reports, is a plain 24-block ViT-L with $16 \times 16 \times 8$ patches on $128 \times 128 \times 64$ crops; for classification, 64 crops of a $512 \times 512 \times 256$ volume at $0.5 \times 0.5 \times 1$ mm are summarized by its pretrained 4-block feature combiner. **Pillar-0** (Agrawal et al., 2025), pretrained on CT and breast magnetic resonance imaging

(MRI) of one academic center, is used through its head-CT encoder, a convolutional stem followed by three stages of multi-scale window-attention blocks over the whole volume, which restricts it to RSNA19 ICH. Inputs follow each model's own orientation, spacing and intensity normalization (Fig. 6 and Appendix A). We are among the authors of TAP-CT.

Classification Methods The methods differ only in which weights of each encoder block they train, among its attention, feed-forward (MLP) and layer-normalization (LN) weights (Figs. 2c and 6c). *Frozen probes*: **Lin**, a linear layer on a pooled feature, and **Attn**, an attentive probe (Bardes et al., 2024; Chen et al., 2024b) in which a learnable CLS token joins the window, crop or scale tokens and two self-attention blocks run over the set. *Low-rank adapters* on the query/key/value projections, with the attentive head: **LoRA**, rank 16 ($\alpha = 32$) on every block (Hu et al., 2022); **LoRA$_{\text{late}}$**, on the last third of the blocks (for Pillar-0, its last stage, i.e., 8 of 12 blocks); **LoRA+ABMIL**, with tokens first pooled by gated attention-based multiple-instance learning (ABMIL; Ilse et al., 2018); **DoRA**, weight-decomposed LoRA (Liu et al., 2024); and **rsLoRA**, rank-stabilized LoRA (Kalajdzievski, 2023), whose $\alpha/\sqrt{r}$ scaling is four times the LoRA scale at $r = 16$. *Two-stage schedules*: **Lin→LoRA** and **Attn→LoRA** train the probe for half the budget and then add late-block LoRA, keeping the probe's head (Kumar et al., 2022). *Full-rank fine-tuning (FT)*: **Attn-only FT** trains every parameter of the attention modules (Touvron et al., 2022) and **Full FT** every parameter of the encoder, both with the attentive head.

Segmentation Methods Segmentation uses TAP-CT and SPECTRE. **Lin** is a $1 \times 1 \times 1$ convolution on the deepest token grid, upsampled trilinearly. **Dec** is a light multi-depth decoder similar to UNETR (Hatamizadeh et al., 2022) but without its skip connections from the input: the token grids of four depths are projected, concatenated, fused and upsampled to the voxel grid by 3^3 convolutions (2.4 to 2.6 M parameters; Figs. 2b and 7). LoRA, LoRA$_{\text{late}}$ and DoRA add the adapters of classification under Dec, Full FT trains the encoder with it, and Dec→LoRA trains Dec for half the budget before adding all-block LoRA. Predictions are independent sigmoids per structure.

Per-task and Joint Training Every cohort and dataset is trained in two modes with identical data and selection (Fig. 2d): *per-task*, one model per label or structure, and *joint*, one model with shared encoder, adapters and head and one output per label or structure. A *label–mode pair* is one label (or structure) on one backbone in one mode. A per-task model has one output; a joint model has one output per label or structure and sums the classification loss over the labels annotated for a case but averages the segmentation loss over structures (Equations 2 and 4, Appendix A).

3. Experiments

Data Classification uses five public CT cohorts with 39 binary labels (Tabs. 23 and 24 and Fig. 17 in Appendix F). The CVPR26 cohort is the release of the CVPR 2026 3D CT foundation-model challenge, which groups AMOS (Ji et al., 2022) with 15 disease labels, COVID-CTset (Rahimzadeh et al., 2021) and LUNA25 nodule malignancy (Peeters et al., 2026). FDG-PET/CT (Gatidis et al., 2022) provides four diagnosis groups, which we predict from the CT alone. The remaining cohorts are RSNA 2022 cervical-spine fracture detection (RSNA22 C-spine; Lin et al., 2023), RSNA 2023 abdominal trauma detection

(RSNA23 trauma; Rudie et al., 2024) and RSNA 2019 intracranial hemorrhage (ICH) detection (RSNA19 ICH; Flanders et al., 2020). Segmentation uses AMOS22 (15 organs; Ji et al., 2022), KiTS23 (Heller et al., 2023), LiTS (Bilic et al., 2023), MSD Pancreas (Antonelli et al., 2022) and TotalSegmentator (Wasserthal et al., 2023; Wasserthal, 2026), whose 117 structures we merge into 15 (Tab. 25), for 37 structures in total. For the CVPR26 cohort and AMOS22, the official validation cases are divided into a selection and a held-out half, whereas the other sources use fixed seeded splits of the released labeled cases, and TotalSegmentator holds out its official test split.

Optimization All runs use AdamW with a 10% linear warm-up, cosine decay to 10^{-6} and bf16 precision. A classification run trains for 8000 steps at batch size 1 on a masked, positive-weighted binary cross-entropy, and a segmentation run trains for 8000 steps of four crops on the sum of soft Dice and binary cross-entropy; two-stage methods split this budget in halves. Consequently, a joint model uses $1/K$ of the steps of the K per-task models it replaces, except for the 15-structure joint segmentation models (AMOS22, TotalSegmentator), which train for 24 000 steps. Each method has one default learning rate, set by what it trains: 10^{-3} for the classification linear probe and the segmentation decoder, 10^{-2} for the segmentation linear probe, 10^{-4} for the attentive probe and the adapters, and 2×10^{-5} for encoder weights. None of these settings was tuned on our data (Tabs. 2 and 3). Every 1000 steps, the model is evaluated on the selection split, and the checkpoint with the best macro AUROC or mean Dice is kept and its held-out metric reported.

Evaluation We report AUROC per label and Dice per structure and case, on each backbone's working grid (Maier-Hein et al., 2024). Methods are compared per label–mode pair with DeLong's test (DeLong et al., 1988) for AUROC or a paired case bootstrap for Dice, both with Benjamini–Hochberg correction (Benjamini and Hochberg, 1995), and are ranked within each pair that has every method; TotalSegmentator per-structure models were trained for Lin, Dec, LoRA and Full FT only. The main comparisons are summarized at label level, where each label counts once with its two modes averaged, with 95% intervals from a bootstrap over cohorts, labels and held-out cases (Appendix B). As the bootstrap draws only four or five cohorts per backbone, which makes its intervals optimistic, and does not include the variation between training seeds, we also count the cohorts with a positive mean difference. We flag a method as not trained when the median standard deviation of its held-out probabilities is below 0.010 (Appendix B). Only SPECTRE Full FT was flagged (0.004, against at least 0.114 for every other method); we report it in the tables but exclude it from ranks and from averages over methods. The joint models of Lin, LoRA, Attn-only FT and Full FT (classification) and of Dec, LoRA and Full FT (segmentation) were trained with two further seeds (Appendix D).

4. Results

Tab. 1 summarizes the main comparisons; per-label results, paired tests, seed repeats and controls, and success and failure cases are given in Appendices E, C, D and G.

Table 1: Main comparisons per backbone. Each entry is the held-out AUROC (classification) or Dice (segmentation) of the first method minus that of the second, averaged over labels or structures, above its 95% bootstrap interval over cohorts, labels and cases, which excludes seed variation (Appendix B). > 0: cohorts (datasets) with a positive mean difference. Joint − per-task compares one model per cohort with one per label, averaged over methods. Per-task and joint values are averaged, except in the two LoRA − Dec sub-rows. Pooled estimates: Tab. 7. †Includes runs that did not train (Section 3).

Classification (AUROC)							*Segmentation* (Dice)				
	TAP-CT		SPECTRE		Pillar-0			TAP-CT		SPECTRE	
Comparison	Mean	> 0	Mean	> 0	Mean	> 0	Comparison	Mean	> 0	Mean	> 0
LoRA − Lin	+0.032 [+0.007, +0.053]	4/5	+0.037 [+0.009, +0.075]	4/4	+0.033 [+0.021, +0.044]	1/1	Dec − Lin	+0.085 [+0.055, +0.109]	5/5	+0.061 [+0.027, +0.088]	5/5
LoRA − Attn	+0.022 [+0.006, +0.042]	5/5	+0.033 [+0.003, +0.078]	4/4	+0.034 [+0.021, +0.052]	1/1	LoRA − Dec	−0.005 [−0.021, +0.024]	4/5	+0.023 [−0.018, +0.076]	4/5
Attn-only FT − LoRA	+0.019 [−0.001, +0.054]	4/5	+0.002 [−0.037, +0.026]	2/4	+0.001 [−0.002, +0.004]	1/1	per-structure	−0.021 [−0.036, +0.002]	2/5	+0.006 [−0.003, +0.031]	5/5
Full FT − Lin	+0.022 [−0.061, +0.068]	4/5	−0.117† [−0.195, −0.059]	0/4	+0.004 [−0.013, +0.021]	1/1	joint	+0.011 [−0.021, +0.066]	4/5	+0.040 [−0.039, +0.147]	4/5
							Full FT − Dec	0.000 [−0.024, +0.016]	4/5	−0.003 [−0.035, +0.009]	2/5
Joint − per-task	+0.012 [−0.014, +0.038]	3/5	−0.003 [−0.044, +0.051]	3/4	−0.008 [−0.013, −0.001]	0/1	Joint − per-task	−0.046 [−0.090, −0.008]	0/5	−0.058 [−0.093, −0.009]	0/5

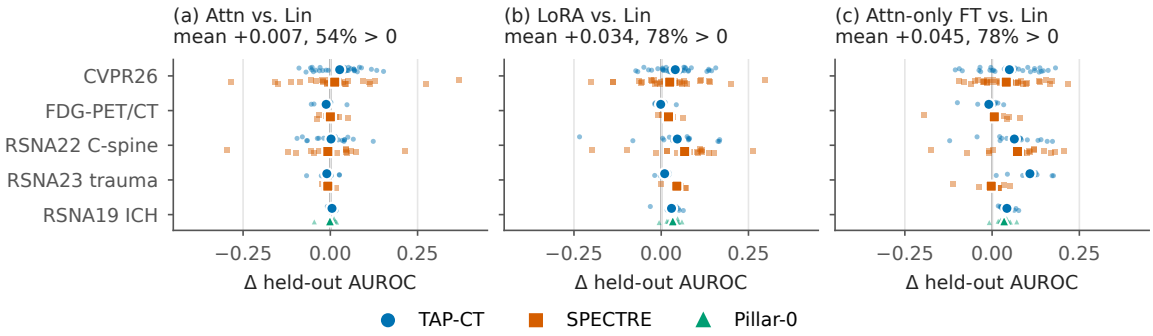


Figure 3: Per-label change in held-out AUROC relative to the linear probe of the same backbone, cohort and mode, for (a) the frozen attentive probe, (b) LoRA and (c) attention-only fine-tuning. Small markers: label–mode pairs; large markers: cohort means. Panel titles give the mean change and the share of pairs above zero over all backbones. All methods are shown in Fig. 10.

4.1. Classification

Across TAP-CT, SPECTRE and Pillar-0, adapting the encoder with LoRA improved held-out AUROC over both frozen probes, by 0.034 over the linear probe and by 0.027 over the frozen attentive probe, which uses the same head (Tab. 7; per backbone in Tab. 1 and Fig. 3). Both 95% intervals exclude zero, and the improvement over the attentive probe was positive in all 10 cohort–backbone pairs. The mean held-out AUROC rose from 0.727 to 0.760 on TAP-CT, from 0.718 to 0.755 on SPECTRE and from 0.887 to 0.920 on Pillar-0, each averaged over that backbone's own labels (Tab. 5). Fine-tuning all parameters of the attention modules showed the same pattern, with gains of 0.045 and 0.038, respectively (Tab. 7). In contrast, the frozen attentive probe improved on the linear probe by only

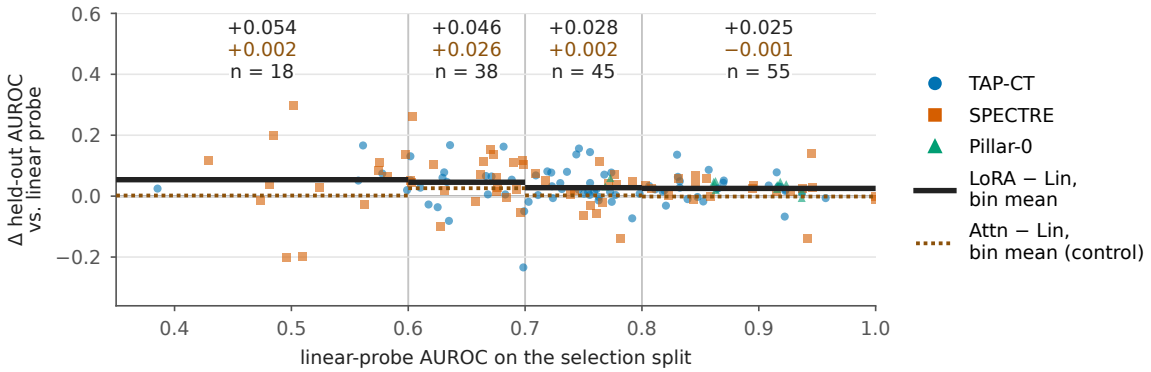


Figure 4: Gain of LoRA over the linear probe in held-out AUROC per label–mode pair, against the linear probe's selection-split AUROC (marker: backbone). Solid segments are bin means over backbones (edges 0.6, 0.7 and 0.8), and dotted segments the same for the frozen attentive probe, a negative control. Above each bin: the LoRA mean, the attentive-probe mean (brown) and the number of pairs.

0.007 (Fig. 3a), which suggests that the gain stems from adapting the encoder rather than from the head. Selecting the learning rate of each probe on the selection data changed the gain of LoRA in the joint models by at most 0.002 pooled over backbones and 0.014 on any backbone (Tab. 17). On Pillar-0, the gain over the linear probe at its selected learning rate was +0.010 (interval 0.000 to +0.021), and the attentive probe selected the lowest learning rate of its grid, 3×10^{-5}, in 8 of 10 pairs. As shown in Fig. 4, the gain was largest for labels on which the linear probe performed worst (0.054 at a selection AUROC of at most 0.6, against 0.025 above 0.8). The gain also persisted after excluding FDG-PET/CT and COVID-CTset, whose labels may be predictable from the acquisition rather than the anatomy (0.037; Tab. 7). LoRA also outperformed the linear probe in 27 of 30 cohort–seed pairs of the repeated joint models (Tab. 13), and when the bootstrap also resampled their seeds, the gain of these joint models was +0.033 (interval +0.007 to +0.065).

At the default settings, fine-tuning all attention parameters achieved the best mean rank over all label–mode pairs (3.91, against 4.76 for LoRA; Fig. 1c) and ranked above LoRA on every backbone. It outperformed LoRA in 25 of 30 cohort–seed pairs of the repeated joint models and by 0.011 on average over the main runs, although this interval includes zero (Tab. 7), as does that of the joint models when their seeds are resampled (−0.010 to +0.040); the difference was largest on TAP-CT (0.019). LoRA updates 19 to 54 times fewer encoder parameters (Fig. 6c), yet this did not make training cheaper. Its peak GPU memory per step was within 1.0 GiB of that of attention-only fine-tuning on every backbone and its steps were not faster (Tab. 4), so the saving lies in the encoder weights stored per model. Fine-tuning the MLP weights instead of the attention weights performed on par with attention-only fine-tuning on TAP-CT and Pillar-0, but remained behind it by 0.060 on SPECTRE (joint models; Tab. 16).

Full fine-tuning required its own setting. With the default setting, it was ahead of the linear probe on TAP-CT by 0.022 on average (interval including zero; Tab. 1), and was the best joint model on RSNA23 trauma (Tab. 20), where Liu et al. (2026) also found full fine-tuning ahead of LoRA for spleen injury. On Pillar-0 it performed on par with the linear probe, whereas on SPECTRE it did not train: its median selection AUROC remained

at 0.529, against 0.743 for attention-only fine-tuning at the same learning rates (Fig. 16). With layer-wise learning-rate decay of 0.75 per block, the best of three alternative settings on selection data, SPECTRE Full FT trained and was ahead of LoRA by 0.017 in the joint models (Tab. 15). However, this comparison favors full fine-tuning, as LoRA was not re-tuned.

On average over backbones, none of the LoRA variants improved on plain LoRA by more than 0.002 (Fig. 10), consistent with the gain of 0.002 AUROC from a mixture of low-rank experts applied to the pooled embedding of a fully fine-tuned Pillar-0 (Yoo et al., 2026), although there the experts act after the encoder rather than inside its blocks. The two-stage schedules and rsLoRA were behind LoRA by 0.017 to 0.022. At a quarter of the learning rate, which matches its update scale to that of LoRA, rsLoRA performed on par with LoRA on SPECTRE and Pillar-0 but remained behind it on TAP-CT by 0.020 (Tab. 16). These rankings hold for one default setting and 8000 steps per method. On CVPR26, a 26 000-step schedule changed the joint models by −0.094 to +0.017 on TAP-CT and by −0.025 to +0.074 on SPECTRE, depending on the method (Tab. 22), which is as large as the differences between methods.

4.2. Segmentation

Replacing the linear probe by the frozen multi-depth decoder improved Dice on every dataset and backbone, by 0.085 on TAP-CT and 0.061 on SPECTRE (Tab. 1; examples in Fig. 18, Appendix G). On TAP-CT, this accounted for all of the gain of LoRA over the linear probe (0.080), and on SPECTRE for most of it (0.085; Fig. 11). The pretrained weights still mattered, as the TAP-CT architecture trained from scratch was behind the fine-tuned pretrained encoder at the same budget on every dataset, by 0.103 to 0.236 Dice (Tab. 19).

On top of the decoder, adapting the encoder changed Dice little on average. LoRA changed Dice relative to the frozen decoder by −0.005 on TAP-CT and +0.023 on SPECTRE, with intervals that include zero (Tab. 1, Fig. 9; per-structure changes in Fig. 8, Appendix C). Its largest gain occurred where the joint frozen decoder fell furthest below the per-structure one, in the SPECTRE AMOS22 joint model (0.658 to 0.784; Tab. 11). Full fine-tuning achieved the best mean rank (2.94, against 3.22 for LoRA; Fig. 1e), driven by its rank on TAP-CT (2.05). However, its mean change on TAP-CT was 0.000, because 2 structure–mode pairs lost more than 0.1, among them the AMOS22 esophagus of the joint model (−0.713; Appendix G). Tripling the steps of the 15-structure joint models also improved their Dice by 0.121 on TAP-CT and 0.147 on SPECTRE (Tab. 11), and 77% of segmentation runs selected one of their last two validations (Fig. 13). These comparisons therefore rank methods at a fixed budget rather than at convergence.

4.3. Per-task and Joint Training

In classification, one joint model per cohort was not detectably different from one model per label on average, at a fraction of the training and with one checkpoint selected for all labels. Joint minus per-task AUROC was +0.004 over TAP-CT, SPECTRE and Pillar-0 (interval −0.021 to +0.030; Tab. 7), whereas a joint model uses $1/K$ of the training steps of the K per-label models it replaces. On Pillar-0 alone, the joint models were slightly behind

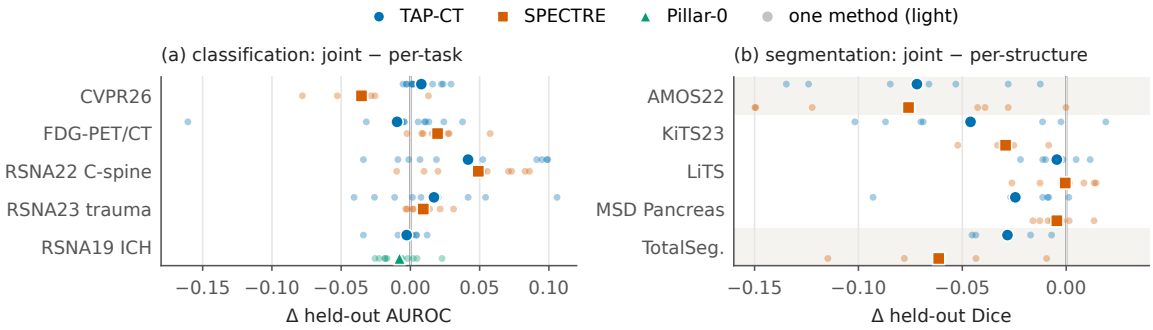


Figure 5: Joint minus per-task training: held-out (a) AUROC per classification cohort and (b) Dice per segmentation dataset. Large markers: mean over methods and labels (structures) per backbone; light markers: individual methods. Shaded rows: 15-structure datasets, whose joint models were trained for 24 000 steps. Values right of zero favor the joint model. Per-label values: Appendix E and Fig. 14.

(−0.008, interval −0.013 to −0.001). Per cohort and averaged over methods, the difference ranged from −0.035 to +0.049 (Fig. 5a; per method in Fig. 14).

In segmentation, in contrast, the joint models lost Dice. The loss was concentrated on small structures and was partly due to the smaller budget per structure. With each structure counted once (Tab. 1), one joint model per dataset changed Dice by −0.046 on TAP-CT and −0.058 on SPECTRE, most on the 15-structure AMOS22 (Fig. 5b). Averaged over structure–method pairs, the joint model lost 0.050 Dice on TAP-CT and 0.058 on SPECTRE. Much of this loss came from 28 of 428 pairs on small structures, namely the adrenal glands and esophagus of AMOS22 and the merged glands of TotalSegmentator, on which the joint model scored below 0.01 while the per-structure model reached at least 0.4 (per-structure tables in Appendix E). Excluding these pairs, the losses were 0.010 and 0.024, respectively. Training crops are centered on a random foreground voxel of all structures together, so small structures rarely lie at the center of a crop. The soft Dice of Equation 3 is also computed per crop with smoothing, so an empty prediction minimizes it for a structure absent from the crop, which may favor empty predictions for small structures in joint models. We tested neither structure-balanced sampling (Isensee et al., 2021) nor a Dice computed over the batch. At matched total compute, with each per-structure model trained for 1600 steps, the joint frozen decoder and joint full fine-tuning remained behind per-structure models in three of the four dataset–backbone pairs (Tab. 18). The joint LoRA model, in contrast, outperformed per-structure LoRA by 0.042 to 0.136 Dice, but per-structure LoRA at 1600 steps reached only 0.617 to 0.691 Dice, against 0.760 to 0.806 for per-structure decoders. The smaller budget per structure therefore explains part, but not all, of the joint models' loss.

5. Discussion and Conclusion

Adapting the encoder helped most on the labels on which the frozen linear probe performed worst (Fig. 4). Part of this trend reflects the smaller headroom of labels above 0.8, but within the lowest bin, where both start from the same linear probe, LoRA gained 0.054 and the frozen attentive probe, which uses the same head, only 0.002; the pretrained features

therefore appear to lack information that a stronger head cannot recover but a small update of the encoder can. LoRA reduces the number of weights stored per model but not the memory or time of training, since, like attention-only fine-tuning, it back-propagates through every encoder block (Tab. 4).

Our study has several limitations. Each method was trained with a single default setting and a fixed budget, with rank-16 adapters on the query, key and value projections, and only full fine-tuning was re-tuned after it failed; the ranking among the stronger methods may therefore change with method-specific tuning or longer training, and the segmentation models had not converged. Segmentation was scored with Dice only, on each backbone's working grid. Finally, SPECTRE saw several of our datasets, without labels, during pretraining, and the AMOS release gives no patient identifiers. We discuss these limitations, and the work that would address each of them, in Appendix H.

Based on our findings, at the default settings and budgets used here, we propose the following guidelines for adapting 3D CT foundation models: (1) For classification, adapt the encoder rather than training a head on frozen features; rank-16 LoRA on every block improved AUROC by 0.034 over the linear probe, and fine-tuning all attention parameters was slightly better at similar memory and time per step (Tab. 4), whereas LoRA updates 19 to 54 times fewer encoder weights (Fig. 6c). (2) For classification, start with one joint model per cohort, which was not detectably different from per-label models on average at $1/K$ of the training, and compare the two on selection data when the budget allows. (3) For segmentation, place a multi-depth decoder (here without input skip connections) on the frozen encoder before adapting the encoder, and inspect Dice per structure when fine-tuning. (4) For segmentation with a per-crop soft Dice and crops centered on the foreground of all structures, train one model per structure when the budget allows, and otherwise check the Dice of small structures, on which our joint models failed. (5) When fully fine-tuning, select a lower learning rate or layer-wise learning-rate decay on selection data and verify that held-out predictions vary between cases.

References

Kumar Krishna Agrawal, Longchao Liu, Long Lian, Michael Nercessian, Natalia Harguindeguy, Yufu Wu, et al. Pillar-0: A new frontier for radiology foundation models. *arXiv preprint arXiv:2511.17803*, 2025.

Michela Antonelli, Annika Reinke, Spyridon Bakas, Keyvan Farahani, Annette Kopp-Schneider, Bennett A. Landman, et al. The medical segmentation decathlon. *Nature Communications*, 13(1):4128, 2022. doi: 10.1038/s41467-022-30695-9.

Mohammed Baharoon, Waseem Qureshi, Jiahong Ouyang, Yanwu Xu, Abdulrhman Aljouie, and Wei Peng. Evaluating general purpose vision foundation models for medical image analysis: An experimental study of DINOv2 on radiology benchmarks. *arXiv preprint arXiv:2312.02366*, 2023.

Adrien Bardes, Quentin Garrido, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, et al. Revisiting feature prediction for learning visual representations from video. *Transactions on Machine Learning Research*, 2024. URL https://openreview.net/forum?id=QaCCuDfBk2. arXiv:2404.08471.

Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.

Patrick Bilic, Patrick Christ, Hongwei Bran Li, Eugene Vorontsov, Avi Ben-Cohen, Georgios Kaissis, et al. The liver tumor segmentation benchmark (LiTS). *Medical Image Analysis*, 84:102680, 2023. doi: 10.1016/j.media.2022.102680.

Louis Blankemeier, Ashwin Kumar, Joseph Paul Cohen, Jiaming Liu, Longchao Liu, Dave Van Veen, et al. Merlin: a computed tomography vision–language foundation model and dataset. *Nature*, 652(8112):1318–1328, 2026. doi: 10.1038/s41586-026-10181-8.

Cheng Chen, Juzheng Miao, Dufan Wu, Aoxiao Zhong, Zhiling Yan, Sekeun Kim, et al. MA-SAM: Modality-agnostic SAM adaptation for 3D medical image segmentation. *Medical Image Analysis*, 98:103310, 2024a. doi: 10.1016/j.media.2024.103310.

Xiaokang Chen, Mingyu Ding, Xiaodi Wang, Ying Xin, Shentong Mo, Yunhao Wang, et al. Context autoencoder for self-supervised representation learning. *International Journal of Computer Vision*, 132(1):208–223, 2024b. doi: 10.1007/s11263-023-01852-4.

Cris Claessens, Christiaan Viviers, Giacomo D'Amicantonio, Egor Bondarev, and Fons van der Sommen. Scaling self-supervised and cross-modal pretraining for volumetric CT transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 13636–13647, 2026. arXiv:2511.17209.

Timothée Darcet, Maxime Oquab, Julien Mairal, and Piotr Bojanowski. Vision transformers need registers. In *International Conference on Learning Representations (ICLR)*, 2024. URL https://openreview.net/forum?id=2dnO3LLiJ1.

Elizabeth R. DeLong, David M. DeLong, and Daniel L. Clarke-Pearson. Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach. *Biometrics*, 44(3):837–845, 1988. doi: 10.2307/2531595.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021. arXiv:2010.11929.

Raman Dutt, Linus Ericsson, Pedro Sanchez, Sotirios A. Tsaftaris, and Timothy Hospedales. Parameter-efficient fine-tuning for medical image analysis: The missed opportunity. In *Proceedings of the 7th International Conference on Medical Imaging with Deep Learning*, volume 250 of *Proceedings of Machine Learning Research*, pages 406–425. PMLR, 2024. URL https://proceedings.mlr.press/v250/dutt24a.html.

Marc Fischer, Alexander Bartler, and Bin Yang. Prompt tuning for parameter-efficient medical image segmentation. *Medical Image Analysis*, 91:103024, 2024. doi: 10.1016/j.media.2023.103024.

Adam E. Flanders, Luciano M. Prevedello, George Shih, Safwan S. Halabi, Jayashree Kalpathy-Cramer, Robyn Ball, et al. Construction of a machine learning dataset through collaboration: The RSNA 2019 brain CT hemorrhage challenge. *Radiology: Artificial Intelligence*, 2(3):e190211, 2020. doi: 10.1148/ryai.2020190211.

Sergios Gatidis, Tobias Hepp, Marcel Früh, Christian La Fougère, Konstantin Nikolaou, Christina Pfannenberg, et al. A whole-body FDG-PET/CT dataset with manually annotated tumor lesions. *Scientific Data*, 9(1):601, 2022. doi: 10.1038/s41597-022-01718-3.

Shizhan Gong, Yuan Zhong, Wenao Ma, Jinpeng Li, Zhao Wang, Jingyang Zhang, et al. 3DSAM-adapter: Holistic adaptation of SAM from 2D to 3D for promptable tumor segmentation. *Medical Image Analysis*, 98:103324, 2024. doi: 10.1016/j.media.2024.103324.

Ibrahim Ethem Hamamci, Sezgin Er, Chenyu Wang, Furkan Almas, Ayse Gulnihan Simsek, Sevval Nil Esirgun, et al. Generalist foundation models from a multimodal dataset for 3D computed tomography. *Nature Biomedical Engineering*, 10(8):1610–1628, 2026. doi: 10.1038/s41551-025-01599-y.

Ali Hatamizadeh, Yucheng Tang, Vishwesh Nath, Dong Yang, Andriy Myronenko, Bennett Landman, et al. UNETR: Transformers for 3D medical image segmentation. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 1748–1758, 2022. doi: 10.1109/WACV51458.2022.00181.

Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15979–15988, 2022. doi: 10.1109/CVPR52688.2022.01553.

Yufan He, Pengfei Guo, Yucheng Tang, Andriy Myronenko, Vishwesh Nath, Ziyue Xu, et al. VISTA3D: A unified segmentation foundation model for 3D medical imaging. In

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pages 20863–20873, 2025. doi: 10.1109/CVPR52734.2025.01943.

Nicholas Heller, Fabian Isensee, Dasha Trofimova, Resha Tejpaul, Zhongchen Zhao, Huai Chen, et al. The KiTS21 challenge: Automatic segmentation of kidneys, renal tumors, and renal cysts in corticomedullary-phase CT. *arXiv preprint arXiv:2307.01984*, 2023.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, et al. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022. URL https://openreview.net/forum?id=nZeVKeeFYf9. arXiv:2106.09685.

Maximilian Ilse, Jakub M. Tomczak, and Max Welling. Attention-based deep multiple instance learning. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 2127–2136. PMLR, 2018. URL https://proceedings.mlr.press/v80/ilse18a.html.

Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl, Jens Petersen, and Klaus H. Maier-Hein. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 18(2):203–211, 2021. doi: 10.1038/s41592-020-01008-z.

Yuanfeng Ji, Haotian Bai, Chongjian Ge, Jie Yang, Ye Zhu, Ruimao Zhang, et al. AMOS: A large-scale abdominal multi-organ benchmark for versatile medical image segmentation. In *Advances in Neural Information Processing Systems*, volume 35, pages 36722–36732, 2022. doi: 10.52202/068431-2661.

Menglin Jia, Luming Tang, Bor-Chun Chen, Claire Cardie, Serge Belongie, Bharath Hariharan, et al. Visual prompt tuning. In *Computer Vision – ECCV 2022*, Lecture Notes in Computer Science, pages 709–727. Springer Nature Switzerland, 2022. doi: 10.1007/978-3-031-19827-4_41.

Damjan Kalajdzievski. A rank stabilization scaling factor for fine-tuning with LoRA. *arXiv preprint arXiv:2312.03732*, 2023.

Febryan Putra Kartika, Cheng-Yu Ma, Ying-Jia Lin, Chi-Tung Cheng, and Kuan-Fu Chen. Efficient self-supervised adaptation of 3D abdominal vision-language model for institution-specific HCC classification via full fine-tuning and PEFT. In *Proceedings of The 9th International Conference on Medical Imaging with Deep Learning*, volume 315 of *Proceedings of Machine Learning Research*, pages 3245–3269. PMLR, 2026. URL https://proceedings.mlr.press/v315/kartika26a.html.

Sangwook Kim, Thomas G. Purdie, and Chris McIntosh. Cross-task attention network: Improving multi-task learning for medical imaging applications. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2023 Workshops*, Lecture Notes in Computer Science, pages 119–128. Springer Nature Switzerland, 2023. doi: 10.1007/978-3-031-47401-9_12.

Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, et al. Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 3992–4003, 2023. doi: 10.1109/ICCV51070.2023.00371.

Ananya Kumar, Aditi Raghunathan, Robbie Jones, Tengyu Ma, and Percy Liang. Fine-tuning can distort pretrained features and underperform out-of-distribution. In *International Conference on Learning Representations (ICLR)*, 2022. arXiv:2202.10054.

Wenxuan Li, Alan Yuille, and Zongwei Zhou. How well do supervised 3D models transfer to medical imaging tasks? In *International Conference on Learning Representations (ICLR)*, 2024. URL https://openreview.net/forum?id=AhizIPytk4.

Hui Ming Lin, Errol Colak, Tyler Richards, Felipe C. Kitamura, Luciano M. Prevedello, Jason Talbott, et al. The RSNA cervical spine fracture CT dataset. *Radiology: Artificial Intelligence*, 5(5):e230034, 2023. doi: 10.1148/ryai.230034.

Han Liu, Bogdan Georgescu, Yanbo Zhang, Youngjin Yoo, Michael Baumgartner, Riqiang Gao, et al. Revisiting 2D foundation models for scalable 3D medical image classification. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 30021–30031, 2026. arXiv:2512.12887.

Shih-Yang Liu, Chien-Yi Wang, Hongxu Yin, Pavlo Molchanov, Yu-Chiang Frank Wang, Kwang-Ting Cheng, et al. DoRA: Weight-decomposed low-rank adaptation. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 32100–32121. PMLR, 2024. URL https://proceedings.mlr.press/v235/liu24bn.html.

Son Thai Ly and Hien V. Nguyen. Self-auditing parameter-efficient fine-tuning for few-shot 3D medical image segmentation. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2026*, Lecture Notes in Computer Science, pages 559–569. Springer Nature Switzerland, 2026. doi: 10.1007/978-3-032-38069-2_53. arXiv:2603.05822.

Jun Ma, Yao Zhang, Song Gu, Cheng Zhu, Cheng Ge, Yichi Zhang, et al. AbdomenCT-1K: Is abdominal organ segmentation a solved problem? *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(10):6695–6714, 2022. doi: 10.1109/TPAMI.2021.3100536.

Lena Maier-Hein, Annika Reinke, Patrick Godau, Minu D. Tizabi, Florian Buettner, Evangelia Christodoulou, et al. Metrics reloaded: recommendations for image analysis validation. *Nature Methods*, 21(2):195–212, 2024. doi: 10.1038/s41592-023-02151-z.

Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024. URL https://openreview.net/forum?id=a68SUt6zFt. arXiv:2304.07193.

Suraj Pai, Ibrahim Hadzic, Dennis Bontempi, Keno Bressem, Benjamin H. Kann, Andriy Fedorov, et al. Vision foundation models for computed tomography. *arXiv preprint arXiv:2501.09001*, 2025.

Dré Peeters, Bogdan Obreja, Noa Antonissen, and Colin Jacobs. The LUNA25 challenge: Public training and development set – imaging data, 2025. URL https://doi.org/10.5281/zenodo.14223624. Version 1.0.0.

Dré Peeters, Bogdan Obreja, Noa Antonissen, Zaigham Saghir, Ugo Pastorino, Mario Silva, et al. Benchmarking of AI and radiologists for indeterminate lung nodule malignancy risk estimation at screening CT: The LUNA25 challenge. *Radiology: Artificial Intelligence*, 8 (5):e260179, 2026. doi: 10.1148/ryai.260179.

Chongyu Qu, Tiezheng Zhang, Hualin Qiao, Jie Liu, Yucheng Tang, Alan Yuille, et al. AbdomenAtlas-8K: Annotating 8,000 CT volumes for multi-organ segmentation in three weeks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2305.09666.

Mohammad Rahimzadeh, Abolfazl Attar, and Seyed Mohammad Sakhaei. A fully automated deep learning-based network for detecting COVID-19 from a new and large lung CT scan dataset. *Biomedical Signal Processing and Control*, 68:102588, 2021. doi: 10.1016/j.bspc.2021.102588.

Jeffrey D. Rudie, Hui-Ming Lin, Robyn L. Ball, Sabeena Jalal, Luciano M. Prevedello, Savvas Nicolaou, et al. The RSNA abdominal traumatic injury CT (RATIC) dataset. *Radiology: Artificial Intelligence*, 6(6):e240101, 2024. doi: 10.1148/ryai.240101.

Julio Silva-Rodríguez, Jose Dolz, and Ismail Ben Ayed. Towards foundation models and few-shot parameter-efficient fine-tuning for volumetric organ segmentation. *Medical Image Analysis*, 103:103596, 2025. doi: 10.1016/j.media.2025.103596.

Hugo Touvron, Matthieu Cord, Alaaeldin El-Nouby, Jakob Verbeek, and Hervé Jégou. Three things everyone should know about vision transformers. In *Computer Vision – ECCV 2022*, Lecture Notes in Computer Science, pages 497–515. Springer Nature Switzerland, 2022. doi: 10.1007/978-3-031-20053-3_29.

Tim Veenboer, George Yiasemis, Eric Marcus, Vivien van Veldhuizen, Cees G. M. Snoek, Jonas Teuwen, et al. TAP-CT: 3D task-agnostic pretraining of computed tomography foundation models. In *Proceedings of The 9th International Conference on Medical Imaging with Deep Learning*, volume 315 of *Proceedings of Machine Learning Research*, pages 726–753. PMLR, 2026. URL https://proceedings.mlr.press/v315/veenboer26a.html. arXiv:2512.00872.

Jakob Wasserthal. Dataset with segmentations of 117 important anatomical structures in 1939 CT images, 2026. URL https://doi.org/10.5281/zenodo.22688904. Version 3.0.0.

Jakob Wasserthal, Hanns-Christian Breit, Manfred T. Meyer, Maurice Pradella, Daniel Hinck, Alexander W. Sauter, et al. TotalSegmentator: Robust segmentation of 104 anatomic structures in CT images. *Radiology: Artificial Intelligence*, 5(5):e230024, 2023. doi: 10.1148/ryai.230024.

Junde Wu, Ziyue Wang, Mingxuan Hong, Wei Ji, Huazhu Fu, Yanwu Xu, et al. Medical SAM adapter: Adapting segment anything model for medical image segmentation. *Medical Image Analysis*, 102:103547, 2025. doi: 10.1016/j.media.2025.103547.

Youngjin Yoo, Han Liu, Bogdan Georgescu, Yanbo Zhang, Sasa Grbic, Michael Baumgartner, et al. Specializing foundation models via mixture of low-rank experts for comprehensive head CT analysis. *arXiv preprint arXiv:2603.00675*, 2026.

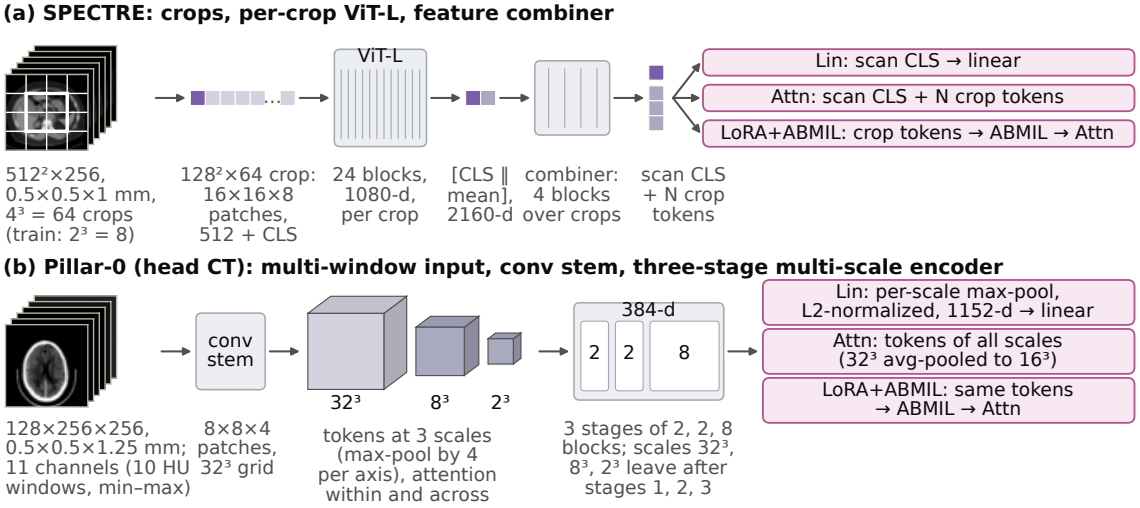
Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. Gradient surgery for multi-task learning. In *Advances in Neural Information Processing Systems*, volume 33, 2020. URL https://papers.nips.cc/paper_files/paper/2020/hash/3fe78a8acf5fda99de95303940a2420c-Abstract.html.

Yan Zhao, Xiuying Wang, Tongtong Che, Guoqing Bao, and Shuyu Li. Multi-task deep learning for medical image computing and analysis: A review. *Computers in Biology and Medicine*, 153:106496, 2023. doi: 10.1016/j.compbiomed.2022.106496.

Weicheng Zhu, Haoxu Huang, Huanze Tang, Rushabh Musthyala, Boyang Yu, Long Chen, et al. 3D foundation model for generalizable disease detection in head computed tomography. *Nature Biomedical Engineering*, 2026. doi: 10.1038/s41551-026-01668-w.

Yitao Zhu, Zhenrong Shen, Zihao Zhao, Sheng Wang, Xin Wang, Xiangyu Zhao, et al. MeLo: Low-rank adaptation is better than fine-tuning for medical image diagnosis. In *IEEE International Symposium on Biomedical Imaging (ISBI)*, pages 1–5, 2024. doi: 10.1109/ISBI56570.2024.10635615.

Appendix A. Training and evaluation details



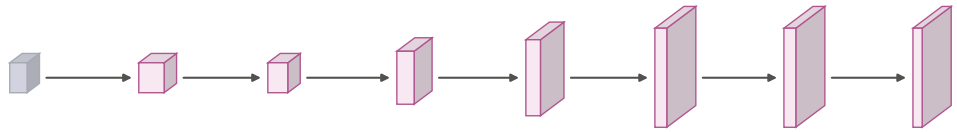
(c) What each classification method trains

	early blocks				late blocks					encoder parameters updated		
	patch emb.	W_{qkv}	W_{proj}	MLP, norms	W_{qkv}	W_{proj}	MLP, norms	head	steps	TAP-CT	SPECTRE	Pillar-0
Lin								pooled → linear	8000	0	0	0
Attn								attentive probe	8000	0	0	0
LoRA								attentive probe	8000	0.59 M	2.4 M	0.55 M*
LoRA$_{late}$								attentive probe	8000	0.20 M	1.0 M	0.25 M*
LoRA+ABMIL								ABMIL → Attn	8000	0.59 M	2.4 M	0.55 M*
DoRA (+ m)		m			m			attentive probe	8000	0.62 M	2.5 M	0.57 M*
rsLoRA (α/√r)								attentive probe	8000	0.59 M	2.4 M	0.55 M*
Lin→LoRA					2			linear	4000 + 4000	0.20 M	1.0 M	0.25 M*
Attn→LoRA					2			attentive probe	4000 + 4000	0.20 M	1.0 M	0.25 M*
Attn-only FT								attentive probe	8000	28 M	131 M	11 M*
Full FT								attentive probe	8000	87 M	397 M	45 M*

frozen rank-16 adapter (m: DoRA magnitude; 2: added in stage 2) trained in full

Figure 6: Classification with SPECTRE and Pillar-0; TAP-CT is shown in Fig. 2. (a, b) How each backbone turns a volume into the tokens that the heads read. SPECTRE encodes crops separately and summarizes them with its pretrained feature combiner, whereas Pillar-0 encodes the whole head CT with a convolutional stem and a three-stage multi-scale transformer. (c) The encoder components that each of the eleven methods trains, its head, its number of steps and the number of encoder parameters that receive a gradient. For Pillar-0, this count excludes per-scale blocks that never run; ratios in the text use unrounded counts. The CVPR26 joint models were also trained for 26 000 steps (13 000 + 13 000).

(a) Segmentation decoder: tensor shapes (channels × depth × height × width)



	block tokens (4 taps)	1×1×1 conv, concatenate	3^3 conv	upsample, 3^3 conv	upsample, 3^3 conv	upsample, 3^3 conv	upsample, 3^3 conv	1×1×1 conv
TAP-CT	$768\times3\times28^2$	$4\cdot128\times3\times28^2$	$128\times3\times28^2$	$64\times6\times56^2$	$32\times12\times112^2$	$16\times12\times224^2$	–	$C\times12\times224^2$
SPECTRE	1080×8^3	$4\cdot128\times8^3$	128×8^3	64×16^3	32×32^3	16×64^3	$16\times64\times128^2$	$C\times64\times128^2$

(b) What each segmentation method trains

	early blocks				late blocks				encoder parameters updated		
	patch emb.	W_{qkv}	W_{proj}	MLP, norms	W_{qkv}	W_{proj}	MLP, norms	head	steps	TAP-CT	SPECTRE
Lin								1×1×1 conv	8000	0	0
Dec								decoder	8000	0	0
LoRA								decoder	8000	0.59 M	2.1 M*
LoRA$_{\text{late}}$								decoder	8000	0.20 M	0.69 M
DoRA (+ m)		m			m			decoder	8000	0.62 M	2.2 M*
Dec→LoRA		2			2			decoder	4000 + 4000	0.59 M	2.1 M*
Full FT								decoder	8000	87 M	339 M*

frozen rank-16 adapter (m: DoRA magnitude; 2: added in stage 2) trained in full

Figure 7: Segmentation decoder and methods. (a) Tensor shapes through the decoder for TAP-CT and SPECTRE, from the four encoder taps to the C output maps, one per structure. (b) The encoder components that each of the seven methods trains and the number of encoder parameters that receive a gradient. The decoder reads blocks 3, 6, 9 and 12 of TAP-CT or 6, 12, 18 and 24 of SPECTRE after the final normalization; each 3^3 convolution is followed by instance normalization and a leaky ReLU, upsampling is trilinear along the axes whose stride is still above 1, and there are no skip connections from the input. The decoder has 2.45 M (TAP-CT) and 2.62 M (SPECTRE) parameters for $C = 15$, and Lin, a $1 \times 1 \times 1$ convolution on the deepest tap followed by trilinear upsampling, has 769 and 1081 parameters per structure; C is 1 per structure or the number of structures in joint training. The AMOS22 and TotalSegmentator joint models were trained for 24 000 steps (12 000 + 12 000 for the two-stage method). On SPECTRE, LoRA, DoRA, Dec→LoRA and Full FT also wrap or unfreeze the feature combiner, which never enters the segmentation features and receives no gradient; the counts exclude it, and including it gives 2.4 M, 2.5 M, 2.4 M and 397 M parameters, respectively.

Input Preprocessing Every volume is reoriented to the frame of the backbone's pre-training and resampled to its grid. TAP-CT: LPS, native slice spacing, in-plane trilinear resize to 224×224, depth padded with air to at least 12 slices and to a multiple of 4; Hounsfield units (HU) clipped to $[-1008, 822]$ and standardized with the model's mean and standard deviation; windows of 12 slices with a stride of 3 slices plus a final window ending at the last slice (at most 32 windows per training step, all windows at evaluation). SPECTRE: RAS, $0.5 \times 0.5 \times 1$ mm, centered $512 \times 512 \times 256$ field of view padded with

the scan minimum, HU in $[-1000, 1000]$ scaled to $[0, 1]$; a $4 \times 4 \times 4$ grid of 64 crops of $128 \times 128 \times 64$ voxels at evaluation and a random contiguous $2 \times 2 \times 2$ sub-grid of 8 crops at training; each crop is summarized by its CLS token concatenated with its mean patch token before the feature combiner. Pillar-0: LPI, $0.5 \times 0.5 \times 1.25$ mm, centered $128 \times 256 \times 256$ volume, eleven input channels (ten fixed center/width windows and a per-volume min–max channel), one forward pass; the linear head reads the model's L2-normalized pooled embedding (three scales concatenated, 1152-d), the attentive head reads the tokens of the three scales, any scale with more than 4096 tokens being average-pooled on its grid. For segmentation, TAP-CT keeps the classification grid (in-plane 224, native depth) and crops of $12 \times 224 \times 224$ voxels; SPECTRE resamples to 1 mm and uses crops of $64 \times 128 \times 128$ voxels. A training crop is centered on a random foreground voxel of the run's structures with probability 0.67 and placed uniformly otherwise; volumes smaller than the crop are padded with the image minimum. At evaluation the whole volume is covered by a sliding window with Gaussian weighting (overlap 0.5 for TAP-CT, 0.25 for SPECTRE) and Dice is computed on the backbone's working grid, without post-processing.

Heads The linear head is dropout (0.2) followed by one linear layer on the pooled feature: the mean of the window CLS tokens for TAP-CT (768-d), the feature combiner's CLS token for SPECTRE (1080-d) and the pooled image embedding for Pillar-0 (1152-d). The attentive probe prepends a learnable CLS token to the token set (the window CLS tokens for TAP-CT, the combiner tokens for SPECTRE and the multi-scale tokens for Pillar-0), applies two pre-norm transformer blocks with eight-head self-attention and a GELU feed-forward layer of ratio 4, normalizes the CLS output and maps it through dropout (0.2) and a linear layer. For one label, the linear head has 769, 1081 and 1153 parameters and the attentive probe 14.2, 28.0 and 31.9 M on TAP-CT, SPECTRE and Pillar-0, respectively. LoRA+ABMIL pools the window or crop tokens by gated attention (Ilse et al., 2018), $a_i = \mathrm{softmax}_i\big(\boldsymbol{w}^\top(\tanh(\boldsymbol{V}\boldsymbol{x}_i) \odot \sigma(\boldsymbol{U}\boldsymbol{x}_i))\big)$ with hidden size 128, and feeds the pooled token to the attentive probe. The segmentation linear probe is a $1 \times 1 \times 1$ convolution on the deepest grid (TAP-CT block 12, SPECTRE block 24), upsampled trilinearly. The decoder projects each of the four grids to 128 channels with a $1 \times 1 \times 1$ convolution, resizes them to a common grid (the token grid), concatenates them (512 channels), fuses them with a 3^3 convolution, instance normalization and leaky ReLU to 128 channels, then applies stages of trilinear $\times 2$ upsampling followed by the same convolution block that halves the channels (to a minimum of 16) until the voxel grid is reached (three stages for TAP-CT, four for SPECTRE), and ends with a $1 \times 1 \times 1$ convolution to one logit per structure. It has 2.45 M (TAP-CT) and 2.62 M (SPECTRE) parameters plus $17C$ or $33C$ for C structures, and no skip connections from the input.

Adapters LoRA (Hu et al., 2022) adds $\frac{\alpha}{r}\boldsymbol{B}\boldsymbol{A}\,\mathrm{drop}(\boldsymbol{x})$ to the output $\boldsymbol{W}_0\boldsymbol{x}$ of a frozen projection, with $r = 16$, $\alpha = 32$ and dropout 0.1; $\boldsymbol{A}$ is initialized Kaiming-uniform and $\boldsymbol{B}$ at zero, so training starts from the pretrained model. It is applied to the fused query/key/value projection of every TAP-CT block and to the separate query and key/value projections of every SPECTRE block, of SPECTRE's feature combiner, and of every Pillar-0 block. Late adapters cover TAP-CT blocks 9–12, SPECTRE blocks 17–24 and its feature combiner and the third stage of Pillar-0 (8 of 12 blocks). rsLoRA scales the update by $\alpha/\sqrt{r}$ instead of α/r. DoRA (Liu et al., 2024) rescales each row of the adapted weight to a learned

magnitude $\boldsymbol{m}$, $\boldsymbol{W} = \boldsymbol{m} \odot (\boldsymbol{W}_0 + \frac{\alpha}{r}\boldsymbol{B}\boldsymbol{A})/\|\boldsymbol{W}_0 + \frac{\alpha}{r}\boldsymbol{B}\boldsymbol{A}\|_{\text{row}}$, with $\boldsymbol{m}$ initialized to the row norms of $\boldsymbol{W}_0$; the magnitude is a two-dimensional parameter and receives weight decay like the adapter matrices. Attention-only fine-tuning unfreezes every parameter of the attention modules that LoRA would adapt (query, key, value and output projections with their biases). On Pillar-0, the per-scale blocks of scales that have already left the encoder and the text projection are also wrapped or unfrozen, but they never run and receive no gradient; counting them, LoRA updates 1.1 M, $\text{LoRA}_{\text{late}}$ 0.74 M, Attn-only FT 21 M and Full FT 94 M parameters (Fig. 6c excludes them). In segmentation the SPECTRE adapters and full fine-tuning also cover the feature combiner, which the segmentation path does not evaluate; those parameters receive no gradient.

Two-stage Methods Lin→LoRA and Attn→LoRA take the probe weights from the parent run's checkpoint at step 4000 of its 8000-step schedule, add late-block LoRA with $\boldsymbol{B} = 0$, and train 4000 further steps with a fresh optimizer and schedule. Dec→LoRA takes the decoder from the Dec run at the midpoint of its schedule (step 4000; step 12 000 for the 24 000-step joint models) and trains all-block LoRA with the decoder for the second half.

Optimization We used AdamW with the default $\beta = (0.9, 0.999)$ and $\epsilon = 10^{-8}$ in all runs. The learning rate rises linearly from 1% of its peak over the first 10% of the steps and then follows a cosine to 10^{-6}. Weight decay is applied to parameters of dimension two or more, excluding biases and normalization weights. Gradients are not clipped. Classification trains at batch size 1 without gradient accumulation; segmentation at a micro-batch of 2 crops with accumulation 2, so four crops per step. A training step uses B cases $b = 1, \dots, B$ ($B = 1$ in classification; $B = 2$ crops per micro-batch in segmentation).

Model outputs. A classification cohort has K binary labels $t = 1, \dots, K$. For case b, a per-task model for label t outputs one logit $z_{bt} \in \mathbb{R}$, and a joint model outputs one logit per label, $\boldsymbol{z}_b = (z_{b1}, \dots, z_{bK})$; the reference is $y_{bt} \in \{0, 1\}$. A segmentation dataset has C structures $c = 1, \dots, C$. For a crop b of V voxels, a per-structure model for structure c outputs one logit map $\boldsymbol{z}_{bc} \in \mathbb{R}^V$, and a joint model outputs one map per structure, $\boldsymbol{z}_{b1}, \dots, \boldsymbol{z}_{bC}$; the reference is the binary mask $\boldsymbol{y}_{bc} \in \{0, 1\}^V$. Both tasks use the binary cross-entropy of a logit z with reference y and positive weight w,

$$\text{BCE}(z, y; w) = -\, w\, y \log \sigma(z) - (1 - y) \log \big(1 - \sigma(z)\big), \tag{1}$$

where σ is the logistic sigmoid.

Classification. A per-task model for label t is trained only on the cases annotated for label t, and a joint model on all cases of the cohort, with the losses

$$\mathcal{L}_{\text{cls}}^{\text{per-task}}(t) = \frac{1}{B} \sum_{b=1}^{B} \text{BCE}(z_{bt}, y_{bt}; w_t), \qquad w_t = \text{clip}\Big(\frac{n_t^-}{n_t^+},\, 0.05,\, 20\Big), \tag{2a}$$

$$\mathcal{L}_{\text{cls}}^{\text{joint}} = \frac{1}{B} \sum_{b=1}^{B} \sum_{t=1}^{K} m_{bt}\, \text{BCE}(z_{bt}, y_{bt}; w_t), \qquad m_{bt} = \mathbf{1}\big[\, y_{bt} \text{ is annotated} \,\big], \tag{2b}$$

where n_t^- and n_t^+ are the numbers of negative and positive training cases of label t, and the mask m_{bt} is 1 if label t is annotated for case b and 0 otherwise. Every case is annotated for every label of its cohort ($m_{bt} = 1$), except in CVPR26 ($K = 17$), where each COVID-CTset

and LUNA25 case is annotated for its own label only and each AMOS case for a subset of the 15 AMOS labels.

Segmentation. For one structure of one crop, the loss is the soft Dice loss plus the voxel-averaged, unweighted binary cross-entropy,

$$\ell(\boldsymbol{z}, \boldsymbol{y}) = 1 - \mathrm{Dice}\big(\sigma(\boldsymbol{z}), \boldsymbol{y}\big) + \frac{1}{V}\sum_{v=1}^{V} \mathrm{BCE}(z_v, y_v; 1), \tag{3}$$

$$\mathrm{Dice}(\boldsymbol{p}, \boldsymbol{y}) = \frac{2\sum_v p_v y_v + 1}{\sum_v p_v + \sum_v y_v + 1},$$

where Dice is the soft Dice coefficient of the voxel probabilities $\boldsymbol{p} = \sigma(\boldsymbol{z})$ with a smoothing constant of 1. A per-structure model for structure c and a joint model are trained on

$$\mathcal{L}_{\mathrm{seg}}^{\text{per-structure}}(c) = \frac{1}{B}\sum_{b=1}^{B} \ell(\boldsymbol{z}_{bc}, \boldsymbol{y}_{bc}), \tag{4a}$$

$$\mathcal{L}_{\mathrm{seg}}^{\text{joint}} = \frac{1}{BC}\sum_{b=1}^{B}\sum_{c=1}^{C} \ell(\boldsymbol{z}_{bc}, \boldsymbol{y}_{bc}). \tag{4b}$$

The joint classification loss thus sums over labels, whereas the joint segmentation loss averages over structures. In the joint CVPR26 models the three sources are sampled in the ratio 10:8:8 (AMOS, COVID-CTset, LUNA25); every other run cycles one shuffled training set.

Table 2: Classification settings per method. LoRA-type adapters have rank 16, $\alpha = 32$ and dropout 0.1. The learning rate and weight decay apply to every trained parameter unless separate values are given for the encoder and the head. All runs use AdamW for 8000 steps (two-stage methods: 4000 + 4000) at batch size 1, with a 10% warm-up, cosine decay to 10^{-6}, bf16 precision and a head dropout of 0.2.

Method	Trained in the encoder	Head	Learning rate / weight decay
Lin	nothing	linear	10^{-3} / 0
Attn	nothing	attentive	10^{-4} / 0.01
LoRA	LoRA on q, k, v of every block	attentive	10^{-4} / 0.01
$\mathrm{LoRA}_{\mathrm{late}}$	LoRA on q, k, v of the late blocks	attentive	10^{-4} / 0.01
LoRA+ABMIL	as LoRA	ABMIL then attentive	10^{-4} / 0.01
DoRA	DoRA on q, k, v of every block	attentive	10^{-4} / 0.01
rsLoRA	LoRA, scale $\alpha/\sqrt{r}$	attentive	10^{-4} / 0.01
Lin→LoRA	late LoRA from step 4000 of Lin	linear (inherited)	10^{-4} / 0.01
Attn→LoRA	late LoRA from step 4000 of Attn	attentive (inherited)	10^{-4} / 0.01
Attn-only FT	all attention parameters	attentive	encoder 2×10^{-5} / 0.05, head 10^{-4} / 0.01
Full FT	every parameter	attentive	encoder 2×10^{-5} / 0.05, head 10^{-4} / 0.01
Repeats and controls (Appendix D)			
Full FT, lower learning rate	every parameter	attentive	encoder 5×10^{-6} / 0.05, head 10^{-4} / 0.01
Full FT, layer decay	every parameter, learning rate ×0.75 per block from the top	attentive	encoder 2×10^{-5} / 0.05, head 10^{-4} / 0.01
Full FT, accumulation	every parameter; accumulation 8, clipping 1.0, 1000 steps	attentive	as Full FT
MLP-only FT	both MLP layers of every block	attentive	encoder 2×10^{-5} / 0.05, head 10^{-4} / 0.01
rsLoRA, matched learning rate	LoRA, scale $\alpha/\sqrt{r}$	attentive	2.5×10^{-5} / 0.01

Table 3: Segmentation settings per method. All runs use AdamW for 8000 steps (24 000 for the AMOS22 and TotalSegmentator joint models; two-stage methods split the budget in halves) with four crops per step, a 10% warm-up, cosine decay to 10^{-6} and bf16 precision. LoRA-type adapters are configured as in Tab. 2.

Method	Trained in the encoder	Head	Learning rate / weight decay
Lin	nothing	linear probe	10^{-2} / 0
Dec	nothing	decoder	10^{-3} / 10^{-4}
LoRA	LoRA on q, k, v of every block	decoder	10^{-4} / 10^{-4}
$\text{LoRA}_{\text{late}}$	LoRA on q, k, v of the late blocks	decoder	10^{-4} / 10^{-4}
DoRA	DoRA on q, k, v of every block	decoder	10^{-4} / 10^{-4}
Full FT	every parameter	decoder	encoder 2×10^{-5}, head 10^{-3} / 10^{-4}
Dec→LoRA	all-block LoRA from the midpoint of Dec	decoder (inherited)	10^{-4} / 10^{-4}
Controls (Appendix D)			
From scratch	every parameter, encoder re-initialized	decoder	as Full FT
Compute-matched per-structure	Dec, LoRA, Full FT at 1600 steps per structure	decoder	as above

Table 4: Training cost per method. GiB is the peak GPU memory of one training step (forward pass, backward pass and AdamW update in bf16, weights included) on one A100 80GB PCIe, measured for the largest of 16 training inputs of FDG-PET/CT (TAP-CT, SPECTRE) or RSNA19 ICH (Pillar-0) at batch size 1, and for a micro-batch of two AMOS22 crops in segmentation. The column s gives the GPU time of that step without data loading. The column min gives the wall-clock minutes per 1000 training steps of our runs, excluding validation and including data loading (so differences between frozen heads reflect data loading rather than the head), converted to one H100 with a factor per GPU model that was estimated from configurations that ran on several models (typical deviation of a run from the fit: 16% in classification and 28% in segmentation). For two-stage methods, the second stage is reported.

	TAP-CT			SPECTRE			Pillar-0		
Method	GiB	s	min	GiB	s	min	GiB	s	min
Classification									
Lin	1.4	0.16	1.7	2.3	0.06	3.7	3.1	0.05	1.2
Attn	1.5	0.17	1.6	4.1	0.07	3.6	6.1	0.09	1.2
LoRA	18.9	0.49	3.1	7.6	0.24	3.8	7.4	0.21	1.6
$\text{LoRA}_{\text{late}}$	7.0	0.27	1.9	5.2	0.14	3.6	6.1	0.12	1.2
LoRA+ABMIL	18.9	0.49	3.0	7.6	0.24	3.9	4.0	0.20	1.6
DoRA	19.1	0.47	3.1	8.0	0.24	3.9	7.4	0.21	1.7
rsLoRA	18.9	0.49	3.4	7.6	0.24	3.7	7.4	0.21	1.6
Lin→LoRA	6.8	0.27	1.7	4.8	0.13	3.5	3.1	0.10	0.9
Attn→LoRA	7.0	0.27	1.8	5.2	0.14	3.6	6.1	0.12	1.1
Attn-only FT	18.5	0.47	3.0	8.6	0.19	3.7	7.4	0.15	1.4
Full FT	28.0	0.55	3.3	12.0	0.23	4.0	10.8	0.26	2.2
Segmentation									
Lin	2.2	0.02	0.6	3.6	0.05	3.3	–	–	–
Dec	2.5	0.03	0.6	3.9	0.06	1.8	–	–	–
LoRA	3.7	0.06	0.9	5.3	0.19	2.3	–	–	–
$\text{LoRA}_{\text{late}}$	2.9	0.04	0.9	4.4	0.10	2.9	–	–	–
DoRA	3.8	0.06	1.1	5.7	0.20	2.9	–	–	–
Full FT	4.9	0.07	1.0	8.2	0.18	2.4	–	–	–
Dec→LoRA	3.7	0.06	0.9	5.3	0.19	2.7	–	–	–

Model Selection and Per-case Outputs Every 1000 steps the selection and held-out splits are evaluated and a checkpoint of the trained parameters is written; the checkpoint with the highest selection macro AUROC (mean over labels with both classes present) or mean Dice (mean over structures of the mean over cases in which the structure is present in the reference or the prediction) is kept, ties keeping the earlier checkpoint. For TotalSegmentator only the selection split is evaluated during training; the held-out split is evaluated afterwards with the selected checkpoint, and the recomputed selection Dice is required to match the recorded value within 0.005. Per case, the held-out probability of every known label (classification) and the Dice of every structure (segmentation) at the selected checkpoint are stored and are the input of every number in this paper. Dice thresholds the logits at 0; a structure enters a case's mean when the reference or the prediction contains it, so a false-positive structure scores 0 and an absent, unpredicted structure is skipped.

Appendix B. Statistical procedure

Every number in this paper is computed from the stored per-case predictions of the selected checkpoint of every run (1516 main runs and 417 repeats and controls, of which the 50 probe learning-rate checks of Tab. 17 are evaluated by a second script with the same functions; the classification files hold the held-out probability of every known label of every case, the segmentation files the Dice of every structure of every case). The 1516 main runs include the 8000-step joint models of AMOS22 and TotalSegmentator (28 runs) and the 26 000-step CVPR26 joint models (22 runs).

Per-run Metrics AUROC is the Mann–Whitney statistic with midranks, where tied scores count one half. Every label has both classes in its selection and held-out splits. Dice per structure is the mean over the held-out cases in which the structure is present in the reference or the prediction; an absent, unpredicted structure is skipped and a false-positive structure counts as 0. Cohort and dataset means (Tabs. 20 and 21) are means over the labels or structures of complete cells. The recomputed selection metrics matched the values logged during training to within 2.01×10^{-3} for every run (108 segmentation runs had their per-case Dice re-inferred from the selected checkpoint).

Did-not-train Rule For each backbone and method, the median over label–mode pairs of the standard deviation of the held-out probabilities is computed; a median below 0.010 flags a method that did not train. As the rule acts per backbone and method, a single collapsed cohort, such as TAP-CT Full FT on FDG-PET/CT (Tab. 15), remains in the averages. Only SPECTRE Full FT is flagged (0.004; every other method on TAP-CT, SPECTRE and Pillar-0 is at least 0.114). It stays in the per-cohort tables, the paired tests and the Full FT − Lin estimate, and leaves the ranks and every average over methods.

Paired Tests Per backbone, cohort, mode (per-task, joint) and label, every method is compared with the baselines Lin, LoRA, Attn and Full FT (classification; two-sided DeLong test of the AUROC difference on the same cases (DeLong et al., 1988)) or Lin, Dec and LoRA (segmentation; paired case bootstrap of the mean Dice difference, 4000 resamples shared by every comparison of the same group, two-sided bootstrap p). All paired methods were evaluated on identical case sets. p-values are Benjamini–Hochberg adjusted (Benjamini and Hochberg, 1995) within each family of comparisons (task and baseline), and "significant" means an adjusted p below 0.05; 6240 DeLong tests and 2394 bootstrap tests were made.

Label-level Estimates The unit is one label (structure) on one backbone. For a comparison A − B its value is the mean over the per-task and joint modes of the difference of the two methods' held-out metric (both modes must exist); for joint minus per-task it is the mean over the methods trained in both modes. The reported mean is over the labels of a backbone, or pooled over TAP-CT, SPECTRE and Pillar-0 (78 label–backbone units in classification, 74 in segmentation). Its 95% interval is a percentile bootstrap with 4000 resamples that draws cohort–backbone clusters with replacement, then labels within each drawn cluster, and then replaces each drawn label's value by one of 1000 case-level resamples of that label (positives and negatives resampled separately in classification, with the same case indices for every term of the difference; all cases in segmentation). Besides the comparisons with the linear probe (Lin) and the frozen decoder (Dec), every method with

the attentive head is compared with the frozen attentive probe (matched head), and LoRA − Lin is also computed without FDG-PET/CT and the COVID-CTset label. Table 7 also gives the interval without the case stage. The number of cohort–backbone clusters with a positive mean is reported next to every estimate. In estimates pooled over backbones, the TAP-CT and SPECTRE clusters of a cohort are drawn independently although they share labels and cases, which narrows these intervals, and training seeds are not resampled (seed repeats in Appendix D).

Ranks Methods are ranked within each label–mode pair (a label or structure on one backbone in one mode) in which every method of the task is present, excluding methods that did not train, by held-out AUROC or Dice, with ties sharing the average rank. Classification has 156 such pairs over the three backbones and segmentation 118, as TotalSegmentator per-structure pairs lack three methods and are not ranked.

Seeds, Controls and Convergence Seed repeats are compared through cohort means per seed and the number of cohort–seed pairs in which one method is ahead of another; full fine-tuning variants through the selection-split AUROC, the number of cohorts whose run trained (median spread at least 0.010) and, for the variant with the highest selection AUROC, its label-level difference to Lin and LoRA; compute-matched models through the mean over structures of joint minus matched Dice. Convergence is the share of runs whose selected checkpoint is the last or one of the last two validations. All random draws use fixed seeds, so the script is deterministic.

Appendix C. Paired comparisons and further analyses

Besides the figures and tables cited in the main text, this appendix gives the classification methods by parameters and rank (Tab. 6), the per-label paired tests (Tabs. 8 and 9), all segmentation methods against the linear probe (Fig. 11), encoder adaptation on top of the decoder per dataset (Tab. 10), the selection Dice of the AMOS22 joint models during training (Fig. 12), the difference between backbones (Fig. 15) and the prediction spread behind the did-not-train rule (Tab. 12).

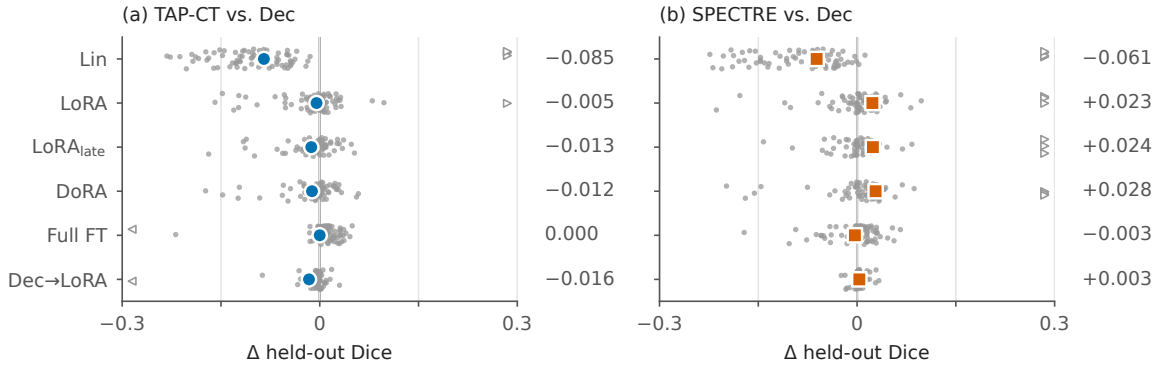


Figure 8: Per-structure change in held-out Dice relative to the frozen decoder of the same backbone, dataset and training mode, for (a) TAP-CT and (b) SPECTRE. Grey markers denote individual structure–mode pairs, with triangles marking values beyond the axis, and colored markers the mean, which is also printed on the right. Means for LoRA$_{\text{late}}$, DoRA and Dec→LoRA exclude TotalSegmentator per-structure models, which were not trained for these methods.

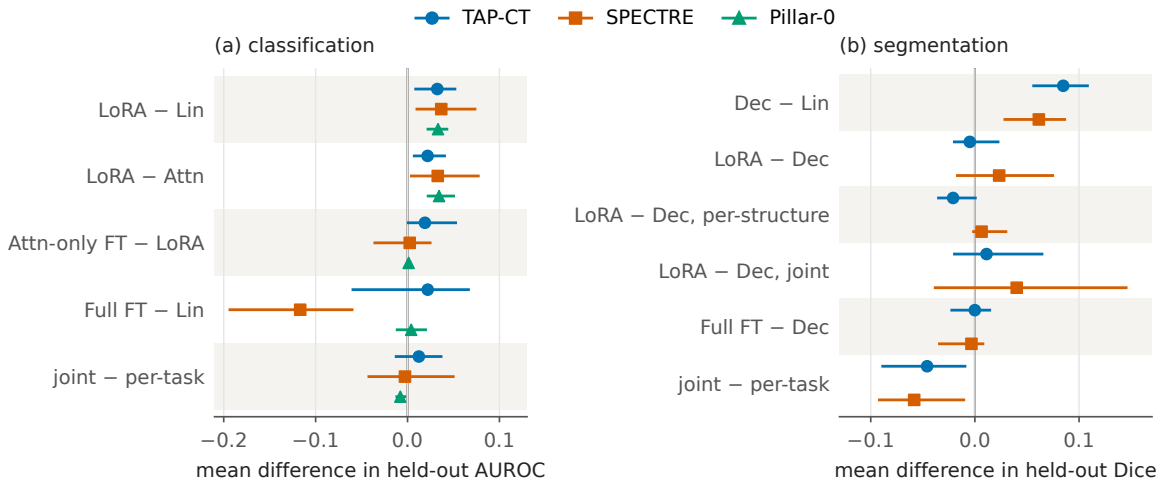


Figure 9: Label-level mean difference in held-out AUROC (a) and Dice (b) for the comparisons of Tab. 1, per backbone, with the 95% interval over cohorts, labels and held-out cases. The SPECTRE Full FT − Lin estimate includes runs that did not train.

Table 5: Classification headline results: pooled held-out AUROC per backbone (mean over all label comparisons, where n = labels × {per-task, joint}) and per-label paired comparisons of LoRA against the linear probe and against full fine-tuning (DeLong test with Benjamini–Hochberg correction), where "sig." denotes a corrected $p < 0.05$ in the stated direction. The mean gain of LoRA over the probe is similar for small and large held-out splits (+0.034 below 100 cases and +0.031 above 400), but it is significant in 0% and 88% of these comparisons, respectively; every comparison above 400 cases is from RSNA19 ICH. Pooled AUROC is averaged over each backbone's own labels and is not comparable between backbones. †Includes runs that did not train (Section 3).

		Pooled AUROC				LoRA vs. linear probe			LoRA vs. full fine-tuning		
Backbone	n	Lin	LoRA	Attn-only FT	Full FT	Mean Δ	LoRA better (%)	sig. (%)	Mean Δ	LoRA better (%)	FT sig. better (%)
TAP-CT	78	0.727	0.760	0.779	0.749	+0.032	78	17	+0.010	55	6
SPECTRE	66	0.718	0.755	0.757	0.601†	+0.037	74	9	+0.153†	94	2
Pillar-0	12	0.887	0.920	0.922	0.891	+0.033	92	83	+0.029	100	0

Table 6: Classification methods compared by trainable parameters, rank and AUROC deficit. Trainable parameters are given for a single-label model, including the head; for Pillar-0, as in Fig. 6c, the count excludes per-scale blocks that never run and receive no gradient. The mean rank (↓) is taken within each of the 156 label–mode pairs, on SPECTRE among the ten methods that trained. The table further gives the share of pairs within 0.01 AUROC of the best method (↑) and the mean AUROC deficit to the best method (↓). **Bold** marks the best and underlining the second-best value of each rank and score column. The pooled values of Full FT exclude SPECTRE, where it did not train, so they are not comparable with those of the other methods and are not marked. Choosing the method per label on the selection split finds the held-out best in 33% of pairs and lowers the mean deficit from 0.036 (always attention-only fine-tuning) to 0.030, at eleven runs per label instead of one.

	Trainable parameters			Mean rank (1 = best)				Within 0.01	Δ to best
Method	TAP-CT	SPECTRE	Pillar-0	TAP-CT	SPECTRE	Pillar-0	All	of best (%)	(AUROC)
Lin	769	1081	1153	7.62	7.15	9.33	7.55	7	0.081
Attn	14.2 M	28.0 M	31.9 M	6.71	6.96	9.75	7.05	10	0.074
LoRA	14.8 M	30.4 M	32.4 M	4.76	4.94	3.83	4.76	23	0.046
$\text{LoRA}_{\text{late}}$	14.4 M	29.1 M	32.1 M	5.46	5.50	7.17	5.61	17	0.054
LoRA+ABMIL	15.0 M	30.7 M	32.7 M	5.61	**4.02**	4.33	4.84	26	0.045
DoRA	14.8 M	30.5 M	32.5 M	5.50	4.72	3.67	5.03	19	0.049
rsLoRA	14.8 M	30.4 M	32.4 M	7.17	5.31	**2.67**	6.04	18	0.065
Lin→LoRA	0.20 M	1.0 M	0.25 M	7.48	5.27	6.58	6.47	11	0.064
Attn→LoRA	14.4 M	29.1 M	32.1 M	6.90	6.47	7.67	6.78	13	0.069
Attn-only FT	42.5 M	159 M	42.5 M	**3.38**	4.67	3.25	**3.91**	**38**	**0.036**
Full FT	101 M	425 M	76.7 M	5.44	–	7.75	5.74	31	0.055

Table 7: Label-level differences with 95% intervals. Each label (or structure) counts once, with its per-task and joint differences averaged, except in the two LoRA − Dec sub-rows. The intervals are percentile bootstraps (4000 resamples) over cohort–backbone groups and labels within them, without and with a third stage that resamples the held-out cases of each drawn label (Appendix B). As Pillar-0 has one cohort, its interval resamples labels and cases only. The last column counts the cohorts (datasets) with a positive mean difference.

				95% interval, resampling		
Comparison	Backbone	n	Mean	cohorts, labels	+ cases	Cohorts > 0
LoRA − Lin	all	78	+0.034	[+0.019, +0.049]	[+0.017, +0.052]	9/10
LoRA − Lin	TAP-CT	39	+0.032	[+0.009, +0.049]	[+0.007, +0.053]	4/5
LoRA − Lin	SPECTRE	33	+0.037	[+0.013, +0.068]	[+0.009, +0.075]	4/4
LoRA − Lin	Pillar-0	6	+0.033	[+0.021, +0.042]	[+0.021, +0.044]	1/1
LoRA − Attn	all	78	+0.027	[+0.013, +0.047]	[+0.012, +0.048]	10/10
LoRA − Attn	TAP-CT	39	+0.022	[+0.009, +0.039]	[+0.006, +0.042]	5/5
LoRA − Attn	SPECTRE	33	+0.033	[+0.007, +0.074]	[+0.003, +0.078]	4/4
LoRA − Attn	Pillar-0	6	+0.034	[+0.021, +0.051]	[+0.021, +0.052]	1/1
Attn-only FT − LoRA	all	78	+0.011	[−0.005, +0.026]	[−0.007, +0.029]	7/10
Attn-only FT − LoRA	TAP-CT	39	+0.019	[+0.002, +0.053]	[−0.001, +0.054]	4/5
Attn-only FT − LoRA	SPECTRE	33	+0.002	[−0.037, +0.023]	[−0.037, +0.026]	2/4
Attn-only FT − LoRA	Pillar-0	6	+0.001	[0.000, +0.003]	[−0.002, +0.004]	1/1
Full FT − Lin	all	78	−0.038	[−0.101, +0.017]	[−0.103, +0.019]	5/10
Full FT − Lin	TAP-CT	39	+0.022	[−0.057, +0.064]	[−0.061, +0.068]	4/5
Full FT − Lin	SPECTRE	33	−0.117	[−0.195, −0.063]	[−0.195, −0.059]	0/4
Full FT − Lin	Pillar-0	6	+0.004	[−0.011, +0.019]	[−0.013, +0.021]	1/1
Joint − per-task	all	78	+0.004	[−0.017, +0.027]	[−0.021, +0.030]	6/10
Joint − per-task	TAP-CT	39	+0.012	[−0.005, +0.032]	[−0.014, +0.038]	3/5
Joint − per-task	SPECTRE	33	−0.003	[−0.035, +0.049]	[−0.044, +0.051]	3/4
Joint − per-task	Pillar-0	6	−0.008	[−0.012, −0.003]	[−0.013, −0.001]	0/1
Attn − Lin	all	78	+0.007	[−0.010, +0.021]	[−0.012, +0.023]	5/10
Attn − Lin	TAP-CT	39	+0.011	[−0.013, +0.027]	[−0.014, +0.029]	3/5
Attn − Lin	SPECTRE	33	+0.004	[−0.022, +0.030]	[−0.030, +0.035]	2/4
Attn − Lin	Pillar-0	6	−0.001	[−0.012, +0.007]	[−0.014, +0.008]	0/1
Attn-only FT − Lin	all	78	+0.045	[+0.027, +0.062]	[+0.024, +0.065]	8/10
Attn-only FT − Lin	TAP-CT	39	+0.051	[+0.026, +0.078]	[+0.023, +0.080]	4/5
Attn-only FT − Lin	SPECTRE	33	+0.039	[−0.001, +0.067]	[−0.003, +0.075]	3/4
Attn-only FT − Lin	Pillar-0	6	+0.034	[+0.022, +0.044]	[+0.021, +0.046]	1/1
Attn-only FT − Attn	all	78	+0.038	[+0.020, +0.060]	[+0.018, +0.062]	10/10
Attn-only FT − Attn	TAP-CT	39	+0.041	[+0.018, +0.078]	[+0.014, +0.082]	5/5
Attn-only FT − Attn	SPECTRE	33	+0.035	[+0.001, +0.069]	[−0.002, +0.075]	4/4
Attn-only FT − Attn	Pillar-0	6	+0.036	[+0.022, +0.052]	[+0.021, +0.053]	1/1
LoRA − Lin, without FDG-PET/CT and COVID-CTset	all	68	+0.037	[+0.023, +0.054]	[+0.019, +0.057]	8/8
LoRA − Lin, without FDG-PET/CT and COVID-CTset	TAP-CT	34	+0.035	[+0.016, +0.053]	[+0.013, +0.056]	4/4
LoRA − Lin, without FDG-PET/CT and COVID-CTset	SPECTRE	28	+0.041	[+0.015, +0.078]	[+0.009, +0.085]	3/3
LoRA − Lin, without FDG-PET/CT and COVID-CTset	Pillar-0	6	+0.033	[+0.021, +0.042]	[+0.020, +0.044]	1/1
Dec − Lin	all	74	+0.073	[+0.049, +0.093]	[+0.049, +0.092]	10/10
Dec − Lin	TAP-CT	37	+0.085	[+0.055, +0.109]	[+0.055, +0.109]	5/5
Dec − Lin	SPECTRE	37	+0.061	[+0.028, +0.089]	[+0.027, +0.088]	5/5
LoRA − Dec	all	74	+0.009	[−0.014, +0.046]	[−0.015, +0.046]	8/10
LoRA − Dec	TAP-CT	37	−0.005	[−0.021, +0.023]	[−0.021, +0.024]	4/5
LoRA − Dec	SPECTRE	37	+0.023	[−0.017, +0.079]	[−0.018, +0.076]	4/5
per-structure models	all	74	−0.007	[−0.021, +0.009]	[−0.022, +0.009]	7/10
per-structure models	TAP-CT	37	−0.021	[−0.035, +0.002]	[−0.036, +0.002]	2/5
per-structure models	SPECTRE	37	+0.006	[−0.001, +0.028]	[−0.003, +0.031]	5/5
joint models	all	74	+0.026	[−0.020, +0.092]	[−0.021, +0.093]	8/10
joint models	TAP-CT	37	+0.011	[−0.019, +0.067]	[−0.021, +0.066]	4/5
joint models	SPECTRE	37	+0.040	[−0.038, +0.148]	[−0.039, +0.147]	4/5
Full FT − Dec	all	74	−0.002	[−0.017, +0.009]	[−0.018, +0.009]	6/10
Full FT − Dec	TAP-CT	37	0.000	[−0.025, +0.015]	[−0.024, +0.016]	4/5
Full FT − Dec	SPECTRE	37	−0.003	[−0.033, +0.008]	[−0.035, +0.009]	2/5
Joint − per-task	all	74	−0.052	[−0.082, −0.021]	[−0.082, −0.022]	0/10
Joint − per-task	TAP-CT	37	−0.046	[−0.088, −0.009]	[−0.090, −0.008]	0/5
Joint − per-task	SPECTRE	37	−0.058	[−0.092, −0.010]	[−0.093, −0.009]	0/5

Table 8: Per-label paired comparisons of held-out AUROC against the linear probe and against full fine-tuning, over the per-task and joint modes. We used DeLong's test with Benjamini–Hochberg correction within each baseline over the comparisons of TAP-CT, SPECTRE and Pillar-0. "Better" is the share of label–mode pairs in which the method's AUROC exceeds that of the baseline, and "Sig." the share in which, in addition, the corrected $p < 0.05$. SPECTRE Full FT did not train.

Backbone	Baseline	Method	Label–mode pairs	Mean ΔAUROC	Better (%)	Sig. better (%)	Sig. worse (%)
TAP-CT	vs Lin	Attn	78	+0.011	59	4	0
		LoRA	78	+0.032	78	17	0
		$\text{LoRA}_{\text{late}}$	78	+0.023	69	14	1
		LoRA+ABMIL	78	+0.024	68	12	1
		DoRA	78	+0.028	68	17	0
		rsLoRA	78	+0.003	55	5	5
		Lin→LoRA	78	+0.001	51	5	1
		Attn→LoRA	78	+0.008	56	9	0
		Attn-only FT	78	+0.051	83	24	0
		Full FT	78	+0.022	72	13	8
TAP-CT	vs Full FT	Lin	78	−0.022	28	12	17
		Attn	78	−0.011	40	10	18
		LoRA	78	+0.010	55	18	6
		$\text{LoRA}_{\text{late}}$	78	+0.001	44	18	6
		LoRA+ABMIL	78	+0.002	45	17	12
		DoRA	78	+0.006	51	17	9
		rsLoRA	78	−0.019	47	13	22
		Lin→LoRA	78	−0.021	31	9	18
		Attn→LoRA	78	−0.014	38	13	15
		Attn-only FT	78	+0.029	63	23	5
SPECTRE	vs Lin	Attn	66	+0.004	50	3	2
		LoRA	66	+0.037	74	9	0
		$\text{LoRA}_{\text{late}}$	66	+0.034	65	5	0
		LoRA+ABMIL	66	+0.052	77	12	2
		DoRA	66	+0.037	77	8	0
		rsLoRA	66	+0.028	70	9	0
		Lin→LoRA	66	+0.035	71	6	0
		Attn→LoRA	66	+0.016	59	6	2
		Attn-only FT	66	+0.039	70	3	3
		Full FT	66	−0.117	21	0	32
SPECTRE	vs Full FT	Lin	66	+0.117	79	33	0
		Attn	66	+0.121	86	35	0
		LoRA	66	+0.153	94	44	2
		$\text{LoRA}_{\text{late}}$	66	+0.151	95	44	0
		LoRA+ABMIL	66	+0.168	94	52	0
		DoRA	66	+0.154	92	44	2
		rsLoRA	66	+0.145	89	39	0
		Lin→LoRA	66	+0.152	94	41	0
		Attn→LoRA	66	+0.133	88	44	0
		Attn-only FT	66	+0.156	89	48	0
Pillar-0	vs Lin	Attn	12	−0.001	42	33	8
		LoRA	12	+0.033	92	83	0
		$\text{LoRA}_{\text{late}}$	12	+0.016	75	67	0
		LoRA+ABMIL	12	+0.026	83	75	8
		DoRA	12	+0.034	92	83	0
		rsLoRA	12	+0.032	100	83	0
		Lin→LoRA	12	+0.022	100	83	0
		Attn→LoRA	12	+0.013	92	58	0
		Attn-only FT	12	+0.034	92	83	0
		Full FT	12	+0.004	67	50	17
Pillar-0	vs Full FT	Lin	12	−0.004	33	17	50
		Attn	12	−0.005	33	17	58
		LoRA	12	+0.029	100	67	0
		$\text{LoRA}_{\text{late}}$	12	+0.012	58	33	25
		LoRA+ABMIL	12	+0.022	58	58	8
		DoRA	12	+0.030	83	67	0
		rsLoRA	12	+0.028	92	67	0
		Lin→LoRA	12	+0.018	67	42	25
		Attn→LoRA	12	+0.009	50	33	25
		Attn-only FT	12	+0.030	100	83	0

Table 9: Per-structure paired comparisons of held-out Dice against the linear probe and the frozen decoder, over the per-structure and joint modes, using a paired bootstrap of the mean per-case difference (4000 resamples) with Benjamini–Hochberg correction within each baseline. Pairing keeps the cases on which a structure enters the Dice of both methods, that is, where it is present in the reference or in either prediction, so the mean differences can differ from the differences of structure means in Tab. 21.

Backbone	Baseline	Method	Pairs	Mean ΔDice	Better (%)	Sig. better (%)	Sig. worse (%)
TAP-CT	vs Lin	Dec	74	+0.086	96	93	4
		LoRA	74	+0.088	96	93	3
		$\text{LoRA}_{\text{late}}$	59	+0.077	95	92	5
		DoRA	59	+0.078	95	90	5
		Full FT	74	+0.085	93	93	7
		Dec→LoRA	59	+0.066	92	88	8
TAP-CT	vs Dec	Lin	74	−0.086	4	4	93
		LoRA	74	+0.002	54	31	30
		$\text{LoRA}_{\text{late}}$	59	−0.006	53	29	32
		DoRA	59	−0.006	53	25	25
		Full FT	74	−0.001	76	53	8
		Dec→LoRA	59	−0.017	29	7	44
SPECTRE	vs Lin	Dec	74	+0.061	92	86	5
		LoRA	74	+0.095	93	93	4
		$\text{LoRA}_{\text{late}}$	59	+0.079	92	90	5
		DoRA	59	+0.088	93	90	7
		Full FT	74	+0.059	89	81	8
		Dec→LoRA	59	+0.051	90	85	7
SPECTRE	vs Dec	Lin	74	−0.061	8	5	86
		LoRA	74	+0.034	80	57	8
		$\text{LoRA}_{\text{late}}$	59	+0.032	69	46	8
		DoRA	59	+0.041	76	54	12
		Full FT	74	−0.002	54	38	20
		Dec→LoRA	59	+0.004	69	20	12

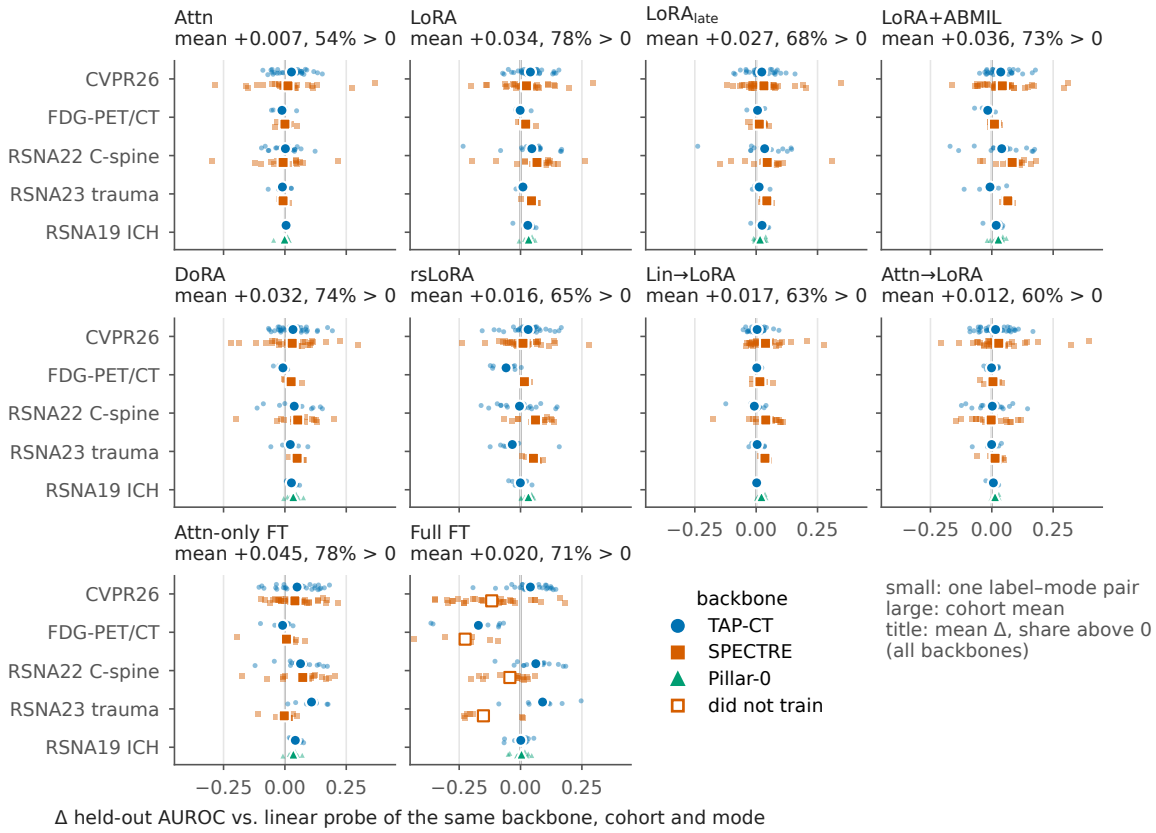


Figure 10: Per-label change in held-out AUROC of each classification method relative to the linear probe of the same backbone, cohort and training mode, with one panel per method. Rows are cohorts and colors denote backbones; small markers denote individual label–mode comparisons and large markers the mean per cohort and backbone. Panel titles give the mean change and the share above zero over the pairs of all three backbones that trained. SPECTRE Full FT did not train, so its cohort means are drawn open and excluded from the title.

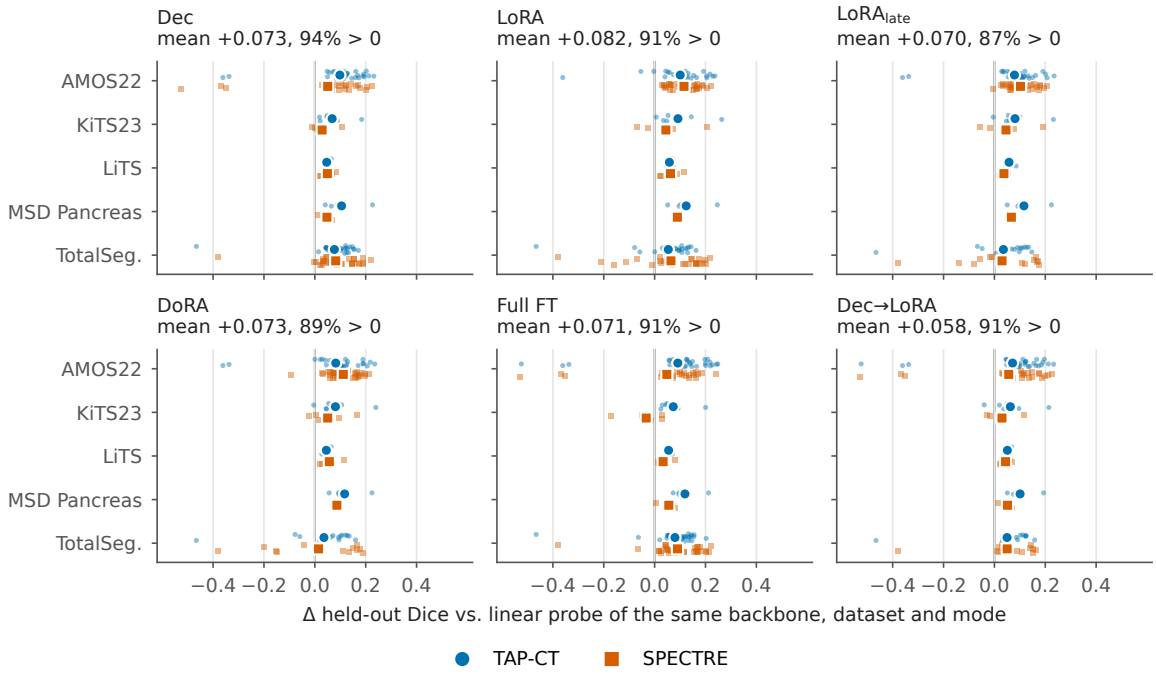


Figure 11: Per-structure change in held-out Dice of each segmentation method relative to the linear probe of the same backbone, dataset and training mode, with one panel per method. Rows are datasets and colors denote backbones; small markers denote individual structure–mode comparisons and large markers the mean per dataset and backbone. Panel titles give the mean change and the share above zero, as in Fig. 10. The values for LoRA$_{\text{late}}$, DoRA and Dec→LoRA exclude TotalSegmentator per-structure models, which were not trained for these methods.

Table 10: Encoder adaptation on top of the frozen decoder, given as the difference of structure-mean held-out Dice per dataset, backbone and training mode. Joint rows of AMOS22 and TotalSegmentator use the 24 000-step schedule, whereas the 8000-step rows show the same models under the shared budget.

		LoRA − Dec		Full FT − Dec	
Dataset	Mode	TAP-CT	SPECTRE	TAP-CT	SPECTRE
AMOS22 (15)	per-structure	−0.026	+0.004	+0.012	−0.016
	joint	+0.031	+0.126	−0.027	+0.011
	joint, 8000 steps	−0.125	−0.071	+0.007	+0.010
KiTS23 (3)	per-structure	−0.021	+0.014	+0.005	−0.071
	joint	+0.068	+0.016	+0.006	−0.052
LiTS (2)	per-structure	+0.007	+0.015	+0.001	−0.002
	joint	+0.016	+0.011	+0.016	−0.030
MSD Pancreas (2)	per-structure	+0.013	+0.047	+0.021	+0.000
	joint	+0.025	+0.037	+0.008	+0.016
TotalSegmentator (15)	per-structure	−0.024	+0.001	−0.011	−0.009
	joint	−0.022	−0.036	+0.017	+0.025
	joint, 8000 steps	−0.199	−0.254	+0.022	+0.037

Table 11: Training budget of the 15-structure joint models (AMOS22, TotalSegmentator). The table gives held-out Dice, averaged over structures, for one model per structure (8000 steps each), one joint model under the shared 8000 steps and one joint model with three times the steps (Dec→LoRA: 12 000 + 12 000).

Dataset	Backbone	Training	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→LoRA
AMOS22	TAP-CT	per-structure, 8000 steps	0.658	0.793	0.767	0.764	0.761	0.805	0.791
AMOS22	TAP-CT	joint, 8000 steps	0.632	0.595	0.470	0.412	0.455	0.602	0.591
AMOS22	TAP-CT	joint, 24 000 steps	0.646	0.708	0.739	0.698	0.708	0.681	0.656
AMOS22	SPECTRE	per-structure, 8000 steps	0.683	0.808	0.812	0.806	0.814	0.792	0.814
AMOS22	SPECTRE	joint, 8000 steps	0.670	0.640	0.568	0.527	0.567	0.649	0.627
AMOS22	SPECTRE	joint, 24 000 steps	0.682	0.658	0.784	0.764	0.775	0.669	0.664
TotalSegmentator	TAP-CT	per-structure, 8000 steps	0.747	0.843	0.819	–	–	0.832	–
TotalSegmentator	TAP-CT	joint, 8000 steps	0.731	0.781	0.583	0.594	0.591	0.803	0.776
TotalSegmentator	TAP-CT	joint, 24 000 steps	0.740	0.798	0.776	0.776	0.776	0.815	0.790
TotalSegmentator	SPECTRE	per-structure, 8000 steps	0.731	0.848	0.848	–	–	0.838	–
TotalSegmentator	SPECTRE	joint, 8000 steps	0.712	0.679	0.425	0.387	0.408	0.716	0.637
TotalSegmentator	SPECTRE	joint, 24 000 steps	0.722	0.770	0.733	0.752	0.736	0.795	0.772

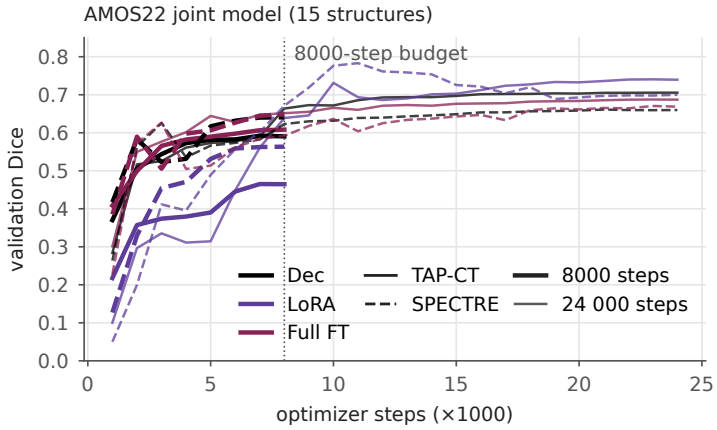


Figure 12: Selection-split Dice during training of the 15-structure AMOS22 joint model with the frozen decoder, LoRA and full fine-tuning, on TAP-CT (solid) and SPECTRE (dashed). Thick lines stop at the shared 8000 steps, whereas thin lines run for 24 000 steps.

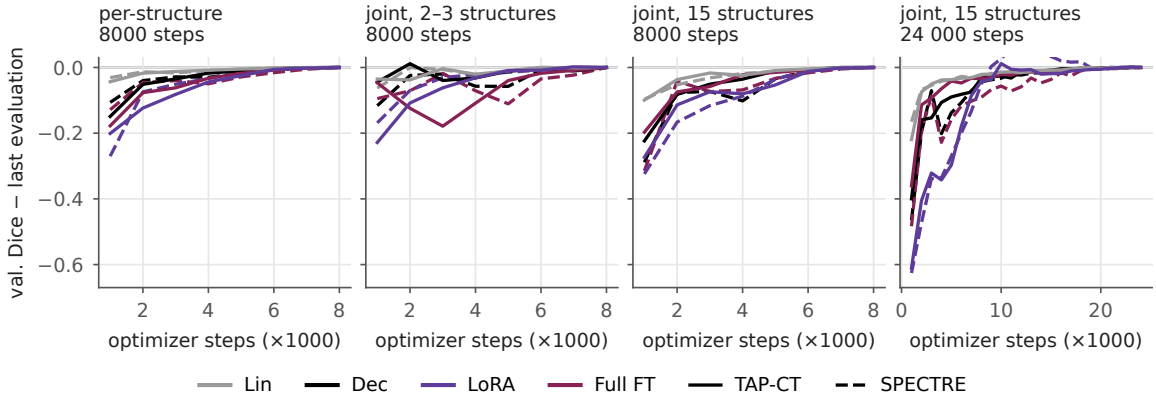


Figure 13: Validation Dice at each evaluation minus that at the run's last evaluation, as the median over the runs of each schedule group, for Lin, Dec, LoRA and Full FT on TAP-CT (solid) and SPECTRE (dashed). Every curve ends at zero by construction, and a curve that is still rising at the right edge had not converged. Overall, 77% of segmentation runs selected one of their last two validations.

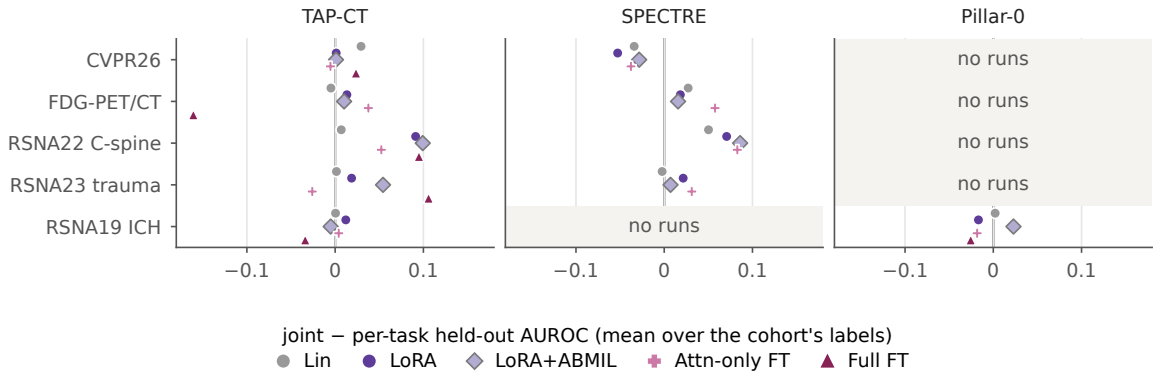


Figure 14: Joint minus per-task held-out AUROC, averaged over the labels of each cohort, for Lin, LoRA, LoRA+ABMIL, Attn-only FT and Full FT per backbone; SPECTRE Full FT, which did not train, is not shown. Values right of zero favor one joint model per cohort.

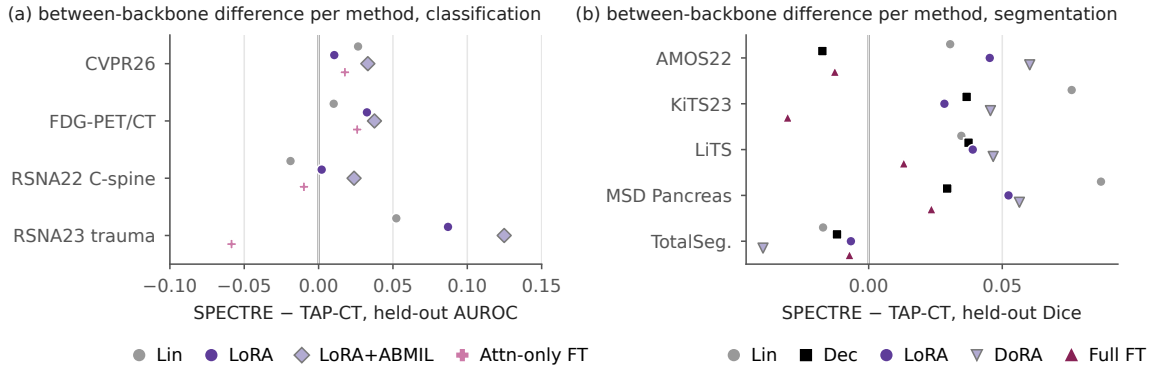


Figure 15: Between-backbone difference per method, given as SPECTRE minus TAP-CT held-out AUROC per cohort (a) and held-out Dice per dataset (b), on the same labels or structures and modes. SPECTRE Full FT, which did not train in classification, is not shown in (a). Dice in (b) is computed on each backbone's own grid, and SPECTRE saw several of these datasets without labels during pretraining (Tab. 26), so these differences do not rank the backbones.

Table 12: Median over label–mode pairs of the standard deviation of the held-out probabilities, per backbone and method. The did-not-train rule flags values below 0.010 and is applied to TAP-CT, SPECTRE and Pillar-0. The flagged value is shown in bold and marked with †, denoting a method that did not train (Section 3).

Backbone	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
TAP-CT	0.227	0.239	0.239	0.235	0.233	0.237	0.167	0.222	0.225	0.247	0.170
SPECTRE	0.187	0.114	0.161	0.185	0.187	0.185	0.150	0.178	0.129	0.148	**0.004**†
Pillar-0	0.294	0.321	0.359	0.350	0.352	0.354	0.359	0.337	0.338	0.355	0.347

Appendix D. Seed repeats, full fine-tuning settings and compute-matched models

Tabs. 13 and 14 give the seed repeats of the joint classification and segmentation models.

Table 13: Held-out AUROC of the joint classification models across seeds 42, 43 and 44, averaged over the cohort's labels and given as mean ± standard deviation over the seeds (their number in brackets), together with the number of seeds in which the mean label difference between two methods is positive. SPECTRE Full FT, which did not train, has seed 42 only. A label-level bootstrap that also draws one seed per method for each cohort gives LoRA − Lin +0.033 [+0.007, +0.065] and Attn-only FT − LoRA +0.016 [−0.010, +0.040] over the 10 cohort–backbone pairs.

Cohort	Lin	LoRA	Attn-only FT	Full FT	LoRA > Lin	Attn-only FT > LoRA
TAP-CT						
CVPR26	0.705 ± .004 (3)	0.719 ± .016 (3)	0.740 ± .009 (3)	0.711 ± .029 (3)	2/3	3/3
FDG-PET/CT	0.860 ± .003 (3)	0.864 ± .008 (3)	0.875 ± .012 (3)	0.661 ± .130 (3)	2/3	3/3
RSNA23 trauma	0.738 ± .008 (3)	0.760 ± .002 (3)	0.842 ± .030 (3)	0.793 ± .157 (3)	3/3	3/3
RSNA22 C-spine	0.623 ± .006 (3)	0.659 ± .050 (3)	0.691 ± .019 (3)	0.695 ± .035 (3)	2/3	2/3
RSNA19 ICH	0.871 ± .002 (3)	0.896 ± .009 (3)	0.904 ± .019 (3)	0.836 ± .086 (3)	3/3	2/3
SPECTRE						
CVPR26	0.711 ± .008 (3)	0.741 ± .024 (3)	0.759 ± .023 (3)	0.575 (1)	3/3	3/3
FDG-PET/CT	0.887 ± .003 (3)	0.901 ± .000 (3)	0.910 ± .006 (3)	0.673 (1)	3/3	3/3
RSNA23 trauma	0.792 ± .004 (3)	0.846 ± .001 (3)	0.834 ± .028 (3)	0.654 (1)	3/3	1/3
RSNA22 C-spine	0.592 ± .046 (3)	0.696 ± .009 (3)	0.683 ± .052 (3)	0.554 (1)	3/3	2/3
Pillar-0						
RSNA19 ICH	0.888 ± .000 (3)	0.916 ± .004 (3)	0.921 ± .007 (3)	0.894 ± .015 (3)	3/3	3/3

Table 14: Held-out Dice of the joint segmentation models across seeds, averaged over structures, as in Tab. 13. The AMOS22 and TotalSegmentator joint models were trained for 24 000 steps.

Dataset	Dec	LoRA	Full FT	LoRA > Dec	Full FT > Dec
TAP-CT					
AMOS22	0.675 ± .029 (3)	0.709 ± .059 (3)	0.680 ± .001 (3)	2/3	2/3
KiTS23	0.516 ± .003 (3)	0.576 ± .005 (3)	0.529 ± .014 (3)	3/3	3/3
LiTS	0.715 ± .016 (3)	0.718 ± .004 (3)	0.717 ± .007 (3)	2/3	2/3
MSD Pancreas	0.480 ± .002 (3)	0.496 ± .009 (3)	0.468 ± .027 (3)	3/3	2/3
TotalSegmentator	0.783 ± .026 (3)	0.773 ± .008 (3)	0.815 ± .000 (3)	1/3	3/3
SPECTRE					
AMOS22	0.658 ± .000 (3)	0.775 ± .011 (3)	0.669 ± .003 (3)	3/3	3/3
KiTS23	0.579 ± .010 (3)	0.616 ± .024 (3)	0.529 ± .015 (3)	3/3	0/3
LiTS	0.741 ± .001 (3)	0.740 ± .012 (3)	0.707 ± .019 (3)	1/3	0/3
MSD Pancreas	0.518 ± .004 (3)	0.551 ± .008 (3)	0.517 ± .011 (3)	3/3	1/3
TotalSegmentator	0.741 ± .025 (3)	0.752 ± .020 (3)	0.762 ± .028 (3)	2/3	3/3

Table 15: Held-out AUROC of the joint classification models under the default and three alternative full fine-tuning settings, averaged over the cohort's labels, next to Lin and LoRA. **Bold** marks the setting selected for each backbone by the best mean selection-split AUROC over its cohorts, which need not have the best held-out value of its row. Grey marks runs whose held-out predictions barely vary, that is, whose median over labels of the standard deviation of the predicted probabilities is below 0.010.

			Full FT			
Cohort	Lin	LoRA	default	lower LR	layer decay	accum. + clip
TAP-CT						
CVPR26	0.707	0.733	0.744	0.730	**0.750**	0.685
FDG-PET/CT	0.858	0.866	0.608	0.861	**0.867**	0.867
RSNA23 trauma	0.739	0.757	0.881	0.860	**0.869**	0.758
RSNA22 C-spine	0.626	0.714	0.732	0.654	**0.618**	0.687
RSNA19 ICH	0.871	0.906	0.854	0.889	**0.917**	0.909
SPECTRE						
CVPR26	0.702	0.716	0.575	0.713	**0.758**	0.511
FDG-PET/CT	0.884	0.901	0.673	0.884	**0.865**	0.883
RSNA23 trauma	0.789	0.846	0.654	0.793	**0.863**	0.578
RSNA22 C-spine	0.629	0.706	0.554	0.568	**0.700**	0.553
Pillar-0						
RSNA19 ICH	0.888	0.912	0.878	**0.924**	0.920	0.918

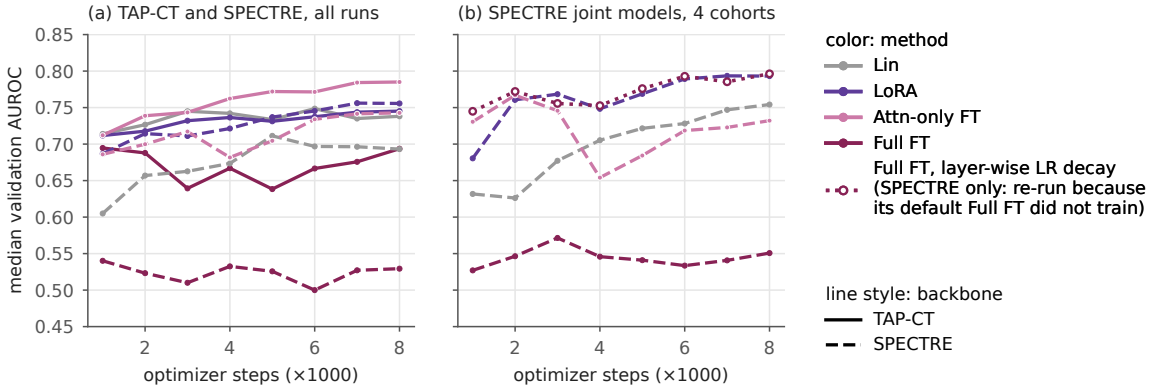


Figure 16: Full fine-tuning on SPECTRE. (a) Median selection-split AUROC at each evaluation under the default settings, over all runs of each method on TAP-CT and SPECTRE. (b) SPECTRE joint models on the four cohorts, including Full FT rerun with layer-wise learning-rate decay, the setting selected in Tab. 15. Table 15 gives all alternative settings on the three backbones.

Table 16: Controls for two confounds of the main comparison: rsLoRA at a quarter of the learning rate, which matches the update scale of LoRA at initialization, and fine-tuning the MLP instead of the attention weights. The table gives the held-out AUROC of the joint models, averaged over the cohort's labels.

Cohort	LoRA	rsLoRA	rsLoRA (LR/4)	Attn-only FT	MLP-only FT
TAP-CT					
CVPR26	0.733	0.722	0.741	0.739	0.745
FDG-PET/CT	0.866	0.785	0.855	0.870	0.866
RSNA23 trauma	0.757	0.684	0.751	0.833	0.894
RSNA22 C-spine	0.714	0.601	0.611	0.712	0.627
RSNA19 ICH	0.906	0.869	0.901	0.915	0.907
SPECTRE					
CVPR26	0.716	0.690	0.739	0.740	0.681
FDG-PET/CT	0.901	0.900	0.908	0.906	0.814
RSNA23 trauma	0.846	0.850	0.834	0.803	0.838
RSNA22 C-spine	0.706	0.691	0.672	0.718	0.626
Pillar-0					
RSNA19 ICH	0.912	0.918	0.920	0.912	0.912

Table 17: LoRA against frozen probes whose learning rate is selected on the selection data, for the joint classification models. Lin was trained at 10^{-4}, 10^{-3} (default) and 10^{-2}, and Attn at 3×10^{-5}, 10^{-4} (default), 3×10^{-4} and 10^{-3}; per cohort and backbone, the rate with the best mean selection-split AUROC over the cohort's labels was kept. It was the default for Lin in 7 of the 10 cohort–backbone pairs and 3×10^{-5} for Attn in 8. Entries are differences in held-out AUROC with the 95% interval over cohorts, labels and cases (Appendix B); unlike Tab. 1, they cover the joint models only.

			Default learning rate		Selected learning rate		
Comparison	Backbone	n	Mean	95% interval	Mean	95% interval	Cohorts > 0
LoRA − Lin	all	78	+0.036	[+0.011, +0.063]	+0.034	[+0.008, +0.065]	10/10
LoRA − Lin	TAP-CT	39	+0.038	[+0.009, +0.074]	+0.038	[+0.008, +0.074]	5/5
LoRA − Lin	SPECTRE	33	+0.035	[−0.010, +0.085]	+0.034	[−0.009, +0.088]	4/4
LoRA − Lin	Pillar-0	6	+0.024	[+0.010, +0.036]	+0.010	[0.000, +0.021]	1/1
LoRA − Attn	all	78	+0.027	[−0.001, +0.066]	+0.025	[−0.004, +0.066]	9/10
LoRA − Attn	TAP-CT	39	+0.031	[+0.004, +0.072]	+0.024	[−0.004, +0.069]	5/5
LoRA − Attn	SPECTRE	33	+0.024	[−0.022, +0.103]	+0.027	[−0.023, +0.115]	3/4
LoRA − Attn	Pillar-0	6	+0.023	[+0.012, +0.034]	+0.023	[+0.010, +0.037]	1/1

Table 18: Joint against compute-matched per-structure segmentation. The table gives held-out Dice, averaged over the 15 structures, of per-structure models trained for 1600 steps (matched) or 8000 steps and of the 24 000-step joint model on the same structures, together with the difference joint − matched and the number of structures in which it is positive.

			per-structure			joint − matched	
Backbone	Method	Struct. finished	matched	8000 steps	joint	mean	> 0
AMOS22 (15 structures)							
TAP-CT	Dec	15	0.763	0.793	0.708	−0.055	11/15
TAP-CT	LoRA	15	0.617	0.767	0.739	+0.122	14/15
TAP-CT	Full FT	15	0.766	0.805	0.681	−0.085	12/15
SPECTRE	Dec	15	0.760	0.808	0.658	−0.101	6/15
SPECTRE	LoRA	15	0.648	0.812	0.784	+0.136	14/15
SPECTRE	Full FT	15	0.793	0.792	0.669	−0.123	5/15
TotalSegmentator (15 structures)							
TAP-CT	Dec	15	0.783	0.843	0.798	+0.015	13/15
TAP-CT	LoRA	15	0.657	0.819	0.776	+0.119	14/15
TAP-CT	Full FT	15	0.803	0.832	0.815	+0.012	14/15
SPECTRE	Dec	15	0.806	0.848	0.770	−0.037	6/15
SPECTRE	LoRA	15	0.691	0.848	0.733	+0.042	11/15
SPECTRE	Full FT	15	0.819	0.838	0.795	−0.024	10/15

Table 19: From-scratch control, in which the TAP-CT architecture with a randomly initialized encoder is trained as the Full FT joint model, next to Full FT and Dec per dataset (held-out Dice, averaged over structures).

Dataset (structures)	From scratch	Dec	Full FT	Scratch − Dec	Scratch − Full FT
AMOS22 (15)	0.496	0.708	0.681	−0.212	−0.185
KiTS23 (3)	0.400	0.514	0.519	−0.114	−0.120
LiTS (2)	0.612	0.699	0.715	−0.087	−0.103
MSD Pancreas (2)	0.308	0.480	0.487	−0.172	−0.180
TotalSegmentator (15)	0.579	0.798	0.815	−0.219	−0.236

Appendix E. Per-cohort, per-label and per-structure results

Table 20: Classification: held-out AUROC, mean over the labels of each cohort (number of labels in brackets), per backbone, training mode and method. Within each row, **bold** is the best and underlined the second-best method. A shaded cell marks the better of per-task and joint training for that method, backbone and cohort. SPECTRE Full FT did not train with the default setting (Section 4.1).

Cohort (labels)	Backbone	Mode	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ ABMIL	DoRA	rsLoRA	Lin→ LoRA	Attn→ LoRA	Attn-only FT	Full FT
CVPR26 (17)	TAP-CT	per-task	0.677	0.718	0.732	0.704	0.728	0.717	0.724	0.697	0.707	**0.744**	0.720
		joint	0.707	0.720	0.733	0.726	0.729	0.733	0.722	0.695	0.708	0.739	**0.744**
CVPR26 (17)	SPECTRE	per-task	0.736	0.725	0.769	0.769	0.776	0.766	0.768	0.769	0.765	**0.778**	0.627
		joint	0.702	0.738	0.716	0.731	**0.747**	0.731	0.690	0.744	0.728	0.740	0.575
FDG-PET/CT (4)	TAP-CT	per-task	0.863	0.836	0.852	**0.868**	0.839	0.847	0.817	0.866	0.856	0.832	0.769
		joint	0.858	0.860	0.866	0.864	0.849	0.858	0.785	0.860	0.862	**0.870**	0.608
FDG-PET/CT (4)	SPECTRE	per-task	0.857	0.862	0.882	0.879	0.874	**0.892**	0.872	0.887	0.867	0.848	0.617
		joint	0.884	0.880	0.901	0.888	0.889	0.900	0.900	0.884	0.884	**0.906**	0.673
RSNA22 C-spine (8)	TAP-CT	per-task	0.619	0.625	0.623	0.608	0.613	0.641	0.635	0.606	0.629	**0.660**	0.637
		joint	0.626	0.624	0.714	0.706	0.713	0.680	0.601	0.624	0.620	0.712	**0.732**
RSNA22 C-spine (8)	SPECTRE	per-task	0.579	0.591	0.635	**0.654**	0.644	0.619	0.638	0.615	0.591	0.635	0.566
		joint	0.629	0.601	0.706	0.644	**0.730**	0.692	0.691	0.670	0.611	0.718	0.554
RSNA23 trauma (4)	TAP-CT	per-task	0.737	0.718	0.739	0.756	0.703	0.752	0.725	0.738	0.716	**0.859**	0.775
		joint	0.739	0.738	0.757	0.744	0.758	0.768	0.684	0.746	0.758	0.833	**0.881**
RSNA23 trauma (4)	SPECTRE	per-task	0.791	0.778	0.824	0.832	**0.852**	0.842	0.836	0.825	0.798	0.772	0.624
		joint	0.789	0.788	0.846	0.835	**0.859**	0.838	0.850	0.827	0.810	0.803	0.654
RSNA19 ICH (6)	TAP-CT	per-task	0.870	0.873	0.894	0.894	0.891	0.897	0.870	0.877	0.878	**0.911**	0.888
		joint	0.871	0.877	0.906	0.895	0.886	0.897	0.869	0.868	0.876	**0.915**	0.854
RSNA19 ICH (6)	Pillar-0	per-task	0.886	0.884	0.929	0.914	0.902	**0.931**	0.920	0.912	0.903	**0.931**	0.904
		joint	0.888	0.888	0.912	0.892	**0.925**	0.912	0.918	0.905	0.898	0.912	0.878

Table 21: Segmentation: held-out Dice, mean over the structures of each dataset (number of structures in brackets), per backbone, training mode and method; – marks methods not run per structure on TotalSegmentator. Within each row, **bold** is the best and underlined the second-best method. A shaded cell marks the better of per-structure and joint training for that method, backbone and dataset. The AMOS22 and TotalSegmentator joint models train 24 000 steps.

Dataset (structures)	Backbone	Mode	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→LoRA
AMOS22 (15)	TAP-CT	per-structure	0.658	0.793	0.767	0.764	0.761	**0.805**	0.791
		joint	0.646	0.708	**0.739**	0.698	0.708	0.681	0.656
AMOS22 (15)	SPECTRE	per-structure	0.683	0.808	0.812	0.806	**0.814**	0.792	**0.814**
		joint	0.682	0.658	**0.784**	0.764	0.775	0.669	0.664
KiTS23 (3)	TAP-CT	per-structure	0.531	0.584	0.563	0.567	0.563	**0.588**	0.587
		joint	0.430	0.514	**0.582**	0.556	0.560	0.519	0.500
KiTS23 (3)	SPECTRE	per-structure	0.583	0.599	0.613	0.617	**0.622**	0.528	0.603
		joint	0.530	0.572	0.588	0.588	**0.593**	0.519	0.570
LiTS (2)	TAP-CT	per-structure	0.659	0.709	**0.716**	0.709	0.714	0.710	0.714
		joint	0.656	0.699	0.715	**0.721**	0.692	0.715	0.703
LiTS (2)	SPECTRE	per-structure	0.688	0.741	**0.756**	0.723	**0.756**	0.739	0.729
		joint	0.696	0.742	**0.753**	0.736	0.743	0.713	0.743
MSD Pancreas (2)	TAP-CT	per-structure	0.427	0.491	0.504	0.501	0.502	**0.512**	0.495
		joint	0.334	0.480	**0.505**	0.492	0.493	0.487	0.468
MSD Pancreas (2)	SPECTRE	per-structure	0.475	0.516	**0.563**	0.538	0.557	0.516	0.518
		joint	0.459	0.514	**0.550**	0.529	**0.550**	0.530	0.520
TotalSegmentator (15)	TAP-CT	per-structure	0.747	**0.843**	0.819	–	–	0.832	–
		joint	0.740	0.798	0.776	0.776	0.776	**0.815**	0.790
TotalSegmentator (15)	SPECTRE	per-structure	0.731	**0.848**	**0.848**	–	–	0.838	–
		joint	0.722	0.770	0.733	0.752	0.736	**0.795**	0.772

Table 22: Held-out AUROC as in Tab. 20, with the longer CVPR26 joint schedule (26 000 steps) as an extra row. Within each row, **bold** is the best and underlined the second-best method. A shaded cell marks the better of per-task and joint training at 8000 steps.

Cohort	Backbone	Mode (labels)	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
CVPR26	TAP-CT	per-task (17)	0.677	0.718	0.732	0.704	0.728	0.717	0.724	0.697	0.707	**0.744**	0.720
CVPR26	TAP-CT	joint (17)	0.707	0.720	0.733	0.726	0.729	0.733	0.722	0.695	0.708	0.739	**0.744**
CVPR26	TAP-CT	joint (26 000 steps) (17)	0.693	0.722	0.692	0.728	0.727	0.720	0.701	0.698	0.726	**0.756**	0.650
CVPR26	SPECTRE	per-task (17)	0.736	0.725	0.769	0.769	0.776	0.766	0.768	0.769	0.765	**0.778**	0.627
CVPR26	SPECTRE	joint (17)	0.702	0.738	0.716	0.731	**0.747**	0.731	0.690	0.744	0.728	0.740	0.575
CVPR26	SPECTRE	joint (26 000 steps) (17)	0.744	0.713	0.755	0.762	0.760	0.752	**0.763**	0.746	0.758	0.756	0.574
FDG-PET/CT	TAP-CT	per-task (4)	0.863	0.836	0.852	**0.868**	0.839	0.847	0.817	0.866	0.856	0.832	0.769
FDG-PET/CT	TAP-CT	joint (4)	0.858	0.860	0.866	0.864	0.849	0.858	0.785	0.860	0.862	**0.870**	0.608
FDG-PET/CT	SPECTRE	per-task (4)	0.857	0.862	0.882	0.879	0.874	**0.892**	0.872	0.887	0.867	0.848	0.617
FDG-PET/CT	SPECTRE	joint (4)	0.884	0.880	0.901	0.888	0.889	0.900	0.900	0.884	0.884	**0.906**	0.673
RSNA22 C-spine	TAP-CT	per-task (8)	0.619	0.625	0.623	0.608	0.613	0.641	0.635	0.606	0.629	**0.660**	0.637
RSNA22 C-spine	TAP-CT	joint (8)	0.626	0.624	0.714	0.706	0.713	0.680	0.601	0.624	0.620	0.712	**0.732**
RSNA22 C-spine	SPECTRE	per-task (8)	0.579	0.591	0.635	**0.654**	0.644	0.619	0.638	0.615	0.591	0.635	0.566
RSNA22 C-spine	SPECTRE	joint (8)	0.629	0.601	0.706	0.644	**0.730**	0.692	0.691	0.670	0.611	0.718	0.554
RSNA23 trauma	TAP-CT	per-task (4)	0.737	0.718	0.739	0.756	0.703	0.752	0.725	0.738	0.716	**0.859**	0.775
RSNA23 trauma	TAP-CT	joint (4)	0.739	0.738	0.757	0.744	0.758	0.768	0.684	0.746	0.758	0.833	**0.881**
RSNA23 trauma	SPECTRE	per-task (4)	0.791	0.778	0.824	0.832	**0.852**	0.842	0.836	0.825	0.798	0.772	0.624
RSNA23 trauma	SPECTRE	joint (4)	0.789	0.788	0.846	0.835	**0.859**	0.838	0.850	0.827	0.810	0.803	0.654
RSNA19 ICH	TAP-CT	per-task (6)	0.870	0.873	0.894	0.894	0.891	0.897	0.870	0.877	0.878	**0.911**	0.888
RSNA19 ICH	TAP-CT	joint (6)	0.871	0.877	0.906	0.895	0.886	0.897	0.869	0.868	0.876	**0.915**	0.854
RSNA19 ICH	Pillar-0	per-task (6)	0.886	0.884	0.929	0.914	0.902	**0.931**	0.920	0.912	0.903	**0.931**	0.904
RSNA19 ICH	Pillar-0	joint (6)	0.888	0.888	0.912	0.892	**0.925**	0.912	0.918	0.905	0.898	0.912	0.878

CVPR26, TAP-CT

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
TAP-CT, per-task											
adrenal_hyperplasia	0.80 [0.62,0.95]	0.84 [0.68,0.97]	0.80 [0.62,0.96]	0.79 [0.62,0.95]	0.81 [0.64,0.95]	0.77 [0.59,0.92]	0.79 [0.60,0.94]	0.80 [0.62,0.95]	0.82 [0.67,0.95]	0.84 [0.68,0.97]	0.87 [0.72,0.98]
ascites	0.78 [0.60,0.92]	0.84 [0.67,0.97]	0.77 [0.59,0.92]	0.74 [0.55,0.90]	0.78 [0.61,0.92]	0.71 [0.50,0.87]	0.77 [0.60,0.93]	0.76 [0.58,0.92]	0.75 [0.56,0.92]	0.68 [0.47,0.85]	0.82 [0.66,0.95]
atherosclerosis	0.75 [0.57,0.90]	0.87 [0.73,0.97]	0.89 [0.76,0.97]	0.83 [0.68,0.96]	0.83 [0.68,0.95]	0.84 [0.68,0.96]	0.86 [0.71,0.97]	0.75 [0.56,0.90]	0.86 [0.71,0.97]	0.81 [0.64,0.93]	0.88 [0.74,0.97]
cholecystitis	0.73 [0.56,0.88]	0.78 [0.62,0.92]	0.74 [0.55,0.90]	0.69 [0.49,0.85]	0.73 [0.54,0.89]	0.75 [0.60,0.90]	0.74 [0.56,0.89]	0.75 [0.56,0.91]	0.72 [0.56,0.88]	0.77 [0.60,0.91]	0.74 [0.57,0.89]
colorectal_cancer	0.56 [0.33,0.77]	0.65 [0.44,0.85]	0.65 [0.43,0.88]	0.65 [0.43,0.85]	0.67 [0.43,0.86]	0.63 [0.40,0.84]	0.62 [0.38,0.84]	0.60 [0.37,0.84]	0.64 [0.42,0.85]	0.70 [0.48,0.88]	0.62 [0.41,0.82]
fatty_liver	0.87 [0.75,0.97]	0.92 [0.82,0.99]	0.95 [0.88,1.00]	0.95 [0.86,1.00]	0.95 [0.88,1.00]	0.94 [0.85,0.99]	0.99 [0.97,1.00]	0.90 [0.78,0.99]	0.92 [0.82,0.98]	0.99 [0.97,1.00]	0.98 [0.94,1.00]
gallstone	0.60 [0.46,0.74]	0.65 [0.49,0.78]	0.66 [0.52,0.80]	0.64 [0.50,0.78]	0.68 [0.54,0.81]	0.68 [0.53,0.82]	0.60 [0.46,0.74]	0.61 [0.46,0.75]	0.69 [0.55,0.82]	0.65 [0.51,0.78]	0.69 [0.56,0.82]
hydronephrosis	0.86 [0.71,0.97]	0.82 [0.64,0.95]	0.90 [0.78,0.98]	0.86 [0.70,0.97]	0.86 [0.71,0.96]	0.85 [0.70,0.96]	0.78 [0.61,0.92]	0.87 [0.71,0.97]	0.90 [0.78,0.98]	0.86 [0.71,0.97]	0.67 [0.47,0.85]
kidney_stone	0.62 [0.49,0.75]	0.69 [0.57,0.80]	0.63 [0.50,0.75]	0.65 [0.52,0.77]	0.64 [0.50,0.76]	0.62 [0.51,0.74]	0.70 [0.58,0.81]	0.70 [0.57,0.81]	0.71 [0.59,0.83]	0.70 [0.58,0.80]	0.67 [0.56,0.79]
liver_calcifications	0.44 [0.25,0.63]	0.37 [0.20,0.56]	0.49 [0.30,0.69]	0.44 [0.26,0.63]	0.47 [0.28,0.67]	0.51 [0.32,0.69]	0.51 [0.31,0.71]	0.43 [0.26,0.63]	0.39 [0.22,0.59]	0.58 [0.38,0.75]	0.57 [0.40,0.75]
liver_cyst	0.64 [0.54,0.74]	0.67 [0.57,0.76]	0.67 [0.56,0.77]	0.64 [0.54,0.74]	0.65 [0.54,0.75]	0.62 [0.53,0.73]	0.68 [0.57,0.77]	0.66 [0.57,0.76]	0.64 [0.55,0.73]	0.69 [0.60,0.78]	0.66 [0.55,0.75]
liver_lesion	0.57 [0.42,0.72]	0.64 [0.48,0.77]	0.71 [0.57,0.83]	0.69 [0.56,0.81]	0.61 [0.47,0.75]	0.66 [0.52,0.79]	0.65 [0.51,0.78]	0.59 [0.45,0.73]	0.64 [0.51,0.77]	0.66 [0.52,0.80]	0.70 [0.57,0.82]
lymphadenopathy	0.60 [0.41,0.78]	0.73 [0.57,0.88]	0.64 [0.48,0.81]	0.67 [0.48,0.85]	0.76 [0.61,0.91]	0.73 [0.55,0.88]	0.66 [0.46,0.82]	0.69 [0.51,0.85]	0.70 [0.52,0.85]	0.76 [0.60,0.90]	0.54 [0.36,0.72]
renal_cyst	0.66 [0.57,0.75]	0.64 [0.55,0.73]	0.67 [0.58,0.76]	0.69 [0.60,0.77]	0.65 [0.56,0.74]	0.64 [0.55,0.73]	0.69 [0.60,0.77]	0.64 [0.55,0.73]	0.64 [0.54,0.73]	0.66 [0.57,0.75]	0.67 [0.57,0.75]
splenomegaly	0.68 [0.52,0.85]	0.75 [0.59,0.88]	0.71 [0.54,0.85]	0.72 [0.57,0.86]	0.71 [0.55,0.88]	0.71 [0.54,0.87]	0.73 [0.57,0.87]	0.70 [0.53,0.87]	0.71 [0.54,0.86]	0.72 [0.55,0.87]	0.71 [0.55,0.86]
covid	0.81 [0.65,0.96]	0.76 [0.58,0.91]	0.97 [0.92,1.00]	0.75 [0.57,0.91]	0.99 [0.96,1.00]	0.99 [0.95,1.00]	0.98 [0.93,1.00]	0.84 [0.68,0.96]	0.77 [0.60,0.93]	1.00 [0.98,1.00]	0.92 [0.78,1.00]
lung_nodule_malignancy	0.53 [0.42,0.63]	0.57 [0.45,0.69]	0.59 [0.47,0.71]	0.56 [0.46,0.67]	0.57 [0.45,0.68]	0.55 [0.43,0.66]	0.57 [0.46,0.69]	0.56 [0.45,0.67]	0.50 [0.39,0.62]	0.59 [0.48,0.70]	0.55 [0.43,0.66]

CVPR26, TAP-CT

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
TAP-CT, joint											
adrenal_hyperplasia	0.84 [0.69,0.96]	0.83 [0.67,0.97]	0.85 [0.69,0.98]	0.84 [0.68,0.96]	0.87 [0.74,0.97]	0.82 [0.63,0.95]	0.82 [0.64,0.94]	0.87 [0.74,0.97]	0.83 [0.68,0.95]	0.80 [0.63,0.95]	0.87 [0.71,0.98]
ascites	0.77 [0.58,0.93]	0.80 [0.62,0.94]	0.84 [0.69,0.97]	0.80 [0.61,0.95]	0.75 [0.56,0.91]	0.81 [0.65,0.96]	0.81 [0.65,0.95]	0.77 [0.58,0.91]	0.68 [0.47,0.85]	0.89 [0.76,0.98]	0.81 [0.64,0.94]
atherosclerosis	0.83 [0.70,0.95]	0.81 [0.64,0.93]	0.87 [0.73,0.97]	0.83 [0.69,0.95]	0.81 [0.64,0.94]	0.81 [0.64,0.94]	0.85 [0.70,0.96]	0.82 [0.65,0.94]	0.78 [0.60,0.91]	0.83 [0.66,0.97]	0.84 [0.67,0.96]
cholecystitis	0.76 [0.57,0.91]	0.70 [0.52,0.87]	0.71 [0.54,0.87]	0.77 [0.58,0.92]	0.76 [0.58,0.89]	0.73 [0.56,0.88]	0.73 [0.53,0.88]	0.71 [0.53,0.88]	0.71 [0.55,0.87]	0.72 [0.56,0.87]	0.72 [0.55,0.88]
colorectal_cancer	0.62 [0.39,0.83]	0.64 [0.40,0.86]	0.70 [0.46,0.89]	0.67 [0.43,0.87]	0.67 [0.44,0.88]	0.66 [0.41,0.88]	0.67 [0.43,0.86]	0.61 [0.38,0.81]	0.68 [0.44,0.88]	0.67 [0.44,0.86]	0.64 [0.39,0.83]
fatty_liver	0.84 [0.69,0.96]	0.80 [0.62,0.94]	0.78 [0.60,0.91]	0.76 [0.57,0.90]	0.82 [0.67,0.93]	0.79 [0.62,0.94]	0.78 [0.61,0.92]	0.82 [0.67,0.94]	0.75 [0.58,0.91]	0.81 [0.65,0.93]	0.88 [0.75,0.97]
gallstone	0.67 [0.53,0.79]	0.66 [0.52,0.80]	0.65 [0.50,0.78]	0.63 [0.48,0.77]	0.66 [0.51,0.79]	0.67 [0.53,0.81]	0.66 [0.52,0.79]	0.67 [0.52,0.80]	0.66 [0.51,0.79]	0.68 [0.54,0.80]	0.67 [0.53,0.80]
hydronephrosis	0.86 [0.71,0.98]	0.83 [0.69,0.95]	0.79 [0.63,0.93]	0.77 [0.59,0.92]	0.81 [0.65,0.94]	0.82 [0.65,0.96]	0.70 [0.49,0.88]	0.83 [0.67,0.96]	0.79 [0.61,0.93]	0.78 [0.58,0.94]	0.76 [0.58,0.91]
kidney_stone	0.74 [0.62,0.84]	0.69 [0.56,0.80]	0.70 [0.59,0.82]	0.71 [0.58,0.82]	0.71 [0.58,0.81]	0.67 [0.56,0.79]	0.63 [0.50,0.74]	0.69 [0.58,0.81]	0.68 [0.56,0.81]	0.63 [0.50,0.74]	0.71 [0.60,0.82]
liver_calcifications	0.43 [0.25,0.62]	0.51 [0.32,0.71]	0.46 [0.26,0.65]	0.51 [0.32,0.72]	0.50 [0.33,0.71]	0.52 [0.34,0.72]	0.47 [0.29,0.68]	0.40 [0.22,0.58]	0.51 [0.33,0.71]	0.54 [0.34,0.76]	0.58 [0.38,0.77]
liver_cyst	0.63 [0.53,0.73]	0.67 [0.57,0.76]	0.65 [0.55,0.75]	0.66 [0.56,0.75]	0.66 [0.56,0.75]	0.66 [0.56,0.76]	0.65 [0.55,0.74]	0.63 [0.52,0.72]	0.63 [0.53,0.72]	0.67 [0.57,0.77]	0.67 [0.56,0.76]
liver_lesion	0.63 [0.48,0.77]	0.71 [0.58,0.83]	0.70 [0.56,0.81]	0.69 [0.55,0.82]	0.70 [0.57,0.82]	0.68 [0.54,0.81]	0.70 [0.56,0.82]	0.66 [0.52,0.79]	0.70 [0.56,0.82]	0.69 [0.55,0.80]	0.66 [0.52,0.79]
lymphadenopathy	0.73 [0.55,0.88]	0.77 [0.60,0.91]	0.78 [0.61,0.92]	0.74 [0.56,0.89]	0.71 [0.55,0.86]	0.75 [0.56,0.89]	0.78 [0.61,0.92]	0.73 [0.58,0.88]	0.68 [0.51,0.85]	0.74 [0.55,0.89]	0.78 [0.59,0.93]
renal_cyst	0.60 [0.50,0.69]	0.64 [0.54,0.72]	0.66 [0.57,0.75]	0.66 [0.57,0.75]	0.68 [0.60,0.78]	0.68 [0.60,0.76]	0.66 [0.56,0.74]	0.61 [0.52,0.70]	0.65 [0.55,0.74]	0.70 [0.61,0.79]	0.67 [0.58,0.76]
splenomegaly	0.72 [0.56,0.86]	0.78 [0.63,0.91]	0.86 [0.73,0.95]	0.82 [0.66,0.95]	0.80 [0.65,0.92]	0.91 [0.81,0.98]	0.87 [0.75,0.97]	0.68 [0.52,0.84]	0.78 [0.64,0.91]	0.86 [0.74,0.96]	0.84 [0.70,0.95]
covid	0.83 [0.66,0.97]	0.74 [0.56,0.89]	0.83 [0.67,0.97]	0.82 [0.64,0.96]	0.84 [0.69,0.97]	0.82 [0.65,0.96]	0.86 [0.69,0.98]	0.82 [0.65,0.95]	0.84 [0.69,0.97]	0.89 [0.75,1.00]	0.93 [0.82,1.00]
lung_nodule_malignancy	0.51 [0.39,0.63]	0.67 [0.55,0.76]	0.65 [0.53,0.75]	0.67 [0.57,0.77]	0.64 [0.53,0.74]	0.65 [0.53,0.75]	0.65 [0.55,0.75]	0.48 [0.37,0.60]	0.68 [0.58,0.78]	0.66 [0.54,0.76]	0.62 [0.51,0.73]

CVPR26, TAP-CT

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
TAP-CT, joint (26 000 steps)											
adrenal_hyperplasia	0.80 [0.61,0.94]	0.76 [0.59,0.92]	0.62 [0.42,0.80]	0.80 [0.62,0.94]	0.76 [0.59,0.92]	0.81 [0.64,0.94]	0.85 [0.70,0.97]	0.81 [0.64,0.95]	0.80 [0.62,0.95]	0.88 [0.75,0.99]	0.88 [0.74,0.98]
ascites	0.79 [0.61,0.93]	0.81 [0.63,0.94]	0.82 [0.65,0.96]	0.85 [0.68,0.97]	0.77 [0.58,0.92]	0.86 [0.69,1.00]	0.82 [0.66,0.96]	0.78 [0.59,0.93]	0.72 [0.52,0.87]	0.76 [0.59,0.91]	0.62 [0.41,0.80]
atherosclerosis	0.80 [0.61,0.93]	0.82 [0.65,0.95]	0.84 [0.69,0.96]	0.82 [0.67,0.95]	0.74 [0.54,0.90]	0.77 [0.59,0.92]	0.80 [0.63,0.93]	0.79 [0.62,0.92]	0.78 [0.60,0.92]	0.85 [0.68,0.97]	0.62 [0.42,0.81]
cholecystitis	0.81 [0.64,0.94]	0.70 [0.53,0.85]	0.69 [0.50,0.83]	0.71 [0.53,0.87]	0.73 [0.57,0.88]	0.71 [0.51,0.87]	0.68 [0.50,0.84]	0.87 [0.75,0.97]	0.84 [0.70,0.94]	0.73 [0.56,0.88]	0.66 [0.48,0.83]
colorectal_cancer	0.60 [0.38,0.85]	0.71 [0.49,0.91]	0.59 [0.37,0.82]	0.70 [0.48,0.91]	0.61 [0.38,0.83]	0.63 [0.40,0.85]	0.71 [0.50,0.89]	0.58 [0.36,0.82]	0.72 [0.47,0.90]	0.68 [0.46,0.88]	0.58 [0.35,0.80]
fatty_liver	0.85 [0.71,0.97]	0.85 [0.72,0.96]	0.79 [0.62,0.93]	0.85 [0.71,0.95]	0.88 [0.75,0.98]	0.80 [0.64,0.93]	0.72 [0.54,0.88]	0.86 [0.73,0.96]	0.85 [0.70,0.96]	0.90 [0.78,0.97]	0.73 [0.56,0.88]
gallstone	0.62 [0.47,0.76]	0.62 [0.47,0.75]	0.66 [0.51,0.80]	0.62 [0.47,0.76]	0.65 [0.51,0.80]	0.68 [0.53,0.82]	0.68 [0.55,0.81]	0.65 [0.51,0.79]	0.63 [0.48,0.77]	0.66 [0.52,0.80]	0.65 [0.52,0.79]
hydronephrosis	0.83 [0.67,0.96]	0.84 [0.69,0.95]	0.76 [0.59,0.92]	0.74 [0.55,0.90]	0.79 [0.62,0.93]	0.76 [0.57,0.91]	0.69 [0.50,0.88]	0.84 [0.68,0.95]	0.80 [0.64,0.93]	0.77 [0.57,0.93]	0.70 [0.49,0.87]
kidney_stone	0.68 [0.55,0.79]	0.72 [0.61,0.83]	0.65 [0.53,0.77]	0.67 [0.56,0.79]	0.67 [0.55,0.79]	0.65 [0.53,0.77]	0.65 [0.53,0.77]	0.69 [0.56,0.80]	0.73 [0.61,0.83]	0.67 [0.54,0.77]	0.59 [0.46,0.72]
liver_calcifications	0.49 [0.30,0.69]	0.43 [0.25,0.62]	0.44 [0.25,0.63]	0.54 [0.34,0.73]	0.44 [0.24,0.62]	0.52 [0.34,0.71]	0.48 [0.30,0.67]	0.39 [0.20,0.58]	0.42 [0.24,0.63]	0.52 [0.33,0.72]	0.40 [0.22,0.58]
liver_cyst	0.65 [0.54,0.75]	0.69 [0.59,0.78]	0.63 [0.53,0.73]	0.71 [0.61,0.80]	0.72 [0.63,0.81]	0.72 [0.63,0.81]	0.63 [0.51,0.72]	0.63 [0.52,0.72]	0.68 [0.58,0.77]	0.67 [0.57,0.76]	0.59 [0.48,0.69]
liver_lesion	0.54 [0.39,0.68]	0.71 [0.58,0.83]	0.70 [0.56,0.82]	0.66 [0.52,0.79]	0.70 [0.57,0.82]	0.70 [0.57,0.82]	0.66 [0.52,0.79]	0.60 [0.46,0.75]	0.71 [0.58,0.82]	0.70 [0.57,0.83]	0.67 [0.54,0.80]
lymphadenopathy	0.69 [0.51,0.86]	0.73 [0.55,0.88]	0.63 [0.45,0.79]	0.74 [0.57,0.90]	0.78 [0.60,0.93]	0.78 [0.60,0.91]	0.68 [0.48,0.84]	0.69 [0.52,0.85]	0.74 [0.56,0.89]	0.77 [0.59,0.92]	0.67 [0.47,0.83]
renal_cyst	0.61 [0.52,0.70]	0.66 [0.57,0.75]	0.63 [0.54,0.73]	0.69 [0.59,0.77]	0.72 [0.64,0.81]	0.66 [0.56,0.74]	0.68 [0.59,0.77]	0.62 [0.53,0.72]	0.65 [0.56,0.75]	0.68 [0.59,0.77]	0.58 [0.48,0.67]
splenomegaly	0.73 [0.58,0.87]	0.83 [0.69,0.93]	0.87 [0.77,0.96]	0.81 [0.65,0.93]	0.81 [0.67,0.92]	0.88 [0.76,0.97]	0.75 [0.60,0.89]	0.73 [0.56,0.88]	0.79 [0.65,0.91]	0.82 [0.67,0.93]	0.76 [0.61,0.90]
covid	0.78 [0.61,0.94]	0.76 [0.59,0.91]	0.81 [0.64,0.95]	0.80 [0.61,0.94]	0.94 [0.80,1.00]	0.78 [0.59,0.95]	0.85 [0.69,0.99]	0.81 [0.66,0.94]	0.82 [0.65,0.96]	0.99 [0.96,1.00]	0.79 [0.62,0.94]
lung_nodule_malignancy	0.52 [0.40,0.64]	0.64 [0.53,0.74]	0.63 [0.52,0.74]	0.68 [0.58,0.78]	0.65 [0.54,0.75]	0.53 [0.42,0.64]	0.58 [0.47,0.69]	0.53 [0.41,0.64]	0.65 [0.54,0.76]	0.78 [0.70,0.87]	0.56 [0.45,0.68]

CVPR26, SPECTRE

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
SPECTRE, per-task											
adrenal_hyperplasia	0.81 [0.64,0.93]	0.52 [0.33,0.75]	0.83 [0.66,0.94]	0.80 [0.62,0.95]	0.86 [0.71,0.96]	0.59 [0.37,0.80]	0.76 [0.58,0.91]	0.80 [0.64,0.94]	0.76 [0.57,0.92]	0.78 [0.60,0.93]	0.76 [0.57,0.93]
ascites	0.89 [0.75,0.99]	0.90 [0.78,0.98]	0.92 [0.80,1.00]	0.88 [0.73,0.98]	0.88 [0.73,0.98]	0.90 [0.76,1.00]	0.88 [0.75,0.98]	0.92 [0.80,1.00]	0.89 [0.75,0.98]	0.80 [0.63,0.93]	0.55 [0.39,0.70]
atherosclerosis	0.76 [0.58,0.92]	0.78 [0.61,0.93]	0.80 [0.63,0.94]	0.83 [0.68,0.95]	0.84 [0.68,0.96]	0.84 [0.69,0.96]	0.79 [0.62,0.94]	0.74 [0.55,0.90]	0.70 [0.52,0.87]	0.78 [0.61,0.93]	0.82 [0.67,0.95]
cholecystitis	0.73 [0.57,0.88]	0.72 [0.55,0.87]	0.80 [0.67,0.92]	0.80 [0.65,0.93]	0.88 [0.77,0.97]	0.84 [0.71,0.94]	0.82 [0.67,0.93]	0.83 [0.70,0.93]	0.79 [0.64,0.93]	0.80 [0.65,0.92]	0.65 [0.48,0.82]
colorectal_cancer	0.60 [0.37,0.83]	0.70 [0.48,0.90]	0.71 [0.50,0.90]	0.65 [0.43,0.86]	0.56 [0.33,0.79]	0.72 [0.51,0.90]	0.67 [0.48,0.82]	0.74 [0.53,0.91]	0.75 [0.54,0.90]	0.74 [0.52,0.90]	0.50 [0.50,0.50]
fatty_liver	0.81 [0.65,0.93]	0.79 [0.63,0.93]	0.95 [0.86,1.00]	0.97 [0.89,1.00]	0.98 [0.93,1.00]	0.92 [0.80,0.99]	0.95 [0.83,1.00]	0.90 [0.79,0.98]	0.94 [0.83,1.00]	0.94 [0.86,1.00]	0.77 [0.60,0.90]
gallstone	0.66 [0.52,0.80]	0.70 [0.55,0.81]	0.64 [0.50,0.78]	0.69 [0.56,0.82]	0.79 [0.66,0.91]	0.69 [0.54,0.82]	0.78 [0.64,0.89]	0.64 [0.49,0.78]	0.75 [0.61,0.87]	0.83 [0.71,0.92]	0.57 [0.42,0.71]
hydronephrosis	0.85 [0.69,0.96]	0.85 [0.70,0.95]	0.91 [0.79,0.99]	0.93 [0.82,1.00]	0.94 [0.85,1.00]	0.90 [0.77,0.99]	0.87 [0.72,0.98]	0.93 [0.82,0.99]	0.90 [0.80,0.98]	0.94 [0.85,1.00]	0.56 [0.34,0.76]
kidney_stone	0.66 [0.54,0.78]	0.68 [0.56,0.81]	0.71 [0.60,0.82]	0.74 [0.63,0.84]	0.72 [0.61,0.83]	0.67 [0.54,0.79]	0.69 [0.56,0.81]	0.67 [0.54,0.78]	0.68 [0.55,0.79]	0.76 [0.66,0.86]	0.59 [0.46,0.70]
liver_calcifications	0.58 [0.39,0.77]	0.69 [0.49,0.86]	0.63 [0.42,0.83]	0.52 [0.32,0.71]	0.52 [0.32,0.71]	0.63 [0.44,0.81]	0.56 [0.37,0.75]	0.62 [0.43,0.81]	0.62 [0.43,0.81]	0.57 [0.41,0.72]	0.55 [0.46,0.65]
liver_cyst	0.58 [0.48,0.69]	0.63 [0.53,0.73]	0.65 [0.54,0.74]	0.65 [0.55,0.75]	0.65 [0.56,0.74]	0.66 [0.55,0.75]	0.64 [0.54,0.74]	0.65 [0.56,0.75]	0.66 [0.56,0.75]	0.66 [0.57,0.75]	0.62 [0.53,0.71]
liver_lesion	0.68 [0.55,0.81]	0.64 [0.51,0.77]	0.62 [0.48,0.76]	0.71 [0.57,0.83]	0.70 [0.57,0.83]	0.70 [0.56,0.82]	0.75 [0.62,0.86]	0.69 [0.57,0.81]	0.69 [0.54,0.82]	0.72 [0.60,0.84]	0.60 [0.47,0.74]
lymphadenopathy	0.76 [0.61,0.89]	0.61 [0.43,0.78]	0.78 [0.62,0.91]	0.74 [0.57,0.88]	0.77 [0.61,0.90]	0.83 [0.69,0.94]	0.78 [0.63,0.91]	0.79 [0.64,0.91]	0.73 [0.56,0.88]	0.70 [0.50,0.85]	0.53 [0.50,0.58]
renal_cyst	0.63 [0.54,0.72]	0.65 [0.55,0.74]	0.57 [0.48,0.67]	0.72 [0.63,0.80]	0.71 [0.62,0.79]	0.58 [0.48,0.67]	0.57 [0.47,0.66]	0.60 [0.51,0.69]	0.67 [0.57,0.75]	0.70 [0.61,0.78]	0.48 [0.41,0.55]
splenomegaly	0.76 [0.59,0.89]	0.70 [0.52,0.85]	0.87 [0.75,0.96]	0.77 [0.61,0.90]	0.70 [0.53,0.83]	0.87 [0.75,0.96]	0.83 [0.68,0.94]	0.82 [0.67,0.93]	0.77 [0.61,0.89]	0.76 [0.62,0.89]	0.57 [0.41,0.75]
covid	1.00 [1.00,1.00]	1.00 [0.98,1.00]	0.99 [0.95,1.00]	0.98 [0.93,1.00]	0.93 [0.81,1.00]	0.98 [0.94,1.00]	0.99 [0.95,1.00]	1.00 [0.98,1.00]	1.00 [0.98,1.00]	0.98 [0.94,1.00]	0.72 [0.54,0.87]
lung_nodule_malignancy	0.74 [0.63,0.84]	0.74 [0.63,0.84]	0.68 [0.57,0.78]	0.71 [0.60,0.80]	0.75 [0.66,0.84]	0.71 [0.60,0.81]	0.74 [0.64,0.83]	0.72 [0.61,0.81]	0.73 [0.63,0.82]	0.76 [0.66,0.85]	0.82 [0.73,0.89]

CVPR26, SPECTRE

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
SPECTRE, joint											
adrenal_hyperplasia	0.72 [0.55,0.89]	0.57 [0.37,0.77]	0.52 [0.31,0.73]	0.63 [0.42,0.80]	0.67 [0.47,0.84]	0.54 [0.32,0.73]	0.69 [0.52,0.86]	0.78 [0.60,0.93]	0.65 [0.46,0.83]	0.74 [0.55,0.89]	0.50 [0.50,0.50]
ascites	0.83 [0.67,0.95]	0.87 [0.72,0.98]	0.84 [0.68,0.96]	0.81 [0.63,0.94]	0.84 [0.67,0.95]	0.86 [0.71,0.98]	0.81 [0.66,0.94]	0.78 [0.62,0.92]	0.82 [0.67,0.95]	0.87 [0.72,0.97]	0.60 [0.43,0.75]
atherosclerosis	0.78 [0.60,0.93]	0.76 [0.58,0.91]	0.76 [0.59,0.91]	0.78 [0.60,0.93]	0.79 [0.62,0.93]	0.74 [0.57,0.89]	0.72 [0.54,0.88]	0.81 [0.64,0.94]	0.77 [0.59,0.91]	0.74 [0.56,0.91]	0.53 [0.50,0.59]
cholecystitis	0.68 [0.51,0.83]	0.80 [0.64,0.92]	0.76 [0.58,0.91]	0.68 [0.50,0.85]	0.77 [0.62,0.90]	0.73 [0.56,0.86]	0.71 [0.53,0.86]	0.71 [0.52,0.87]	0.69 [0.53,0.85]	0.78 [0.63,0.91]	0.66 [0.51,0.80]
colorectal_cancer	0.69 [0.49,0.89]	0.71 [0.48,0.89]	0.67 [0.46,0.86]	0.67 [0.43,0.86]	0.71 [0.49,0.89]	0.71 [0.50,0.89]	0.66 [0.43,0.86]	0.72 [0.51,0.92]	0.70 [0.49,0.89]	0.65 [0.42,0.85]	0.35 [0.18,0.53]
fatty_liver	0.83 [0.68,0.95]	0.71 [0.53,0.88]	0.69 [0.51,0.85]	0.77 [0.60,0.91]	0.80 [0.63,0.93]	0.75 [0.57,0.90]	0.59 [0.40,0.77]	0.80 [0.64,0.93]	0.70 [0.52,0.86]	0.80 [0.65,0.93]	0.67 [0.52,0.80]
gallstone	0.48 [0.33,0.63]	0.75 [0.62,0.87]	0.68 [0.53,0.80]	0.68 [0.53,0.81]	0.77 [0.65,0.88]	0.77 [0.65,0.89]	0.75 [0.63,0.87]	0.75 [0.63,0.87]	0.80 [0.69,0.90]	0.69 [0.55,0.82]	0.66 [0.53,0.78]
hydronephrosis	0.90 [0.78,0.98]	0.81 [0.66,0.94]	0.76 [0.58,0.92]	0.78 [0.60,0.93]	0.74 [0.54,0.89]	0.78 [0.61,0.93]	0.75 [0.59,0.90]	0.86 [0.72,0.97]	0.69 [0.51,0.85]	0.82 [0.66,0.95]	0.74 [0.59,0.88]
kidney_stone	0.63 [0.51,0.74]	0.67 [0.54,0.78]	0.66 [0.55,0.78]	0.70 [0.59,0.81]	0.70 [0.58,0.82]	0.66 [0.56,0.78]	0.66 [0.54,0.77]	0.61 [0.48,0.73]	0.70 [0.58,0.81]	0.68 [0.56,0.79]	0.53 [0.45,0.60]
liver_calcifications	0.34 [0.16,0.53]	0.70 [0.52,0.86]	0.63 [0.44,0.81]	0.68 [0.49,0.85]	0.65 [0.47,0.82]	0.56 [0.37,0.76]	0.47 [0.28,0.66]	0.54 [0.36,0.76]	0.73 [0.55,0.88]	0.49 [0.29,0.68]	0.50 [0.43,0.58]
liver_cyst	0.62 [0.52,0.71]	0.65 [0.55,0.75]	0.66 [0.54,0.75]	0.65 [0.55,0.75]	0.69 [0.59,0.77]	0.71 [0.62,0.80]	0.62 [0.51,0.72]	0.65 [0.55,0.76]	0.63 [0.53,0.73]	0.69 [0.59,0.78]	0.55 [0.47,0.63]
liver_lesion	0.61 [0.46,0.76]	0.63 [0.48,0.76]	0.64 [0.50,0.77]	0.61 [0.46,0.75]	0.65 [0.51,0.79]	0.63 [0.49,0.76]	0.68 [0.55,0.80]	0.66 [0.52,0.79]	0.59 [0.44,0.73]	0.65 [0.50,0.78]	0.52 [0.42,0.61]
lymphadenopathy	0.80 [0.64,0.93]	0.78 [0.63,0.91]	0.81 [0.65,0.94]	0.75 [0.58,0.90]	0.79 [0.63,0.92]	0.78 [0.62,0.92]	0.73 [0.54,0.88]	0.80 [0.65,0.93]	0.74 [0.57,0.89]	0.78 [0.61,0.91]	0.50 [0.50,0.50]
renal_cyst	0.64 [0.54,0.73]	0.63 [0.53,0.72]	0.63 [0.54,0.72]	0.70 [0.61,0.78]	0.64 [0.55,0.73]	0.68 [0.60,0.77]	0.62 [0.52,0.71]	0.67 [0.58,0.77]	0.60 [0.51,0.69]	0.67 [0.59,0.77]	0.52 [0.44,0.60]
splenomegaly	0.66 [0.49,0.81]	0.78 [0.63,0.91]	0.79 [0.64,0.92]	0.85 [0.71,0.95]	0.82 [0.68,0.93]	0.82 [0.68,0.93]	0.61 [0.44,0.78]	0.76 [0.60,0.89]	0.85 [0.71,0.95]	0.83 [0.68,0.94]	0.72 [0.58,0.84]
covid	0.99 [0.96,1.00]	1.00 [1.00,1.00]	0.99 [0.95,1.00]	1.00 [0.98,1.00]	1.00 [0.98,1.00]	1.00 [0.98,1.00]	1.00 [0.98,1.00]	0.99 [0.96,1.00]	1.00 [1.00,1.00]	0.98 [0.95,1.00]	0.71 [0.53,0.89]
lung_nodule_malignancy	0.74 [0.64,0.83]	0.73 [0.62,0.82]	0.71 [0.60,0.80]	0.71 [0.61,0.81]	0.70 [0.60,0.81]	0.69 [0.58,0.80]	0.65 [0.54,0.76]	0.74 [0.63,0.83]	0.72 [0.61,0.83]	0.72 [0.61,0.81]	0.54 [0.44,0.63]

CVPR26, SPECTRE

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
SPECTRE, joint (26 000 steps)											
adrenal_hyperplasia	0.73 [0.53,0.90]	0.44 [0.24,0.66]	0.70 [0.52,0.88]	0.72 [0.53,0.89]	0.71 [0.52,0.87]	0.64 [0.44,0.84]	0.65 [0.45,0.83]	0.70 [0.50,0.86]	0.67 [0.49,0.85]	0.74 [0.55,0.90]	0.29 [0.15,0.46]
ascites	0.88 [0.76,0.99]	0.89 [0.77,0.98]	0.85 [0.70,0.97]	0.88 [0.75,0.98]	0.89 [0.76,0.98]	0.87 [0.73,0.98]	0.87 [0.74,0.97]	0.85 [0.69,0.96]	0.86 [0.71,0.96]	0.87 [0.72,0.97]	0.49 [0.31,0.69]
atherosclerosis	0.83 [0.67,0.96]	0.74 [0.55,0.89]	0.82 [0.67,0.95]	0.79 [0.63,0.92]	0.72 [0.54,0.88]	0.77 [0.59,0.92]	0.81 [0.65,0.94]	0.79 [0.63,0.93]	0.85 [0.69,0.96]	0.81 [0.64,0.94]	0.67 [0.47,0.84]
cholecystitis	0.76 [0.59,0.90]	0.70 [0.52,0.85]	0.77 [0.61,0.90]	0.78 [0.65,0.91]	0.76 [0.60,0.89]	0.71 [0.54,0.85]	0.76 [0.62,0.90]	0.62 [0.43,0.80]	0.72 [0.56,0.87]	0.66 [0.49,0.82]	0.47 [0.33,0.60]
colorectal_cancer	0.71 [0.51,0.89]	0.58 [0.35,0.83]	0.64 [0.42,0.83]	0.74 [0.54,0.91]	0.73 [0.50,0.91]	0.71 [0.51,0.90]	0.77 [0.57,0.93]	0.81 [0.61,0.96]	0.77 [0.57,0.93]	0.67 [0.44,0.88]	0.70 [0.49,0.88]
fatty_liver	0.82 [0.69,0.96]	0.64 [0.46,0.81]	0.90 [0.76,0.99]	0.92 [0.82,1.00]	0.92 [0.82,0.99]	0.91 [0.79,0.99]	0.88 [0.73,0.99]	0.84 [0.70,0.96]	0.85 [0.69,0.97]	0.84 [0.68,0.96]	0.79 [0.64,0.92]
gallstone	0.70 [0.56,0.83]	0.78 [0.65,0.90]	0.66 [0.51,0.79]	0.69 [0.55,0.82]	0.77 [0.64,0.88]	0.67 [0.52,0.80]	0.66 [0.53,0.80]	0.69 [0.54,0.81]	0.69 [0.55,0.81]	0.71 [0.56,0.83]	0.60 [0.45,0.72]
hydronephrosis	0.81 [0.65,0.95]	0.77 [0.60,0.92]	0.84 [0.69,0.95]	0.88 [0.74,0.97]	0.83 [0.68,0.96]	0.78 [0.60,0.92]	0.85 [0.70,0.96]	0.83 [0.69,0.95]	0.89 [0.76,0.98]	0.82 [0.65,0.95]	0.56 [0.35,0.75]
kidney_stone	0.61 [0.47,0.73]	0.68 [0.55,0.80]	0.71 [0.58,0.81]	0.69 [0.57,0.80]	0.70 [0.57,0.80]	0.73 [0.62,0.84]	0.74 [0.63,0.84]	0.72 [0.60,0.83]	0.73 [0.61,0.83]	0.69 [0.57,0.80]	0.54 [0.45,0.63]
liver_calcifications	0.51 [0.32,0.70]	0.74 [0.57,0.90]	0.54 [0.36,0.73]	0.51 [0.33,0.69]	0.52 [0.32,0.70]	0.60 [0.42,0.78]	0.56 [0.37,0.75]	0.54 [0.34,0.74]	0.63 [0.46,0.80]	0.59 [0.39,0.79]	0.52 [0.34,0.70]
liver_cyst	0.64 [0.55,0.74]	0.64 [0.54,0.74]	0.72 [0.62,0.81]	0.71 [0.61,0.79]	0.69 [0.59,0.78]	0.70 [0.61,0.79]	0.70 [0.60,0.79]	0.67 [0.57,0.76]	0.68 [0.58,0.77]	0.68 [0.58,0.77]	0.43 [0.33,0.52]
liver_lesion	0.62 [0.48,0.75]	0.62 [0.48,0.76]	0.68 [0.53,0.80]	0.67 [0.54,0.80]	0.67 [0.53,0.80]	0.66 [0.52,0.80]	0.66 [0.53,0.79]	0.66 [0.51,0.79]	0.62 [0.47,0.76]	0.68 [0.54,0.80]	0.62 [0.48,0.75]
lymphadenopathy	0.76 [0.58,0.90]	0.73 [0.57,0.88]	0.82 [0.68,0.94]	0.75 [0.59,0.90]	0.75 [0.57,0.89]	0.80 [0.63,0.94]	0.79 [0.63,0.92]	0.76 [0.58,0.90]	0.78 [0.61,0.91]	0.80 [0.63,0.94]	0.47 [0.28,0.66]
renal_cyst	0.61 [0.52,0.71]	0.64 [0.54,0.73]	0.65 [0.56,0.75]	0.69 [0.60,0.78]	0.67 [0.58,0.76]	0.70 [0.62,0.79]	0.68 [0.59,0.78]	0.60 [0.50,0.69]	0.66 [0.56,0.75]	0.68 [0.59,0.77]	0.52 [0.44,0.59]
splenomegaly	0.84 [0.71,0.95]	0.78 [0.62,0.90]	0.87 [0.75,0.96]	0.82 [0.68,0.94]	0.82 [0.68,0.93]	0.83 [0.70,0.94]	0.88 [0.77,0.97]	0.83 [0.68,0.94]	0.80 [0.65,0.93]	0.87 [0.74,0.96]	0.50 [0.33,0.68]
covid	1.00 [0.98,1.00]	0.99 [0.96,1.00]	1.00 [0.98,1.00]	0.99 [0.96,1.00]	0.99 [0.96,1.00]	0.99 [0.95,1.00]	0.99 [0.95,1.00]	1.00 [1.00,1.00]	1.00 [0.98,1.00]	0.98 [0.95,1.00]	0.95 [0.84,1.00]
lung_nodule_malignancy	0.79 [0.70,0.87]	0.74 [0.64,0.82]	0.68 [0.58,0.78]	0.74 [0.63,0.84]	0.77 [0.66,0.86]	0.70 [0.60,0.80]	0.70 [0.60,0.81]	0.77 [0.67,0.86]	0.69 [0.58,0.80]	0.76 [0.66,0.85]	0.67 [0.57,0.77]

FDG-PET/CT

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
TAP-CT, per-task											
LUNG_CANCER	0.93 [0.87,0.98]	0.93 [0.87,0.97]	0.92 [0.86,0.97]	0.93 [0.88,0.97]	0.86 [0.76,0.93]	0.90 [0.82,0.96]	0.90 [0.83,0.96]	0.92 [0.86,0.97]	0.91 [0.85,0.96]	0.94 [0.88,0.98]	0.86 [0.77,0.93]
LYMPHOMA	0.93 [0.86,0.99]	0.89 [0.78,0.96]	0.93 [0.86,0.98]	0.89 [0.80,0.96]	0.92 [0.84,0.99]	0.93 [0.86,0.99]	0.94 [0.86,0.99]	0.93 [0.85,0.99]	0.96 [0.91,0.99]	0.94 [0.85,1.00]	0.80 [0.67,0.90]
MELANOMA	0.81 [0.72,0.89]	0.76 [0.64,0.86]	0.79 [0.68,0.90]	0.83 [0.74,0.90]	0.81 [0.71,0.90]	0.82 [0.73,0.89]	0.76 [0.63,0.88]	0.84 [0.75,0.90]	0.81 [0.72,0.90]	0.77 [0.65,0.88]	0.72 [0.58,0.85]
NEGATIVE	0.78 [0.69,0.86]	0.77 [0.68,0.85]	0.77 [0.68,0.85]	0.81 [0.73,0.89]	0.77 [0.68,0.85]	0.73 [0.64,0.82]	0.67 [0.57,0.76]	0.77 [0.68,0.85]	0.75 [0.65,0.83]	0.68 [0.59,0.77]	0.70 [0.60,0.79]
TAP-CT, joint											
LUNG_CANCER	0.93 [0.88,0.98]	0.92 [0.85,0.97]	0.94 [0.89,0.98]	0.94 [0.90,0.98]	0.89 [0.80,0.97]	0.93 [0.86,0.97]	0.81 [0.72,0.89]	0.93 [0.87,0.97]	0.92 [0.85,0.97]	0.93 [0.88,0.97]	0.57 [0.44,0.70]
LYMPHOMA	0.94 [0.85,1.00]	0.92 [0.82,0.99]	0.92 [0.83,0.98]	0.95 [0.87,1.00]	0.93 [0.82,1.00]	0.93 [0.84,0.99]	0.91 [0.81,0.97]	0.93 [0.85,0.99]	0.96 [0.92,1.00]	0.95 [0.88,1.00]	0.58 [0.47,0.68]
MELANOMA	0.80 [0.70,0.88]	0.79 [0.69,0.87]	0.79 [0.69,0.89]	0.80 [0.71,0.88]	0.81 [0.72,0.89]	0.81 [0.71,0.89]	0.78 [0.66,0.87]	0.81 [0.71,0.89]	0.78 [0.66,0.88]	0.80 [0.67,0.90]	0.71 [0.56,0.84]
NEGATIVE	0.76 [0.67,0.84]	0.81 [0.73,0.88]	0.81 [0.72,0.88]	0.76 [0.68,0.84]	0.76 [0.67,0.84]	0.76 [0.68,0.84]	0.65 [0.55,0.74]	0.77 [0.68,0.84]	0.79 [0.70,0.86]	0.80 [0.71,0.87]	0.58 [0.47,0.69]
SPECTRE, per-task											
LUNG_CANCER	0.93 [0.87,0.97]	0.94 [0.90,0.98]	0.95 [0.90,0.98]	0.96 [0.92,0.99]	0.96 [0.92,0.99]	0.96 [0.92,0.99]	0.95 [0.91,0.98]	0.96 [0.92,0.99]	0.96 [0.92,0.99]	0.97 [0.94,0.99]	0.80 [0.70,0.89]
LYMPHOMA	0.94 [0.85,0.99]	0.92 [0.84,0.98]	0.96 [0.89,1.00]	0.97 [0.91,1.00]	0.97 [0.91,1.00]	0.97 [0.91,1.00]	0.94 [0.86,0.99]	0.95 [0.87,0.99]	0.95 [0.88,1.00]	0.97 [0.91,1.00]	0.50 [0.50,0.50]
MELANOMA	0.72 [0.60,0.84]	0.78 [0.65,0.88]	0.79 [0.68,0.89]	0.78 [0.66,0.88]	0.74 [0.61,0.86]	0.79 [0.69,0.89]	0.75 [0.63,0.86]	0.80 [0.70,0.88]	0.77 [0.65,0.87]	0.81 [0.71,0.89]	0.63 [0.50,0.75]
NEGATIVE	0.84 [0.77,0.91]	0.81 [0.73,0.88]	0.83 [0.74,0.90]	0.81 [0.72,0.88]	0.83 [0.75,0.90]	0.84 [0.77,0.91]	0.85 [0.77,0.92]	0.84 [0.76,0.91]	0.79 [0.71,0.87]	0.65 [0.55,0.74]	0.53 [0.44,0.63]
SPECTRE, joint											
LUNG_CANCER	0.97 [0.95,0.99]	0.97 [0.93,0.99]	0.99 [0.97,1.00]	0.98 [0.95,1.00]	0.99 [0.96,1.00]	0.99 [0.97,1.00]	0.99 [0.97,1.00]	0.96 [0.92,0.99]	0.98 [0.94,1.00]	0.99 [0.97,1.00]	0.76 [0.64,0.86]
LYMPHOMA	0.93 [0.84,0.99]	0.95 [0.87,0.99]	0.96 [0.90,1.00]	0.96 [0.91,1.00]	0.95 [0.88,1.00]	0.96 [0.91,1.00]	0.96 [0.91,1.00]	0.95 [0.87,0.99]	0.94 [0.87,0.99]	0.94 [0.88,0.99]	0.69 [0.58,0.80]
MELANOMA	0.80 [0.69,0.90]	0.77 [0.65,0.88]	0.82 [0.73,0.90]	0.78 [0.68,0.88]	0.80 [0.69,0.89]	0.82 [0.73,0.90]	0.81 [0.71,0.91]	0.82 [0.72,0.90]	0.77 [0.66,0.87]	0.85 [0.75,0.93]	0.60 [0.45,0.75]
NEGATIVE	0.83 [0.75,0.90]	0.84 [0.76,0.90]	0.83 [0.75,0.90]	0.82 [0.74,0.90]	0.82 [0.73,0.90]	0.83 [0.75,0.90]	0.84 [0.75,0.91]	0.82 [0.74,0.89]	0.84 [0.76,0.91]	0.85 [0.77,0.91]	0.64 [0.53,0.74]

RSNA22 C-spine

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
TAP-CT, per-task											
C1	0.67 [0.54,0.79]	0.71 [0.59,0.83]	0.68 [0.57,0.79]	0.72 [0.60,0.83]	0.70 [0.58,0.82]	0.72 [0.59,0.84]	0.74 [0.62,0.83]	0.69 [0.56,0.79]	0.71 [0.60,0.83]	0.77 [0.65,0.88]	0.70 [0.57,0.79]
C2	0.71 [0.60,0.80]	0.76 [0.67,0.85]	0.73 [0.62,0.83]	0.76 [0.66,0.84]	0.75 [0.64,0.84]	0.74 [0.64,0.84]	0.72 [0.62,0.82]	0.70 [0.59,0.80]	0.73 [0.62,0.82]	0.77 [0.66,0.86]	0.70 [0.58,0.79]
C3	0.64 [0.42,0.84]	0.55 [0.24,0.84]	0.40 [0.17,0.63]	0.40 [0.19,0.59]	0.46 [0.20,0.71]	0.55 [0.25,0.85]	0.54 [0.23,0.88]	0.51 [0.29,0.76]	0.53 [0.22,0.82]	0.59 [0.30,0.89]	0.53 [0.24,0.84]
C4	0.64 [0.40,0.84]	0.58 [0.38,0.75]	0.56 [0.34,0.78]	0.62 [0.41,0.79]	0.51 [0.31,0.71]	0.53 [0.33,0.73]	0.51 [0.32,0.72]	0.64 [0.41,0.85]	0.56 [0.39,0.75]	0.52 [0.35,0.70]	0.66 [0.48,0.82]
C5	0.51 [0.36,0.67]	0.53 [0.38,0.68]	0.59 [0.46,0.72]	0.50 [0.34,0.66]	0.44 [0.31,0.58]	0.51 [0.36,0.66]	0.60 [0.45,0.74]	0.49 [0.34,0.66]	0.60 [0.46,0.74]	0.54 [0.41,0.66]	0.57 [0.43,0.68]
C6	0.50 [0.37,0.61]	0.62 [0.51,0.73]	0.66 [0.55,0.76]	0.59 [0.48,0.69]	0.66 [0.55,0.77]	0.65 [0.56,0.75]	0.64 [0.53,0.74]	0.50 [0.38,0.63]	0.64 [0.54,0.74]	0.65 [0.54,0.77]	0.68 [0.56,0.78]
C7	0.63 [0.53,0.73]	0.57 [0.46,0.67]	0.68 [0.58,0.77]	0.61 [0.50,0.72]	0.63 [0.53,0.72]	0.66 [0.57,0.75]	0.65 [0.55,0.74]	0.63 [0.53,0.73]	0.59 [0.48,0.68]	0.66 [0.55,0.75]	0.65 [0.54,0.74]
patient_overall	0.65 [0.57,0.73]	0.68 [0.60,0.75]	0.68 [0.61,0.75]	0.67 [0.59,0.74]	0.75 [0.68,0.82]	0.76 [0.70,0.83]	0.68 [0.61,0.75]	0.68 [0.61,0.75]	0.67 [0.59,0.74]	0.78 [0.72,0.84]	0.62 [0.55,0.69]
TAP-CT, joint											
C1	0.64 [0.51,0.76]	0.71 [0.59,0.82]	0.72 [0.60,0.84]	0.75 [0.63,0.85]	0.72 [0.60,0.83]	0.71 [0.58,0.83]	0.68 [0.56,0.80]	0.62 [0.49,0.75]	0.71 [0.59,0.83]	0.70 [0.57,0.82]	0.76 [0.64,0.86]
C2	0.76 [0.66,0.85]	0.73 [0.63,0.83]	0.76 [0.67,0.85]	0.75 [0.65,0.85]	0.77 [0.67,0.85]	0.74 [0.62,0.83]	0.70 [0.60,0.80]	0.74 [0.65,0.85]	0.72 [0.61,0.82]	0.79 [0.71,0.87]	0.81 [0.72,0.89]
C3	0.55 [0.28,0.84]	0.45 [0.21,0.72]	0.63 [0.35,0.89]	0.69 [0.42,0.91]	0.69 [0.43,0.92]	0.59 [0.32,0.85]	0.47 [0.16,0.80]	0.59 [0.34,0.84]	0.43 [0.20,0.67]	0.63 [0.28,0.97]	0.62 [0.28,0.95]
C4	0.69 [0.46,0.89]	0.65 [0.44,0.84]	0.74 [0.53,0.91]	0.71 [0.51,0.88]	0.70 [0.51,0.86]	0.71 [0.53,0.88]	0.53 [0.32,0.73]	0.67 [0.46,0.85]	0.65 [0.46,0.82]	0.73 [0.50,0.91]	0.78 [0.59,0.92]
C5	0.52 [0.36,0.68]	0.56 [0.41,0.70]	0.69 [0.57,0.79]	0.65 [0.53,0.76]	0.68 [0.57,0.79]	0.63 [0.51,0.75]	0.56 [0.45,0.68]	0.51 [0.36,0.66]	0.57 [0.41,0.70]	0.69 [0.57,0.80]	0.69 [0.57,0.81]
C6	0.54 [0.42,0.65]	0.58 [0.46,0.69]	0.71 [0.61,0.80]	0.68 [0.58,0.78]	0.71 [0.62,0.81]	0.68 [0.58,0.79]	0.69 [0.57,0.80]	0.53 [0.41,0.66]	0.57 [0.47,0.68]	0.68 [0.57,0.78]	0.69 [0.58,0.80]
C7	0.64 [0.53,0.74]	0.66 [0.57,0.75]	0.71 [0.62,0.80]	0.70 [0.61,0.78]	0.70 [0.62,0.77]	0.68 [0.59,0.77]	0.58 [0.49,0.68]	0.65 [0.54,0.74]	0.64 [0.55,0.73]	0.72 [0.64,0.79]	0.73 [0.64,0.81]
patient_overall	0.67 [0.60,0.75]	0.65 [0.58,0.73]	0.75 [0.68,0.81]	0.72 [0.65,0.79]	0.73 [0.66,0.80]	0.70 [0.62,0.77]	0.59 [0.52,0.67]	0.68 [0.61,0.75]	0.67 [0.60,0.74]	0.75 [0.69,0.81]	0.77 [0.70,0.83]
SPECTRE, per-task											
C1	0.55 [0.42,0.67]	0.56 [0.45,0.69]	0.68 [0.56,0.79]	0.64 [0.53,0.75]	0.61 [0.48,0.73]	0.59 [0.47,0.71]	0.59 [0.46,0.72]	0.66 [0.54,0.76]	0.61 [0.48,0.73]	0.54 [0.41,0.66]	0.58 [0.43,0.72]
C2	0.67 [0.57,0.77]	0.62 [0.50,0.72]	0.69 [0.60,0.78]	0.73 [0.62,0.82]	0.77 [0.67,0.86]	0.67 [0.57,0.77]	0.77 [0.68,0.86]	0.74 [0.65,0.83]	0.64 [0.53,0.74]	0.66 [0.56,0.75]	0.68 [0.57,0.78]
C3	0.51 [0.39,0.64]	0.72 [0.54,0.89]	0.77 [0.61,0.89]	0.82 [0.61,0.95]	0.60 [0.29,0.88]	0.71 [0.54,0.84]	0.63 [0.46,0.79]	0.57 [0.35,0.73]	0.43 [0.11,0.78]	0.71 [0.46,0.88]	0.56 [0.35,0.77]
C4	0.65 [0.44,0.84]	0.35 [0.16,0.54]	0.45 [0.25,0.64]	0.54 [0.41,0.69]	0.61 [0.48,0.74]	0.44 [0.26,0.64]	0.64 [0.47,0.78]	0.47 [0.33,0.60]	0.58 [0.38,0.76]	0.47 [0.29,0.64]	0.48 [0.35,0.64]
C5	0.56 [0.43,0.68]	0.60 [0.48,0.72]	0.46 [0.34,0.58]	0.59 [0.48,0.69]	0.51 [0.39,0.64]	0.54 [0.40,0.67]	0.49 [0.35,0.63]	0.53 [0.43,0.64]	0.49 [0.38,0.61]	0.56 [0.42,0.68]	0.55 [0.42,0.67]
C6	0.55 [0.44,0.66]	0.61 [0.50,0.72]	0.67 [0.56,0.76]	0.58 [0.47,0.67]	0.67 [0.56,0.77]	0.68 [0.57,0.79]	0.61 [0.50,0.72]	0.63 [0.51,0.74]	0.61 [0.50,0.72]	0.70 [0.59,0.81]	0.58 [0.47,0.69]
C7	0.55 [0.44,0.65]	0.62 [0.52,0.71]	0.65 [0.56,0.74]	0.66 [0.55,0.75]	0.66 [0.56,0.76]	0.67 [0.57,0.77]	0.68 [0.58,0.78]	0.64 [0.54,0.75]	0.67 [0.58,0.76]	0.66 [0.56,0.75]	0.52 [0.50,0.54]
patient_overall	0.60 [0.52,0.68]	0.64 [0.56,0.72]	0.72 [0.65,0.78]	0.68 [0.60,0.75]	0.72 [0.64,0.79]	0.66 [0.58,0.73]	0.70 [0.62,0.77]	0.68 [0.62,0.76]	0.71 [0.63,0.78]	0.78 [0.72,0.85]	0.58 [0.50,0.66]
SPECTRE, joint											
C1	0.65 [0.52,0.77]	0.53 [0.41,0.65]	0.73 [0.61,0.85]	0.70 [0.58,0.81]	0.72 [0.58,0.83]	0.72 [0.59,0.83]	0.72 [0.59,0.83]	0.70 [0.59,0.79]	0.69 [0.56,0.79]	0.73 [0.62,0.83]	0.64 [0.50,0.76]
C2	0.67 [0.56,0.77]	0.67 [0.56,0.76]	0.77 [0.69,0.86]	0.75 [0.66,0.84]	0.78 [0.70,0.86]	0.75 [0.66,0.83]	0.76 [0.67,0.85]	0.77 [0.67,0.85]	0.74 [0.64,0.83]	0.75 [0.66,0.85]	0.68 [0.58,0.78]
C3	0.66 [0.46,0.86]	0.56 [0.30,0.81]	0.65 [0.41,0.89]	0.51 [0.38,0.65]	0.68 [0.44,0.91]	0.63 [0.42,0.82]	0.63 [0.35,0.91]	0.72 [0.54,0.87]	0.52 [0.40,0.65]	0.59 [0.31,0.86]	0.46 [0.18,0.75]
C4	0.63 [0.50,0.75]	0.61 [0.39,0.79]	0.75 [0.58,0.87]	0.68 [0.57,0.78]	0.81 [0.68,0.92]	0.74 [0.58,0.86]	0.70 [0.54,0.86]	0.63 [0.49,0.75]	0.58 [0.46,0.70]	0.75 [0.56,0.90]	0.61 [0.46,0.75]
C5	0.60 [0.48,0.72]	0.57 [0.47,0.67]	0.62 [0.51,0.71]	0.55 [0.45,0.65]	0.69 [0.59,0.77]	0.63 [0.54,0.72]	0.59 [0.47,0.71]	0.59 [0.48,0.69]	0.47 [0.38,0.56]	0.70 [0.58,0.83]	0.46 [0.33,0.62]
C6	0.64 [0.54,0.74]	0.59 [0.48,0.70]	0.68 [0.56,0.79]	0.63 [0.52,0.74]	0.68 [0.58,0.78]	0.67 [0.55,0.79]	0.69 [0.57,0.79]	0.63 [0.52,0.74]	0.61 [0.51,0.71]	0.74 [0.63,0.83]	0.48 [0.36,0.61]
C7	0.60 [0.51,0.70]	0.65 [0.56,0.74]	0.71 [0.63,0.80]	0.66 [0.55,0.75]	0.74 [0.64,0.82]	0.69 [0.58,0.79]	0.71 [0.61,0.80]	0.64 [0.54,0.74]	0.62 [0.53,0.72]	0.72 [0.64,0.81]	0.52 [0.42,0.62]
patient_overall	0.58 [0.50,0.66]	0.63 [0.56,0.70]	0.74 [0.67,0.80]	0.67 [0.60,0.74]	0.74 [0.68,0.81]	0.71 [0.64,0.78]	0.72 [0.65,0.79]	0.68 [0.61,0.75]	0.66 [0.59,0.73]	0.75 [0.68,0.82]	0.59 [0.51,0.66]

RSNA23 trauma

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
TAP-CT, per-task											
any_injury	0.75 [0.69,0.81]	0.78 [0.72,0.83]	0.77 [0.71,0.82]	0.76 [0.71,0.81]	0.78 [0.73,0.83]	0.77 [0.71,0.82]	0.71 [0.65,0.77]	0.76 [0.70,0.81]	0.75 [0.69,0.80]	0.88 [0.84,0.92]	0.87 [0.82,0.91]
kidney_injury	0.70 [0.59,0.81]	0.64 [0.52,0.75]	0.71 [0.59,0.81]	0.69 [0.55,0.82]	0.65 [0.52,0.78]	0.80 [0.66,0.90]	0.86 [0.76,0.95]	0.67 [0.55,0.78]	0.69 [0.57,0.80]	0.88 [0.76,0.96]	0.79 [0.69,0.89]
liver_injury	0.78 [0.69,0.85]	0.77 [0.70,0.83]	0.76 [0.67,0.83]	0.80 [0.73,0.86]	0.78 [0.71,0.84]	0.78 [0.72,0.85]	0.74 [0.66,0.81]	0.77 [0.69,0.85]	0.79 [0.72,0.85]	0.79 [0.72,0.85]	0.69 [0.60,0.76]
spleen_injury	0.72 [0.64,0.79]	0.69 [0.60,0.77]	0.71 [0.64,0.79]	0.77 [0.69,0.85]	0.60 [0.52,0.69]	0.66 [0.57,0.74]	0.59 [0.50,0.68]	0.75 [0.67,0.83]	0.64 [0.54,0.74]	0.89 [0.83,0.94]	0.75 [0.67,0.83]
TAP-CT, joint											
any_injury	0.75 [0.70,0.81]	0.76 [0.70,0.81]	0.77 [0.72,0.82]	0.75 [0.70,0.81]	0.76 [0.70,0.82]	0.79 [0.73,0.84]	0.69 [0.63,0.75]	0.76 [0.71,0.81]	0.77 [0.71,0.82]	0.84 [0.80,0.89]	0.87 [0.83,0.91]
kidney_injury	0.72 [0.59,0.82]	0.70 [0.57,0.81]	0.74 [0.61,0.85]	0.71 [0.58,0.84]	0.78 [0.67,0.86]	0.75 [0.64,0.85]	0.71 [0.61,0.80]	0.70 [0.59,0.80]	0.76 [0.63,0.86]	0.84 [0.76,0.92]	0.96 [0.93,0.99]
liver_injury	0.80 [0.73,0.86]	0.79 [0.71,0.86]	0.81 [0.75,0.87]	0.81 [0.74,0.86]	0.81 [0.74,0.88]	0.79 [0.72,0.86]	0.72 [0.65,0.79]	0.80 [0.74,0.87]	0.80 [0.73,0.86]	0.84 [0.79,0.90]	0.83 [0.76,0.89]
spleen_injury	0.68 [0.60,0.76]	0.71 [0.64,0.79]	0.71 [0.62,0.79]	0.71 [0.62,0.78]	0.68 [0.60,0.76]	0.74 [0.66,0.82]	0.62 [0.53,0.69]	0.72 [0.64,0.80]	0.71 [0.63,0.80]	0.81 [0.73,0.88]	0.85 [0.80,0.90]
SPECTRE, per-task											
any_injury	0.80 [0.75,0.86]	0.79 [0.74,0.84]	0.86 [0.81,0.90]	0.86 [0.82,0.90]	0.87 [0.84,0.91]	0.85 [0.80,0.89]	0.86 [0.81,0.90]	0.84 [0.79,0.89]	0.85 [0.80,0.89]	0.85 [0.81,0.90]	0.57 [0.51,0.64]
kidney_injury	0.85 [0.74,0.93]	0.83 [0.73,0.90]	0.85 [0.74,0.93]	0.90 [0.82,0.96]	0.91 [0.82,0.97]	0.90 [0.80,0.96]	0.86 [0.76,0.93]	0.87 [0.78,0.93]	0.79 [0.68,0.88]	0.74 [0.63,0.83]	0.63 [0.60,0.65]
liver_injury	0.73 [0.65,0.80]	0.73 [0.65,0.81]	0.77 [0.70,0.83]	0.75 [0.68,0.83]	0.77 [0.68,0.86]	0.77 [0.69,0.85]	0.78 [0.69,0.85]	0.77 [0.70,0.83]	0.77 [0.69,0.84]	0.70 [0.59,0.79]	0.74 [0.66,0.80]
spleen_injury	0.78 [0.70,0.85]	0.76 [0.67,0.84]	0.82 [0.75,0.89]	0.81 [0.73,0.89]	0.85 [0.77,0.91]	0.85 [0.79,0.91]	0.85 [0.79,0.91]	0.83 [0.75,0.89]	0.78 [0.69,0.86]	0.80 [0.73,0.87]	0.56 [0.47,0.63]
SPECTRE, joint											
any_injury	0.80 [0.74,0.85]	0.79 [0.74,0.85]	0.87 [0.82,0.91]	0.85 [0.81,0.90]	0.87 [0.83,0.91]	0.86 [0.81,0.90]	0.87 [0.83,0.91]	0.85 [0.80,0.89]	0.83 [0.77,0.87]	0.82 [0.76,0.87]	0.65 [0.58,0.71]
kidney_injury	0.84 [0.71,0.92]	0.83 [0.72,0.91]	0.88 [0.81,0.94]	0.88 [0.79,0.94]	0.91 [0.83,0.96]	0.86 [0.78,0.92]	0.86 [0.79,0.92]	0.87 [0.79,0.94]	0.85 [0.77,0.93]	0.87 [0.80,0.93]	0.64 [0.53,0.77]
liver_injury	0.73 [0.65,0.80]	0.74 [0.66,0.82]	0.80 [0.71,0.88]	0.79 [0.72,0.85]	0.81 [0.73,0.88]	0.80 [0.72,0.87]	0.82 [0.75,0.88]	0.78 [0.70,0.85]	0.78 [0.70,0.85]	0.74 [0.65,0.82]	0.74 [0.65,0.81]
spleen_injury	0.79 [0.72,0.87]	0.79 [0.71,0.86]	0.84 [0.77,0.90]	0.82 [0.74,0.88]	0.84 [0.78,0.90]	0.84 [0.77,0.90]	0.85 [0.79,0.91]	0.81 [0.73,0.88]	0.79 [0.69,0.86]	0.78 [0.69,0.86]	0.59 [0.50,0.67]

RSNA19 ICH

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
TAP-CT, per-task											
any	0.92 [0.91,0.93]	0.92 [0.91,0.93]	0.95 [0.94,0.95]	0.94 [0.93,0.95]	0.93 [0.92,0.94]	0.94 [0.93,0.95]	0.92 [0.91,0.93]	0.92 [0.91,0.93]	0.92 [0.91,0.93]	0.95 [0.94,0.95]	0.94 [0.93,0.95]
epidural	0.81 [0.76,0.86]	0.80 [0.75,0.85]	0.78 [0.73,0.83]	0.79 [0.74,0.83]	0.78 [0.73,0.82]	0.82 [0.77,0.86]	0.76 [0.71,0.82]	0.82 [0.77,0.87]	0.79 [0.74,0.84]	0.84 [0.79,0.88]	0.87 [0.83,0.90]
intraparenchymal	0.87 [0.86,0.89]	0.88 [0.87,0.89]	0.93 [0.92,0.94]	0.93 [0.91,0.94]	0.92 [0.90,0.93]	0.92 [0.91,0.93]	0.92 [0.91,0.93]	0.89 [0.88,0.90]	0.90 [0.89,0.91]	0.94 [0.93,0.95]	0.88 [0.87,0.90]
intraventricular	0.93 [0.92,0.94]	0.94 [0.93,0.95]	0.94 [0.94,0.95]	0.95 [0.94,0.96]	0.95 [0.94,0.96]	0.95 [0.94,0.96]	0.93 [0.92,0.94]	0.93 [0.92,0.94]	0.94 [0.93,0.95]	0.96 [0.96,0.97]	0.95 [0.94,0.96]
subarachnoid	0.85 [0.84,0.87]	0.86 [0.84,0.87]	0.88 [0.86,0.89]	0.88 [0.87,0.90]	0.88 [0.87,0.90]	0.88 [0.86,0.89]	0.86 [0.84,0.87]	0.86 [0.84,0.87]	0.86 [0.85,0.88]	0.87 [0.85,0.89]	0.88 [0.86,0.89]
subdural	0.84 [0.82,0.86]	0.84 [0.82,0.86]	0.89 [0.87,0.90]	0.87 [0.86,0.89]	0.88 [0.87,0.90]	0.87 [0.85,0.89]	0.83 [0.81,0.85]	0.84 [0.83,0.86]	0.85 [0.83,0.86]	0.91 [0.89,0.92]	0.81 [0.79,0.83]
TAP-CT, joint											
any	0.91 [0.90,0.92]	0.92 [0.91,0.93]	0.95 [0.94,0.95]	0.94 [0.93,0.95]	0.94 [0.93,0.95]	0.94 [0.93,0.95]	0.93 [0.92,0.94]	0.91 [0.90,0.92]	0.92 [0.91,0.93]	0.95 [0.95,0.96]	0.91 [0.90,0.92]
epidural	0.81 [0.76,0.86]	0.83 [0.78,0.87]	0.88 [0.84,0.91]	0.86 [0.82,0.90]	0.82 [0.78,0.87]	0.87 [0.83,0.91]	0.79 [0.75,0.83]	0.80 [0.75,0.85]	0.82 [0.77,0.86]	0.89 [0.85,0.92]	0.75 [0.70,0.79]
intraparenchymal	0.87 [0.86,0.89]	0.88 [0.87,0.89]	0.92 [0.91,0.93]	0.91 [0.90,0.92]	0.91 [0.90,0.92]	0.91 [0.90,0.92]	0.90 [0.89,0.91]	0.88 [0.86,0.89]	0.88 [0.87,0.90]	0.92 [0.91,0.93]	0.91 [0.89,0.92]
intraventricular	0.93 [0.91,0.94]	0.93 [0.92,0.94]	0.95 [0.94,0.96]	0.94 [0.94,0.95]	0.93 [0.92,0.94]	0.94 [0.93,0.95]	0.92 [0.91,0.93]	0.92 [0.91,0.93]	0.93 [0.92,0.94]	0.95 [0.94,0.96]	0.93 [0.92,0.94]
subarachnoid	0.85 [0.84,0.87]	0.86 [0.85,0.88]	0.88 [0.86,0.89]	0.87 [0.85,0.88]	0.86 [0.84,0.87]	0.87 [0.85,0.88]	0.87 [0.85,0.88]	0.85 [0.84,0.87]	0.86 [0.85,0.88]	0.88 [0.87,0.89]	0.86 [0.84,0.87]
subdural	0.84 [0.82,0.86]	0.84 [0.82,0.86]	0.87 [0.85,0.88]	0.85 [0.83,0.87]	0.85 [0.83,0.86]	0.85 [0.83,0.86]	0.81 [0.79,0.83]	0.84 [0.82,0.85]	0.83 [0.82,0.85]	0.89 [0.88,0.91]	0.78 [0.76,0.80]
Pillar-0, per-task											
any	0.91 [0.90,0.92]	0.93 [0.92,0.94]	0.96 [0.95,0.96]	0.94 [0.94,0.95]	0.95 [0.94,0.96]	0.95 [0.95,0.96]	0.96 [0.95,0.96]	0.94 [0.94,0.95]	0.94 [0.93,0.95]	0.96 [0.95,0.96]	0.96 [0.95,0.96]
epidural	0.82 [0.77,0.86]	0.77 [0.72,0.82]	0.88 [0.84,0.91]	0.86 [0.81,0.90]	0.81 [0.76,0.85]	0.89 [0.85,0.92]	0.82 [0.77,0.86]	0.86 [0.81,0.90]	0.83 [0.78,0.87]	0.89 [0.85,0.92]	0.82 [0.77,0.87]
intraparenchymal	0.91 [0.90,0.92]	0.92 [0.91,0.93]	0.95 [0.94,0.96]	0.93 [0.92,0.94]	0.93 [0.92,0.94]	0.95 [0.94,0.95]	0.95 [0.94,0.96]	0.93 [0.92,0.94]	0.92 [0.91,0.93]	0.95 [0.94,0.96]	0.94 [0.93,0.95]
intraventricular	0.95 [0.94,0.95]	0.94 [0.93,0.95]	0.96 [0.95,0.97]	0.95 [0.95,0.96]	0.96 [0.95,0.97]	0.96 [0.95,0.96]	0.96 [0.96,0.97]	0.95 [0.94,0.96]	0.95 [0.94,0.96]	0.96 [0.96,0.97]	0.96 [0.95,0.97]
subarachnoid	0.86 [0.85,0.88]	0.86 [0.84,0.88]	0.91 [0.90,0.92]	0.89 [0.87,0.90]	0.84 [0.83,0.86]	0.92 [0.91,0.93]	0.92 [0.90,0.93]	0.88 [0.87,0.90]	0.88 [0.86,0.89]	0.92 [0.90,0.93]	0.85 [0.84,0.87]
subdural	0.87 [0.86,0.89]	0.88 [0.87,0.90]	0.92 [0.90,0.93]	0.91 [0.90,0.92]	0.92 [0.90,0.93]	0.92 [0.91,0.93]	0.92 [0.90,0.93]	0.91 [0.89,0.92]	0.90 [0.89,0.91]	0.91 [0.90,0.93]	0.89 [0.87,0.90]
Pillar-0, joint											
any	0.91 [0.90,0.92]	0.92 [0.91,0.93]	0.95 [0.94,0.96]	0.94 [0.93,0.95]	0.95 [0.95,0.96]	0.95 [0.95,0.96]	0.95 [0.95,0.96]	0.94 [0.93,0.95]	0.93 [0.92,0.94]	0.96 [0.95,0.96]	0.94 [0.93,0.95]
epidural	0.83 [0.78,0.86]	0.82 [0.77,0.86]	0.86 [0.82,0.89]	0.82 [0.77,0.85]	0.88 [0.85,0.92]	0.86 [0.82,0.89]	0.87 [0.84,0.90]	0.85 [0.81,0.89]	0.84 [0.80,0.88]	0.85 [0.81,0.88]	0.78 [0.74,0.82]
intraparenchymal	0.91 [0.90,0.92]	0.91 [0.90,0.92]	0.93 [0.92,0.94]	0.92 [0.91,0.93]	0.94 [0.93,0.95]	0.93 [0.92,0.94]	0.93 [0.92,0.94]	0.92 [0.91,0.93]	0.92 [0.90,0.93]	0.94 [0.93,0.95]	0.92 [0.91,0.93]
intraventricular	0.95 [0.94,0.95]	0.94 [0.93,0.95]	0.94 [0.93,0.95]	0.94 [0.93,0.95]	0.95 [0.94,0.96]	0.94 [0.93,0.95]	0.95 [0.94,0.96]	0.95 [0.94,0.95]	0.94 [0.93,0.95]	0.94 [0.93,0.95]	0.93 [0.92,0.94]
subarachnoid	0.86 [0.85,0.88]	0.86 [0.85,0.88]	0.88 [0.87,0.90]	0.87 [0.85,0.88]	0.90 [0.89,0.92]	0.89 [0.88,0.90]	0.90 [0.88,0.91]	0.88 [0.86,0.89]	0.87 [0.85,0.88]	0.89 [0.87,0.90]	0.87 [0.86,0.89]
subdural	0.87 [0.86,0.89]	0.89 [0.87,0.90]	0.91 [0.89,0.92]	0.87 [0.85,0.88]	0.92 [0.90,0.93]	0.90 [0.89,0.92]	0.91 [0.89,0.92]	0.90 [0.88,0.91]	0.89 [0.87,0.90]	0.91 [0.89,0.92]	0.82 [0.80,0.84]

AMOS22

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→LoRA
TAP-CT, per-structure							
aorta	0.79 [0.77,0.81]	0.92 [0.91,0.92]	0.89 [0.88,0.90]	0.89 [0.88,0.90]	0.89 [0.88,0.90]	0.92 [0.91,0.93]	0.91 [0.91,0.92]
bladder	0.74 [0.69,0.78]	0.85 [0.82,0.88]	0.86 [0.83,0.88]	0.86 [0.82,0.88]	0.86 [0.83,0.88]	0.86 [0.83,0.89]	0.85 [0.81,0.88]
duodenum	0.53 [0.50,0.56]	0.68 [0.64,0.71]	0.68 [0.65,0.70]	0.67 [0.64,0.70]	0.65 [0.63,0.68]	0.70 [0.66,0.73]	0.67 [0.64,0.70]
esophagus	0.55 [0.51,0.58]	0.74 [0.71,0.76]	0.65 [0.62,0.67]	0.64 [0.62,0.67]	0.65 [0.62,0.68]	0.77 [0.75,0.79]	0.73 [0.70,0.76]
gallbladder	0.55 [0.48,0.61]	0.66 [0.60,0.73]	0.67 [0.60,0.74]	0.66 [0.58,0.73]	0.65 [0.58,0.72]	0.69 [0.61,0.76]	0.68 [0.61,0.73]
left_adrenal	0.38 [0.34,0.41]	0.58 [0.54,0.61]	0.42 [0.39,0.45]	0.41 [0.38,0.44]	0.41 [0.37,0.44]	0.59 [0.55,0.62]	0.57 [0.54,0.60]
left_kidney	0.72 [0.69,0.75]	0.88 [0.84,0.91]	0.91 [0.90,0.93]	0.91 [0.90,0.93]	0.91 [0.89,0.93]	0.92 [0.90,0.93]	0.89 [0.85,0.91]
liver	0.89 [0.87,0.91]	0.95 [0.94,0.96]	0.95 [0.94,0.96]	0.95 [0.94,0.96]	0.95 [0.94,0.96]	0.95 [0.94,0.96]	0.95 [0.93,0.96]
pancreas	0.68 [0.66,0.70]	0.78 [0.76,0.80]	0.78 [0.75,0.80]	0.78 [0.75,0.80]	0.77 [0.75,0.79]	0.78 [0.76,0.80]	0.78 [0.76,0.79]
postcava	0.70 [0.68,0.72]	0.82 [0.81,0.84]	0.77 [0.75,0.79]	0.77 [0.75,0.79]	0.77 [0.75,0.78]	0.84 [0.82,0.85]	0.82 [0.81,0.83]
prostate_uterus	0.73 [0.67,0.77]	0.80 [0.74,0.85]	0.78 [0.72,0.83]	0.76 [0.70,0.82]	0.77 [0.70,0.82]	0.78 [0.73,0.83]	0.79 [0.74,0.84]
right_adrenal	0.33 [0.29,0.37]	0.56 [0.52,0.60]	0.42 [0.39,0.44]	0.45 [0.42,0.47]	0.42 [0.39,0.44]	0.57 [0.54,0.61]	0.56 [0.52,0.60]
right_kidney	0.70 [0.67,0.74]	0.90 [0.87,0.92]	0.92 [0.90,0.94]	0.91 [0.88,0.92]	0.92 [0.91,0.93]	0.92 [0.91,0.93]	0.89 [0.87,0.92]
spleen	0.82 [0.77,0.85]	0.90 [0.87,0.93]	0.92 [0.90,0.94]	0.92 [0.88,0.94]	0.92 [0.90,0.94]	0.91 [0.88,0.93]	0.90 [0.87,0.92]
stomach	0.78 [0.74,0.81]	0.87 [0.84,0.89]	0.88 [0.86,0.90]	0.88 [0.86,0.90]	0.87 [0.85,0.90]	0.87 [0.84,0.90]	0.87 [0.84,0.89]
TAP-CT, joint							
aorta	0.78 [0.76,0.80]	0.90 [0.89,0.91]	0.91 [0.90,0.92]	0.90 [0.89,0.91]	0.91 [0.90,0.91]	0.92 [0.92,0.93]	0.90 [0.89,0.91]
bladder	0.73 [0.68,0.78]	0.81 [0.77,0.85]	0.68 [0.64,0.71]	0.76 [0.70,0.81]	0.75 [0.70,0.79]	0.86 [0.83,0.88]	0.78 [0.72,0.82]
duodenum	0.52 [0.48,0.55]	0.68 [0.65,0.71]	0.68 [0.65,0.71]	0.67 [0.64,0.69]	0.69 [0.65,0.72]	0.72 [0.68,0.75]	0.68 [0.66,0.71]
esophagus	0.52 [0.49,0.56]	0.71 [0.69,0.74]	0.73 [0.70,0.75]	0.66 [0.63,0.68]	0.71 [0.69,0.74]	0.00 [0.00,0.00]	0.00 [0.00,0.00]
gallbladder	0.50 [0.42,0.57]	0.67 [0.60,0.73]	0.65 [0.58,0.72]	0.66 [0.58,0.73]	0.69 [0.62,0.75]	0.70 [0.62,0.77]	0.65 [0.58,0.72]
left_adrenal	0.36 [0.32,0.40]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]
left_kidney	0.71 [0.67,0.74]	0.88 [0.84,0.91]	0.91 [0.89,0.93]	0.91 [0.90,0.92]	0.91 [0.89,0.93]	0.93 [0.91,0.94]	0.88 [0.84,0.91]
liver	0.89 [0.87,0.91]	0.94 [0.93,0.95]	0.93 [0.92,0.94]	0.95 [0.94,0.95]	0.94 [0.93,0.95]	0.95 [0.94,0.96]	0.94 [0.93,0.95]
pancreas	0.67 [0.64,0.69]	0.77 [0.75,0.79]	0.77 [0.75,0.79]	0.76 [0.74,0.77]	0.77 [0.75,0.79]	0.79 [0.76,0.81]	0.77 [0.75,0.78]
postcava	0.70 [0.68,0.71]	0.82 [0.81,0.83]	0.83 [0.81,0.84]	0.76 [0.74,0.77]	0.82 [0.81,0.84]	0.84 [0.83,0.86]	0.82 [0.80,0.83]
prostate_uterus	0.72 [0.67,0.76]	0.77 [0.70,0.81]	0.71 [0.65,0.76]	0.77 [0.73,0.81]	0.72 [0.66,0.77]	0.79 [0.73,0.84]	0.76 [0.71,0.81]
right_adrenal	0.34 [0.30,0.37]	0.00 [0.00,0.00]	0.57 [0.53,0.60]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]
right_kidney	0.68 [0.64,0.71]	0.89 [0.86,0.91]	0.92 [0.90,0.93]	0.91 [0.90,0.92]	0.91 [0.89,0.93]	0.93 [0.91,0.94]	0.89 [0.87,0.91]
spleen	0.81 [0.76,0.84]	0.91 [0.87,0.93]	0.92 [0.89,0.94]	0.92 [0.90,0.93]	0.92 [0.90,0.94]	0.93 [0.90,0.94]	0.90 [0.87,0.93]
stomach	0.78 [0.74,0.81]	0.87 [0.85,0.89]	0.87 [0.85,0.89]	0.85 [0.83,0.88]	0.87 [0.84,0.89]	0.88 [0.83,0.91]	0.87 [0.84,0.89]
SPECTRE, per-structure							
aorta	0.78 [0.77,0.80]	0.92 [0.91,0.92]	0.92 [0.91,0.92]	0.92 [0.91,0.92]	0.92 [0.91,0.92]	0.92 [0.91,0.93]	0.92 [0.91,0.92]
bladder	0.81 [0.78,0.84]	0.88 [0.86,0.90]	0.88 [0.86,0.91]	0.88 [0.85,0.90]	0.88 [0.86,0.91]	0.86 [0.84,0.89]	0.88 [0.86,0.90]
duodenum	0.55 [0.51,0.59]	0.71 [0.68,0.75]	0.74 [0.71,0.77]	0.73 [0.70,0.76]	0.75 [0.71,0.77]	0.72 [0.68,0.76]	0.73 [0.69,0.76]
esophagus	0.54 [0.51,0.57]	0.77 [0.75,0.79]	0.76 [0.75,0.78]	0.75 [0.73,0.77]	0.76 [0.74,0.78]	0.78 [0.76,0.80]	0.77 [0.75,0.79]
gallbladder	0.63 [0.56,0.69]	0.73 [0.66,0.79]	0.74 [0.66,0.81]	0.72 [0.65,0.79]	0.76 [0.69,0.82]	0.73 [0.65,0.79]	0.73 [0.67,0.79]
left_adrenal	0.37 [0.33,0.40]	0.57 [0.54,0.60]	0.53 [0.51,0.56]	0.53 [0.50,0.56]	0.54 [0.51,0.56]	0.52 [0.48,0.56]	0.58 [0.54,0.60]
left_kidney	0.76 [0.74,0.77]	0.87 [0.83,0.90]	0.92 [0.90,0.93]	0.91 [0.89,0.92]	0.91 [0.90,0.93]	0.78 [0.76,0.80]	0.90 [0.87,0.92]
liver	0.92 [0.91,0.93]	0.95 [0.94,0.96]	0.96 [0.95,0.96]	0.95 [0.94,0.96]	0.95 [0.94,0.96]	0.95 [0.94,0.96]	0.96 [0.95,0.96]
pancreas	0.70 [0.68,0.72]	0.79 [0.77,0.82]	0.81 [0.79,0.83]	0.80 [0.78,0.82]	0.81 [0.79,0.83]	0.80 [0.78,0.82]	0.81 [0.79,0.83]
postcava	0.65 [0.63,0.68]	0.82 [0.81,0.84]	0.84 [0.82,0.85]	0.83 [0.82,0.85]	0.84 [0.83,0.85]	0.84 [0.83,0.85]	0.83 [0.82,0.84]
prostate_uterus	0.74 [0.69,0.78]	0.80 [0.74,0.84]	0.80 [0.74,0.85]	0.78 [0.71,0.84]	0.81 [0.76,0.85]	0.78 [0.73,0.82]	0.80 [0.75,0.85]
right_adrenal	0.37 [0.33,0.40]	0.57 [0.52,0.61]	0.52 [0.50,0.54]	0.55 [0.51,0.58]	0.53 [0.50,0.56]	0.53 [0.49,0.57]	0.56 [0.52,0.60]
right_kidney	0.70 [0.69,0.72]	0.88 [0.86,0.90]	0.91 [0.89,0.93]	0.90 [0.88,0.92]	0.90 [0.88,0.92]	0.83 [0.80,0.86]	0.90 [0.88,0.92]
spleen	0.87 [0.85,0.88]	0.93 [0.91,0.94]	0.93 [0.92,0.94]	0.93 [0.92,0.95]	0.94 [0.92,0.95]	0.93 [0.91,0.94]	0.93 [0.92,0.94]
stomach	0.85 [0.82,0.87]	0.91 [0.89,0.92]	0.91 [0.89,0.92]	0.91 [0.89,0.92]	0.91 [0.89,0.92]	0.90 [0.87,0.92]	0.91 [0.88,0.92]
SPECTRE, joint							
aorta	0.77 [0.76,0.79]	0.89 [0.88,0.89]	0.90 [0.89,0.90]	0.89 [0.88,0.89]	0.90 [0.89,0.90]	0.91 [0.90,0.91]	0.89 [0.88,0.90]
bladder	0.80 [0.76,0.83]	0.83 [0.80,0.86]	0.86 [0.83,0.88]	0.83 [0.80,0.86]	0.85 [0.82,0.88]	0.84 [0.80,0.87]	0.84 [0.81,0.87]
duodenum	0.55 [0.52,0.59]	0.68 [0.64,0.71]	0.71 [0.69,0.74]	0.70 [0.67,0.72]	0.71 [0.68,0.74]	0.71 [0.67,0.74]	0.69 [0.65,0.72]
esophagus	0.53 [0.50,0.56]	0.00 [0.00,0.00]	0.70 [0.67,0.72]	0.52 [0.48,0.57]	0.70 [0.67,0.72]	0.00 [0.00,0.00]	0.00 [0.00,0.00]
gallbladder	0.65 [0.58,0.70]	0.70 [0.63,0.76]	0.74 [0.67,0.80]	0.72 [0.64,0.78]	0.55 [0.49,0.62]	0.75 [0.70,0.80]	0.70 [0.62,0.76]
left_adrenal	0.35 [0.31,0.39]	0.00 [0.00,0.00]	0.53 [0.49,0.55]	0.43 [0.39,0.47]	0.53 [0.49,0.55]	0.00 [0.00,0.00]	0.00 [0.00,0.00]
left_kidney	0.76 [0.72,0.79]	0.84 [0.80,0.88]	0.80 [0.76,0.83]	0.87 [0.83,0.90]	0.84 [0.83,0.86]	0.88 [0.85,0.90]	0.88 [0.86,0.89]
liver	0.92 [0.91,0.93]	0.95 [0.94,0.95]	0.95 [0.94,0.95]	0.94 [0.94,0.95]	0.95 [0.94,0.95]	0.95 [0.94,0.96]	0.95 [0.95,0.95]
pancreas	0.70 [0.68,0.72]	0.77 [0.75,0.79]	0.79 [0.77,0.81]	0.78 [0.75,0.79]	0.78 [0.76,0.80]	0.79 [0.77,0.81]	0.77 [0.75,0.79]
postcava	0.66 [0.64,0.68]	0.80 [0.79,0.81]	0.83 [0.81,0.84]	0.82 [0.81,0.83]	0.83 [0.81,0.84]	0.83 [0.81,0.84]	0.81 [0.79,0.82]
prostate_uterus	0.70 [0.64,0.76]	0.75 [0.70,0.79]	0.75 [0.69,0.79]	0.77 [0.72,0.82]	0.78 [0.74,0.83]	0.78 [0.72,0.82]	0.76 [0.72,0.80]
right_adrenal	0.37 [0.33,0.40]	0.00 [0.00,0.00]	0.52 [0.49,0.56]	0.50 [0.46,0.53]	0.53 [0.49,0.57]	0.00 [0.00,0.00]	0.00 [0.00,0.00]
right_kidney	0.77 [0.73,0.80]	0.87 [0.84,0.89]	0.89 [0.87,0.91]	0.88 [0.85,0.90]	0.86 [0.84,0.88]	0.84 [0.81,0.87]	0.88 [0.85,0.90]
spleen	0.86 [0.84,0.87]	0.91 [0.90,0.92]	0.90 [0.89,0.92]	0.92 [0.90,0.93]	0.92 [0.91,0.93]	0.90 [0.88,0.92]	0.91 [0.90,0.92]
stomach	0.84 [0.81,0.86]	0.89 [0.86,0.91]	0.89 [0.87,0.91]	0.89 [0.87,0.91]	0.90 [0.88,0.92]	0.87 [0.83,0.91]	0.89 [0.87,0.91]

KiTS23

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→LoRA
TAP-CT, per-structure							
cyst	0.17 [0.09,0.26]	0.19 [0.10,0.29]	0.18 [0.11,0.26]	0.17 [0.10,0.24]	0.17 [0.10,0.24]	0.20 [0.11,0.30]	0.19 [0.10,0.29]
kidney	0.89 [0.87,0.90]	0.94 [0.93,0.94]	0.94 [0.93,0.95]	0.94 [0.93,0.95]	0.94 [0.93,0.95]	0.94 [0.92,0.95]	0.94 [0.93,0.94]
tumor	0.54 [0.48,0.59]	0.62 [0.56,0.68]	0.57 [0.50,0.64]	0.59 [0.53,0.66]	0.58 [0.52,0.65]	0.63 [0.57,0.69]	0.63 [0.58,0.69]
TAP-CT, joint							
cyst	0.04 [0.01,0.08]	0.09 [0.05,0.14]	0.18 [0.11,0.27]	0.14 [0.08,0.21]	0.15 [0.08,0.21]	0.08 [0.04,0.13]	0.00 [0.00,0.00]
kidney	0.89 [0.88,0.90]	0.91 [0.90,0.92]	0.94 [0.93,0.94]	0.94 [0.93,0.95]	0.93 [0.92,0.94]	0.91 [0.90,0.92]	0.93 [0.91,0.94]
tumor	0.36 [0.28,0.44]	0.54 [0.47,0.61]	0.62 [0.56,0.68]	0.59 [0.53,0.66]	0.60 [0.53,0.66]	0.56 [0.49,0.62]	0.57 [0.50,0.65]
SPECTRE, per-structure							
cyst	0.22 [0.14,0.31]	0.22 [0.14,0.31]	0.19 [0.12,0.27]	0.21 [0.13,0.29]	0.20 [0.13,0.27]	0.05 [0.02,0.08]	0.21 [0.13,0.29]
kidney	0.90 [0.89,0.91]	0.94 [0.93,0.95]	0.95 [0.94,0.95]	0.94 [0.93,0.95]	0.95 [0.94,0.95]	0.93 [0.91,0.95]	0.94 [0.94,0.95]
tumor	0.62 [0.56,0.68]	0.63 [0.56,0.70]	0.70 [0.64,0.75]	0.70 [0.64,0.75]	0.72 [0.66,0.77]	0.60 [0.52,0.67]	0.66 [0.59,0.72]
SPECTRE, joint							
cyst	0.16 [0.07,0.27]	0.15 [0.07,0.23]	0.09 [0.04,0.15]	0.10 [0.06,0.16]	0.17 [0.09,0.24]	0.11 [0.07,0.16]	0.13 [0.06,0.21]
kidney	0.90 [0.89,0.91]	0.93 [0.92,0.94]	0.94 [0.94,0.95]	0.94 [0.93,0.95]	0.92 [0.91,0.93]	0.89 [0.87,0.91]	0.94 [0.93,0.94]
tumor	0.53 [0.45,0.61]	0.63 [0.56,0.70]	0.73 [0.68,0.78]	0.72 [0.66,0.77]	0.69 [0.63,0.75]	0.56 [0.49,0.62]	0.64 [0.57,0.71]

LiTS

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→LoRA
TAP-CT, per-structure							
liver	0.92 [0.90,0.93]	0.95 [0.93,0.97]	0.96 [0.95,0.97]	0.96 [0.95,0.97]	0.96 [0.95,0.97]	0.96 [0.96,0.97]	0.96 [0.95,0.97]
tumor	0.40 [0.26,0.53]	0.47 [0.33,0.59]	0.47 [0.32,0.62]	0.46 [0.32,0.59]	0.47 [0.31,0.61]	0.46 [0.31,0.61]	0.47 [0.34,0.60]
TAP-CT, joint							
liver	0.91 [0.89,0.93]	0.95 [0.94,0.96]	0.96 [0.94,0.97]	0.96 [0.94,0.97]	0.96 [0.94,0.97]	0.96 [0.95,0.97]	0.95 [0.94,0.96]
tumor	0.40 [0.25,0.53]	0.45 [0.30,0.60]	0.47 [0.33,0.62]	0.48 [0.35,0.63]	0.43 [0.28,0.59]	0.47 [0.33,0.60]	0.45 [0.30,0.60]
SPECTRE, per-structure							
liver	0.94 [0.94,0.95]	0.97 [0.96,0.97]	0.97 [0.96,0.97]	0.97 [0.96,0.97]	0.97 [0.96,0.97]	0.97 [0.96,0.97]	0.97 [0.96,0.97]
tumor	0.43 [0.28,0.57]	0.52 [0.36,0.66]	0.55 [0.41,0.67]	0.48 [0.33,0.62]	0.55 [0.40,0.67]	0.51 [0.36,0.66]	0.49 [0.33,0.63]
SPECTRE, joint							
liver	0.94 [0.93,0.94]	0.96 [0.96,0.96]	0.96 [0.95,0.97]	0.96 [0.96,0.97]	0.96 [0.96,0.97]	0.96 [0.94,0.97]	0.96 [0.96,0.96]
tumor	0.46 [0.31,0.61]	0.52 [0.35,0.66]	0.55 [0.40,0.67]	0.51 [0.37,0.65]	0.52 [0.37,0.65]	0.47 [0.31,0.63]	0.53 [0.37,0.67]

MSD Pancreas

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→LoRA
TAP-CT, per-structure							
pancreas	0.66 [0.62,0.70]	0.75 [0.71,0.78]	0.77 [0.73,0.79]	0.76 [0.73,0.79]	0.76 [0.72,0.79]	0.76 [0.72,0.79]	0.75 [0.71,0.78]
tumor	0.19 [0.14,0.25]	0.23 [0.16,0.31]	0.24 [0.18,0.31]	0.24 [0.18,0.30]	0.25 [0.18,0.31]	0.26 [0.19,0.34]	0.24 [0.18,0.31]
TAP-CT, joint							
pancreas	0.66 [0.62,0.69]	0.72 [0.69,0.75]	0.75 [0.72,0.78]	0.75 [0.71,0.78]	0.75 [0.72,0.78]	0.75 [0.72,0.78]	0.73 [0.69,0.77]
tumor	0.01 [0.00,0.02]	0.24 [0.17,0.31]	0.26 [0.19,0.32]	0.23 [0.17,0.30]	0.23 [0.18,0.29]	0.22 [0.15,0.30]	0.20 [0.12,0.29]
SPECTRE, per-structure							
pancreas	0.71 [0.68,0.74]	0.78 [0.74,0.80]	0.79 [0.76,0.82]	0.78 [0.74,0.81]	0.79 [0.76,0.82]	0.79 [0.75,0.81]	0.78 [0.75,0.81]
tumor	0.24 [0.18,0.31]	0.26 [0.18,0.32]	0.34 [0.26,0.41]	0.30 [0.22,0.38]	0.32 [0.24,0.40]	0.25 [0.17,0.33]	0.26 [0.18,0.33]
SPECTRE, joint							
pancreas	0.70 [0.67,0.73]	0.76 [0.72,0.78]	0.79 [0.75,0.81]	0.77 [0.73,0.79]	0.79 [0.76,0.81]	0.77 [0.74,0.80]	0.76 [0.73,0.78]
tumor	0.22 [0.15,0.29]	0.27 [0.19,0.35]	0.32 [0.24,0.40]	0.29 [0.21,0.38]	0.31 [0.24,0.40]	0.29 [0.21,0.38]	0.28 [0.20,0.36]

TotalSegmentator

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→LoRA
TAP-CT, per-structure							
arteries	0.66 [0.63,0.70]	0.78 [0.74,0.81]	0.73 [0.68,0.77]	–	–	0.78 [0.74,0.81]	–
bladder_prostate	0.83 [0.79,0.86]	0.84 [0.81,0.88]	0.83 [0.78,0.87]	–	–	0.85 [0.81,0.88]	–
brain_spinal_cord	0.78 [0.76,0.81]	0.91 [0.90,0.92]	0.84 [0.82,0.86]	–	–	0.91 [0.91,0.92]	–
gi_tract	0.76 [0.73,0.79]	0.86 [0.84,0.88]	0.84 [0.81,0.87]	–	–	0.86 [0.84,0.88]	–
glands	0.47 [0.44,0.51]	0.63 [0.59,0.66]	0.52 [0.48,0.56]	–	–	0.41 [0.38,0.44]	–
heart	0.85 [0.80,0.88]	0.89 [0.85,0.93]	0.88 [0.83,0.93]	–	–	0.90 [0.86,0.93]	–
kidneys	0.79 [0.74,0.83]	0.85 [0.80,0.89]	0.85 [0.80,0.89]	–	–	0.84 [0.80,0.89]	–
liver	0.87 [0.83,0.90]	0.92 [0.89,0.95]	0.93 [0.90,0.96]	–	–	0.93 [0.90,0.96]	–
lungs_airway	0.90 [0.88,0.92]	0.95 [0.94,0.96]	0.94 [0.91,0.96]	–	–	0.95 [0.92,0.97]	–
muscles	0.80 [0.78,0.81]	0.90 [0.89,0.91]	0.90 [0.87,0.91]	–	–	0.91 [0.90,0.92]	–
other_bones	0.73 [0.72,0.75]	0.88 [0.87,0.89]	0.88 [0.87,0.89]	–	–	0.89 [0.88,0.90]	–
pancreas	0.64 [0.60,0.68]	0.72 [0.68,0.76]	0.70 [0.64,0.75]	–	–	0.73 [0.68,0.77]	–
spine	0.73 [0.71,0.75]	0.87 [0.86,0.88]	0.87 [0.85,0.88]	–	–	0.88 [0.86,0.89]	–
spleen	0.79 [0.76,0.82]	0.88 [0.85,0.91]	0.89 [0.86,0.92]	–	–	0.91 [0.89,0.93]	–
veins	0.59 [0.56,0.61]	0.74 [0.71,0.76]	0.70 [0.67,0.72]	–	–	0.72 [0.70,0.75]	–
TAP-CT, joint							
arteries	0.67 [0.63,0.70]	0.79 [0.75,0.82]	0.80 [0.76,0.83]	0.77 [0.74,0.80]	0.80 [0.77,0.83]	0.81 [0.77,0.85]	0.79 [0.75,0.82]
bladder_prostate	0.81 [0.76,0.85]	0.85 [0.82,0.88]	0.84 [0.80,0.88]	0.82 [0.78,0.85]	0.83 [0.79,0.86]	0.87 [0.84,0.90]	0.84 [0.79,0.87]
brain_spinal_cord	0.79 [0.76,0.81]	0.91 [0.90,0.92]	0.89 [0.87,0.91]	0.89 [0.87,0.91]	0.89 [0.87,0.90]	0.92 [0.91,0.92]	0.91 [0.90,0.92]
gi_tract	0.76 [0.73,0.78]	0.86 [0.84,0.87]	0.84 [0.81,0.86]	0.85 [0.84,0.87]	0.85 [0.83,0.87]	0.88 [0.86,0.89]	0.85 [0.83,0.87]
glands	0.47 [0.43,0.50]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]
heart	0.84 [0.81,0.88]	0.89 [0.85,0.92]	0.89 [0.85,0.92]	0.88 [0.83,0.92]	0.89 [0.85,0.92]	0.90 [0.85,0.93]	0.88 [0.84,0.92]
kidneys	0.77 [0.72,0.82]	0.83 [0.78,0.88]	0.84 [0.79,0.88]	0.84 [0.80,0.88]	0.83 [0.79,0.87]	0.86 [0.82,0.90]	0.83 [0.78,0.87]
liver	0.86 [0.83,0.89]	0.92 [0.89,0.95]	0.80 [0.73,0.87]	0.81 [0.75,0.87]	0.80 [0.73,0.87]	0.94 [0.90,0.96]	0.92 [0.88,0.95]
lungs_airway	0.90 [0.87,0.92]	0.95 [0.93,0.96]	0.82 [0.76,0.89]	0.83 [0.77,0.89]	0.82 [0.76,0.88]	0.95 [0.94,0.96]	0.94 [0.93,0.95]
muscles	0.80 [0.79,0.82]	0.89 [0.88,0.90]	0.90 [0.89,0.91]	0.88 [0.85,0.90]	0.89 [0.88,0.90]	0.91 [0.90,0.92]	0.89 [0.88,0.90]
other_bones	0.73 [0.71,0.75]	0.86 [0.85,0.87]	0.86 [0.85,0.87]	0.86 [0.84,0.87]	0.86 [0.85,0.87]	0.88 [0.86,0.89]	0.85 [0.84,0.86]
pancreas	0.62 [0.57,0.66]	0.71 [0.67,0.75]	0.68 [0.63,0.73]	0.71 [0.67,0.75]	0.69 [0.64,0.74]	0.74 [0.68,0.78]	0.68 [0.63,0.72]
spine	0.73 [0.71,0.75]	0.85 [0.84,0.87]	0.84 [0.82,0.85]	0.85 [0.83,0.86]	0.84 [0.82,0.85]	0.86 [0.85,0.88]	0.84 [0.83,0.86]
spleen	0.77 [0.74,0.81]	0.89 [0.86,0.91]	0.89 [0.87,0.91]	0.91 [0.88,0.93]	0.89 [0.87,0.92]	0.93 [0.90,0.94]	0.87 [0.85,0.90]
veins	0.58 [0.56,0.61]	0.75 [0.73,0.78]	0.75 [0.72,0.77]	0.73 [0.70,0.75]	0.74 [0.72,0.76]	0.79 [0.76,0.80]	0.74 [0.71,0.76]
SPECTRE, per-structure							
arteries	0.63 [0.59,0.67]	0.79 [0.75,0.82]	0.77 [0.73,0.81]	–	–	0.80 [0.77,0.83]	–
bladder_prostate	0.81 [0.77,0.85]	0.86 [0.83,0.90]	0.87 [0.82,0.91]	–	–	0.87 [0.84,0.90]	–
brain_spinal_cord	0.74 [0.72,0.76]	0.91 [0.91,0.92]	0.90 [0.88,0.92]	–	–	0.91 [0.89,0.92]	–
gi_tract	0.75 [0.71,0.78]	0.85 [0.83,0.88]	0.87 [0.84,0.89]	–	–	0.87 [0.84,0.89]	–
glands	0.41 [0.37,0.44]	0.63 [0.59,0.66]	0.61 [0.57,0.64]	–	–	0.57 [0.53,0.60]	–
heart	0.86 [0.83,0.89]	0.91 [0.87,0.94]	0.91 [0.88,0.94]	–	–	0.91 [0.87,0.94]	–
kidneys	0.79 [0.74,0.83]	0.85 [0.81,0.89]	0.86 [0.80,0.90]	–	–	0.80 [0.75,0.85]	–
liver	0.91 [0.87,0.93]	0.94 [0.91,0.96]	0.93 [0.89,0.96]	–	–	0.92 [0.89,0.95]	–
lungs_airway	0.90 [0.88,0.92]	0.95 [0.94,0.96]	0.95 [0.94,0.96]	–	–	0.95 [0.94,0.96]	–
muscles	0.75 [0.72,0.77]	0.90 [0.88,0.91]	0.91 [0.90,0.92]	–	–	0.91 [0.90,0.92]	–
other_bones	0.68 [0.66,0.70]	0.87 [0.85,0.88]	0.87 [0.86,0.88]	–	–	0.88 [0.87,0.89]	–
pancreas	0.66 [0.62,0.69]	0.75 [0.71,0.79]	0.73 [0.68,0.78]	–	–	0.74 [0.69,0.78]	–
spine	0.67 [0.65,0.69]	0.86 [0.84,0.87]	0.87 [0.85,0.88]	–	–	0.87 [0.86,0.88]	–
spleen	0.87 [0.84,0.89]	0.91 [0.88,0.93]	0.91 [0.87,0.93]	–	–	0.80 [0.77,0.84]	–
veins	0.55 [0.52,0.57]	0.74 [0.71,0.76]	0.77 [0.74,0.79]	–	–	0.77 [0.75,0.79]	–
SPECTRE, joint							
arteries	0.63 [0.60,0.66]	0.74 [0.70,0.77]	0.77 [0.73,0.80]	0.75 [0.71,0.79]	0.76 [0.72,0.80]	0.78 [0.75,0.82]	0.74 [0.70,0.77]
bladder_prostate	0.80 [0.75,0.84]	0.80 [0.76,0.83]	0.69 [0.63,0.74]	0.74 [0.70,0.78]	0.60 [0.52,0.67]	0.85 [0.82,0.88]	0.81 [0.79,0.84]
brain_spinal_cord	0.74 [0.71,0.76]	0.90 [0.89,0.91]	0.90 [0.88,0.92]	0.90 [0.88,0.92]	0.90 [0.88,0.91]	0.91 [0.91,0.92]	0.90 [0.89,0.91]
gi_tract	0.74 [0.71,0.78]	0.83 [0.81,0.86]	0.85 [0.83,0.87]	0.85 [0.83,0.87]	0.85 [0.83,0.87]	0.86 [0.84,0.88]	0.83 [0.81,0.86]
glands	0.38 [0.34,0.42]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]
heart	0.86 [0.82,0.89]	0.89 [0.86,0.92]	0.88 [0.83,0.92]	0.88 [0.84,0.92]	0.88 [0.83,0.92]	0.91 [0.87,0.93]	0.89 [0.86,0.92]
kidneys	0.77 [0.72,0.81]	0.81 [0.77,0.85]	0.79 [0.73,0.84]	0.81 [0.77,0.85]	0.79 [0.74,0.84]	0.81 [0.76,0.86]	0.80 [0.76,0.84]
liver	0.90 [0.87,0.93]	0.92 [0.89,0.95]	0.75 [0.67,0.82]	0.83 [0.76,0.89]	0.75 [0.68,0.82]	0.93 [0.89,0.95]	0.92 [0.89,0.95]
lungs_airway	0.90 [0.87,0.92]	0.93 [0.90,0.95]	0.83 [0.77,0.88]	0.88 [0.83,0.92]	0.85 [0.80,0.90]	0.94 [0.93,0.95]	0.93 [0.90,0.94]
muscles	0.74 [0.71,0.76]	0.87 [0.86,0.88]	0.88 [0.86,0.90]	0.89 [0.88,0.90]	0.89 [0.86,0.90]	0.90 [0.89,0.91]	0.88 [0.86,0.89]
other_bones	0.67 [0.65,0.69]	0.82 [0.81,0.84]	0.85 [0.83,0.86]	0.84 [0.83,0.85]	0.85 [0.84,0.86]	0.85 [0.84,0.86]	0.82 [0.81,0.84]
pancreas	0.64 [0.60,0.68]	0.67 [0.63,0.72]	0.63 [0.59,0.67]	0.64 [0.59,0.68]	0.66 [0.61,0.70]	0.70 [0.66,0.74]	0.69 [0.64,0.73]
spine	0.66 [0.64,0.68]	0.81 [0.79,0.82]	0.82 [0.81,0.84]	0.82 [0.81,0.84]	0.83 [0.81,0.84]	0.83 [0.81,0.84]	0.81 [0.79,0.82]
spleen	0.86 [0.84,0.88]	0.86 [0.84,0.89]	0.65 [0.58,0.71]	0.72 [0.66,0.77]	0.71 [0.64,0.76]	0.88 [0.85,0.91]	0.87 [0.84,0.89]
veins	0.54 [0.52,0.56]	0.69 [0.66,0.71]	0.72 [0.70,0.74]	0.72 [0.70,0.73]	0.73 [0.71,0.75]	0.75 [0.73,0.77]	0.69 [0.67,0.71]

Appendix F. Cohorts and examples

Table 23: Cohorts and datasets. Labels are binary labels (classification) or structures (segmentation), and counts are cases in the training, selection and held-out splits. In the split rule, "official val, halves" denotes the official validation cases sorted by identifier and alternately assigned to the selection and held-out splits, and "seeded" one fixed seeded division of the released labeled cases (80/10/10 of the released training data for RSNA22 C-spine, RSNA23 trauma and FDG-PET/CT, 70/15/15 for KiTS23 and MSD Pancreas, and 91/20/20 volumes for LiTS). TotalSegmentator holds out its official test split and takes an equally large selection split from the official training split. FDG-PET/CT cases are studies (1014 studies of 900 patients, with no patient spanning splits), and the RSNA19 ICH training cases are a stratified subset of the released training patients. Backbones: T = TAP-CT, S = SPECTRE, P = Pillar-0.

Cohort / dataset	Labels	Split rule	Backbones	Train	Sel.	Held-out
Classification						
CVPR26: AMOS	15 disease labels	official val, halves	T S	1107[†]	178	178
CVPR26: COVID-CTset	COVID-19	official val, halves	T S	308	32	32
CVPR26: LUNA25	nodule malignancy	official val, halves	T S	1000	100	100
FDG-PET/CT	4 diagnosis groups	seeded, patient level	T S	776	116	122
RSNA22 C-spine	patient, C1–C7 (8)	seeded, patient level	T S	1615	202	202
RSNA23 trauma	any, kidney, liver, spleen (4)	seeded, one series per patient	T S	2517	315	315
RSNA19 ICH	any + 5 subtypes (6)	stratified, patient level	T P	5000	3269	3264
Segmentation						
AMOS22	15 organs	official val, halves	T S	200	50	50
KiTS23	kidney, tumor, cyst	seeded	T S	343	73	73
LiTS	liver, tumor	seeded	T S	91	20	20
MSD Pancreas	pancreas, tumor	seeded	T S	197	42	42
TotalSegmentator	15 merged structures	official test held out	T S	1721	109	109

[†] distinct AMOS cases over the 15 per-label files; each label is known for a subset of cases.

Table 24: Selection and held-out cases and positives per classification label, identical in every run. FDG-PET/CT (held-out) and RSNA22 C-spine (selection) each list one case fewer than Tab. 23, whose image was missing from our copy of the data. Except for AMOS, whose release gives no patient identifiers, no patient appears in two splits: FDG-PET/CT and RSNA19 ICH are split by patient, COVID-CTset, LUNA25 and RSNA23 trauma contain one scan per patient, and RSNA22 C-spine one study per patient (Lin et al., 2023).

	Selection		Held-out	
Label	Cases	Positives	Cases	Positives
CVPR26				
adrenal hyperplasia	37	21	33	14
ascites	36	17	32	17
atherosclerosis	35	17	33	17
cholecystitis	60	27	42	24
colorectal cancer	26	14	26	12
fatty liver	42	23	36	16
gallstone	55	26	61	32
hydronephrosis	32	12	32	20
kidney stone	79	37	79	42
liver calcifications	36	19	36	17
liver cyst	126	65	118	57
liver lesion	72	36	62	31
lymphadenopathy	34	17	38	19
renal cyst	131	66	135	67
splenomegaly	27	15	43	20
COVID-19 (COVID-CTset)	32	16	32	16
nodule malignancy (LUNA25)	100	51	100	49
FDG-PET/CT				
lung cancer	116	17	121	16
lymphoma	116	15	121	15
melanoma	116	21	121	23
negative	116	63	121	67
RSNA22 C-spine				
C1	201	15	202	20
C2	201	27	202	30
C3	201	9	202	5
C4	201	10	202	10
C5	201	19	202	19
C6	201	30	202	24
C7	201	40	202	40
any fracture (patient)	201	95	202	96
RSNA23 trauma				
any injury	315	91	315	91
kidney injury	315	25	315	19
liver injury	315	38	315	34
spleen injury	315	31	315	39
RSNA19 ICH				
any	3269	1331	3264	1326
epidural	3269	34	3264	73
intraparenchymal	3269	798	3264	814
intraventricular	3269	553	3264	516
subarachnoid	3269	602	3264	568
subdural	3269	583	3264	586

Table 25: The 117 released TotalSegmentator structures merged into 15 within anatomical systems. Where two source masks overlap, the voxel is assigned to the smaller structure.

Structure	Source structures
lungs and airway	five lung lobes, trachea
heart	heart, left atrial appendage
arteries	aorta, brachiocephalic trunk, left and right subclavian, common carotid and iliac arteries
veins	superior and inferior vena cava, left and right brachiocephalic and iliac veins, portal and splenic vein, pulmonary vein
liver	liver, gallbladder
spleen	spleen
pancreas	pancreas
kidneys	left and right kidney, left and right kidney cysts
GI tract	esophagus, stomach, duodenum, small bowel, colon
bladder and prostate	urinary bladder, prostate
glands	thyroid, left and right adrenal gland
brain and spinal cord	brain, spinal cord
spine	25 vertebrae (C1–L6), sacrum
other bones	24 ribs, sternum, costal cartilages, clavicles, scapulae, humeri, hip bones, femurs, skull
muscles	gluteus maximus, medius and minimus, iliopsoas, autochthon (left and right)

Table 26: Evaluation data that appear in a backbone's pretraining data, as listed in the backbones' papers (images only; no labels of our tasks were used in pretraining). TAP-CT was pretrained on in-house oncological CT (Veenboer et al., 2026) and Pillar-0 on CT and breast MRI of one academic center (Agrawal et al., 2025), neither listing a public dataset. SPECTRE lists, among others, AMOS, NLST, AbdomenAtlas 1.0 Mini (5195 scans) and AbdomenCT-1K (1062 scans) (Claessens et al., 2026), and removed only AMOS from its own evaluation. The last two aggregate public datasets: AbdomenCT-1K contains the 201 LiTS, 300 KiTS19 and 420 MSD Pancreas scans (Ma et al., 2022), and the 5195 scans of AbdomenAtlas include 131 LiTS, 300 KiTS, 945 MSD CT, 200 AMOS22 and 2200 FLARE'23 scans but no TotalSegmentator scans (Qu et al., 2023).

Evaluation data	TAP-CT	SPECTRE	Pillar-0
AMOS (CVPR26) and AMOS22	no	yes	no
LUNA25	no	yes[a]	no
LiTS, MSD Pancreas	no	yes[b]	no
KiTS23	no	partly[c]	no
TotalSegmentator, FDG-PET/CT	no	not confirmed[d]	no
COVID-CTset, RSNA19 ICH, RSNA22 C-spine, RSNA23 trauma	no	not listed	no

[a] through NLST, which is in the pretraining data; the public LUNA25 training and development scans are NLST scans (Peeters et al., 2025). [b] every case of our splits, through AbdomenCT-1K and AbdomenAtlas. [c] the 300 KiTS19 cases (45 of our 73 held-out cases); the 189 cases added for KiTS23 are not included. [d] neither is a source of AbdomenCT-1K or AbdomenAtlas, but AbdomenAtlas includes 2200 FLARE'23 scans whose sources we could not access, so we could not confirm that they are absent.

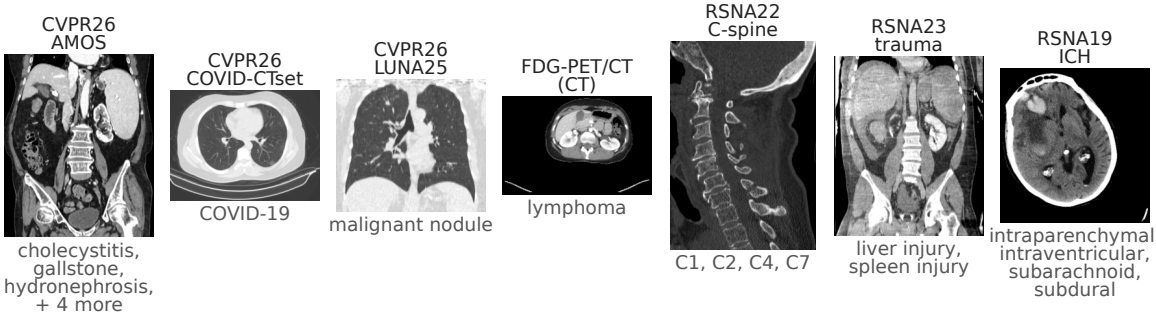


Figure 17: One held-out case per classification source, namely the case with the most positive labels, which are listed below it. The slices show anatomy, not necessarily the findings. Views are fixed per source and pass through the body's center of mass: coronal for AMOS, LUNA25 and RSNA23 trauma, sagittal for RSNA22 C-spine, and axial for COVID-CTset (thick slices), FDG-PET/CT (whole body) and RSNA19 ICH. Windows are soft tissue, lung (COVID-CTset, LUNA25), bone (RSNA22 C-spine) and brain (RSNA19 ICH).

Appendix G. Success and failure cases

The cases in Fig. 19 (classification) and Figs. 20, 21, 22, 23 and 24 (segmentation) were chosen by a fixed rule from the held-out per-case results. For segmentation, each held-out case was scored by its Dice averaged over structures and over the joint models of Lin, Dec, LoRA and Full FT on TAP-CT and SPECTRE, where the AMOS22 and TotalSegmentator models are those trained for 24 000 steps. Among the cases scored by all eight models, we show the case at the 90th percentile of this score as the success case and the lowest-scoring case as the failure case. For classification, we used, per source, the label with the most held-out positives, excluding the negative group of FDG-PET/CT. Each held-out positive was scored by the share of held-out negatives it outranks, averaged over the per-task models of Lin, LoRA, Attn-only FT and Full FT on every backbone evaluated on that cohort. The success case is the positive at the 90th percentile of this score, and the weakest positive is the one with the lowest score. For LiTS, Fig. 23 instead shows one representative case with a reference tumor, chosen by the rule of Fig. 18.

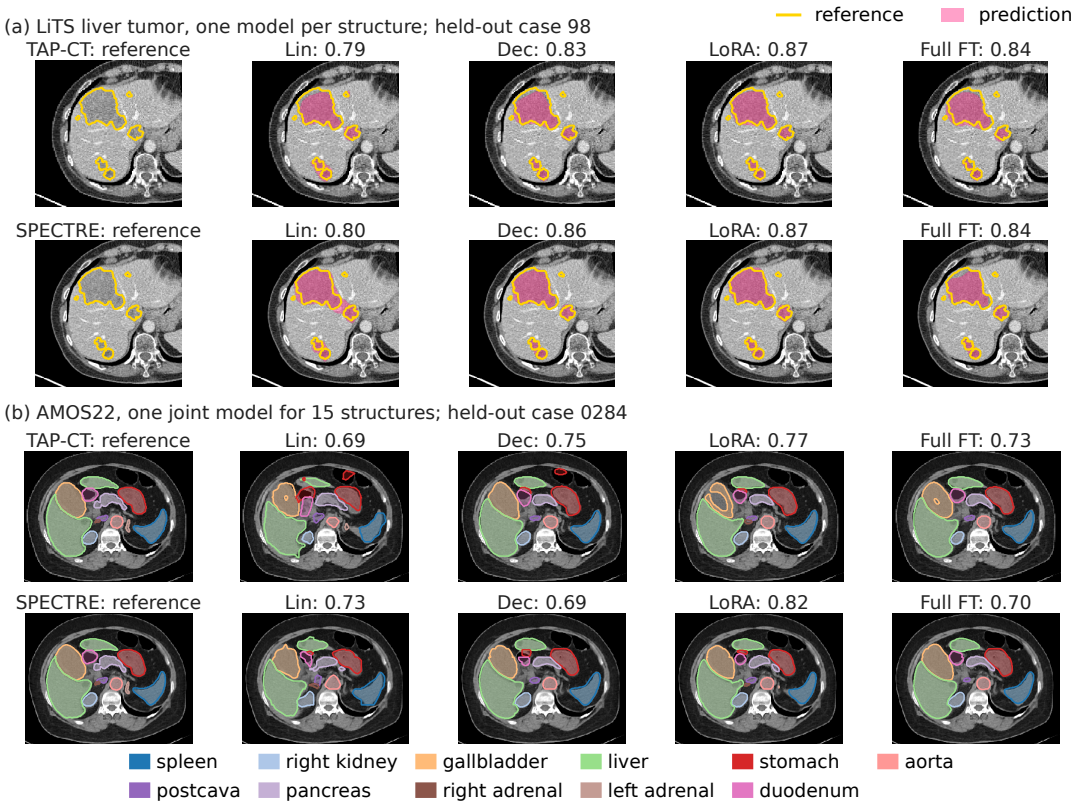


Figure 18: Segmentation examples on held-out cases, each shown on the backbone's own grid at the soft-tissue window, with the case's Dice in the panel titles (mean over structures in (b)). (a) A LiTS liver tumor segmented with one model per structure, where the reference is outlined in yellow and the prediction is shown in pink. (b) An AMOS22 case segmented with one joint model for all 15 structures, where colors denote structures. The decoder improves on the linear probe in every panel except SPECTRE in (b), where the frozen decoder underperforms on AMOS22 (Section 4.2); LoRA has the highest Dice in all four, and full fine-tuning does not improve on LoRA. The LiTS case has a large tumor and a higher tumor Dice than the dataset mean. The cases were chosen by a fixed rule among held-out cases with a valid Dice in every run on TAP-CT and SPECTRE: a case had to agree with each backbone's mean held-out Dice on the best and the worst method (a) or on the full ranking (b), and among such cases we show the one whose pairwise Dice differences are closest to the mean differences. 3 of 18 cases qualify for (a) and 23 of 50 for (b).

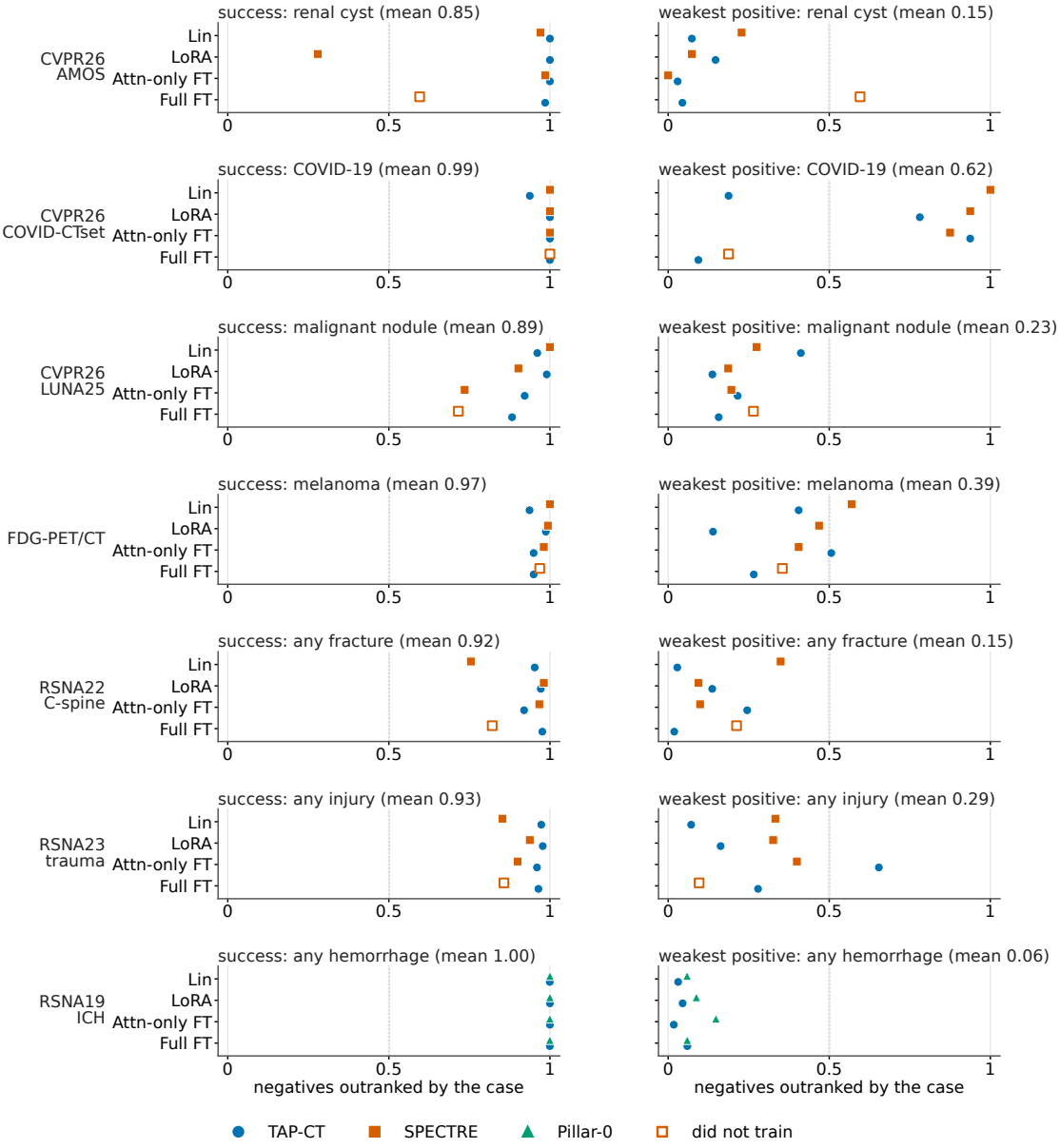


Figure 19: Classification success (left) and weakest positive (right) cases, with one positive held-out case per source. Markers give the share of held-out negatives that the case outranks, for four methods and each backbone, where 1 means that the case is ranked above every negative and the dotted line marks 0.5; SPECTRE Full FT, which did not train, is drawn with open markers. The title gives the label and the case's mean over methods and backbones. Rows are the three CVPR26 sources, FDG-PET/CT, RSNA22 C-spine, RSNA23 trauma and RSNA19 ICH.

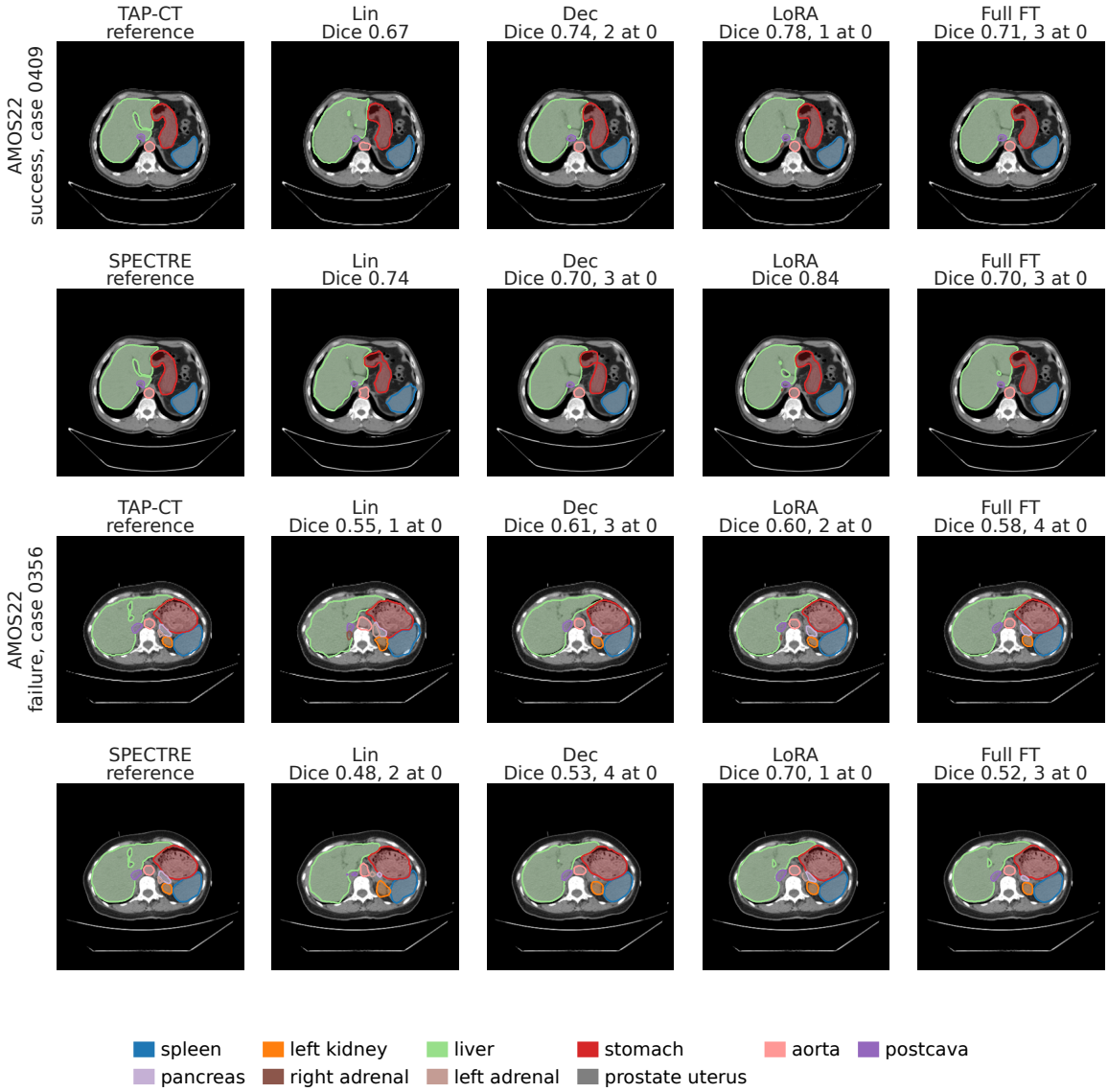


Figure 20: Segmentation success and failure cases on AMOS22 (15 structures) for the joint model. Rows show the success case (top two) and the failure case (bottom two), each on TAP-CT and SPECTRE, and columns the reference and the predictions of Lin, Dec, LoRA and Full FT on the axial slice with the most reference foreground, at the soft-tissue window. Panel titles give the method's Dice for the whole case, averaged over its structures, and how many structures score 0; these are the adrenal glands for most adapted methods and, for some, the esophagus or the gallbladder. For comparison, the dataset-mean Dice of the joint models is 0.646, 0.708, 0.739 and 0.681 for Lin, Dec, LoRA and Full FT on TAP-CT, and 0.682, 0.658, 0.784 and 0.669 on SPECTRE (Tab. 21).

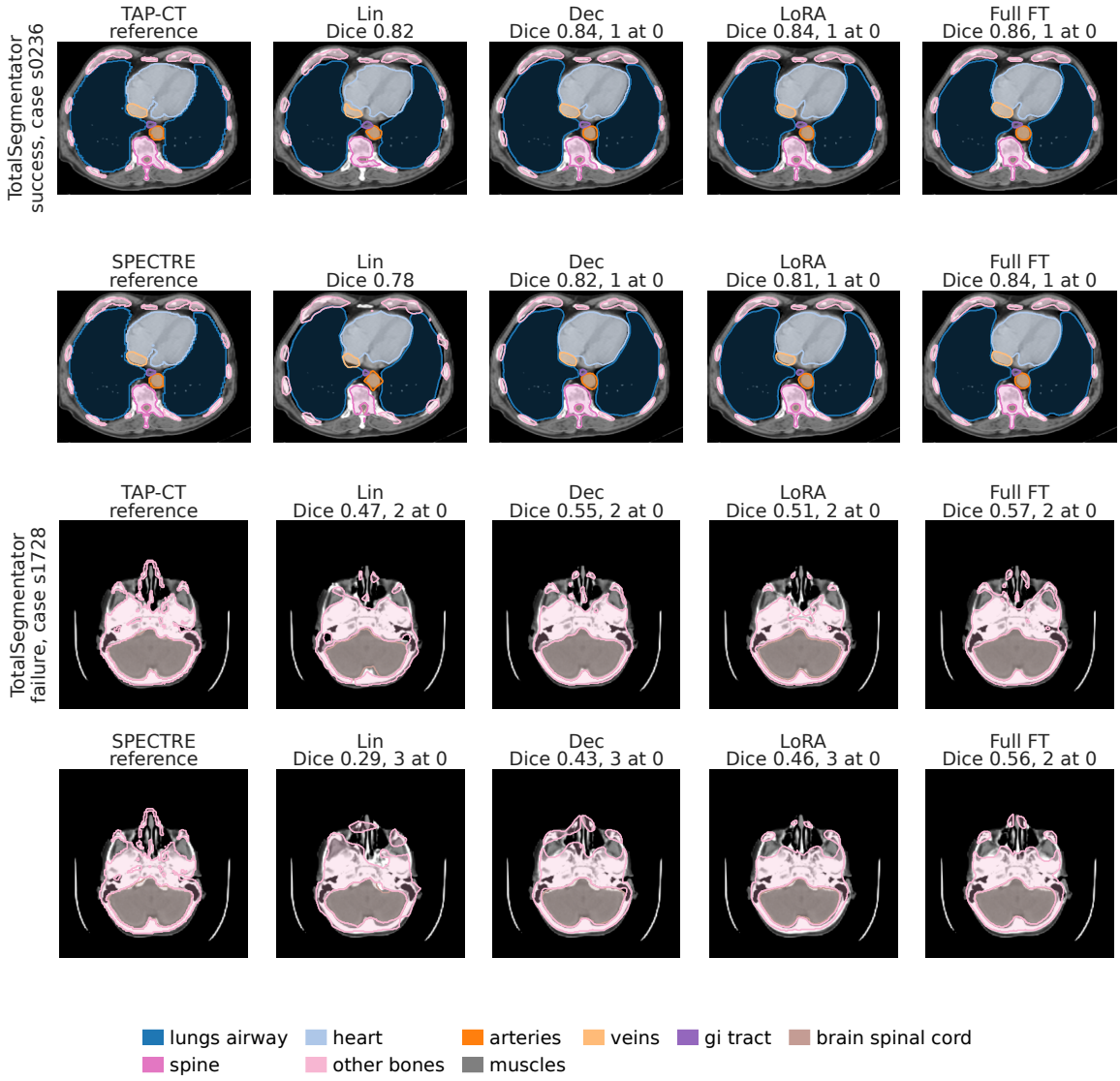


Figure 21: Segmentation success and failure cases on TotalSegmentator (15 merged structures), shown as in Fig. 20. Every model except the linear probe scores 0 on the merged glands in the success case, and every model except the TAP-CT linear probe does so in the failure case. In the failure case, some models also score 0 on the arteries, the liver, the lungs and airway or the gastrointestinal tract.

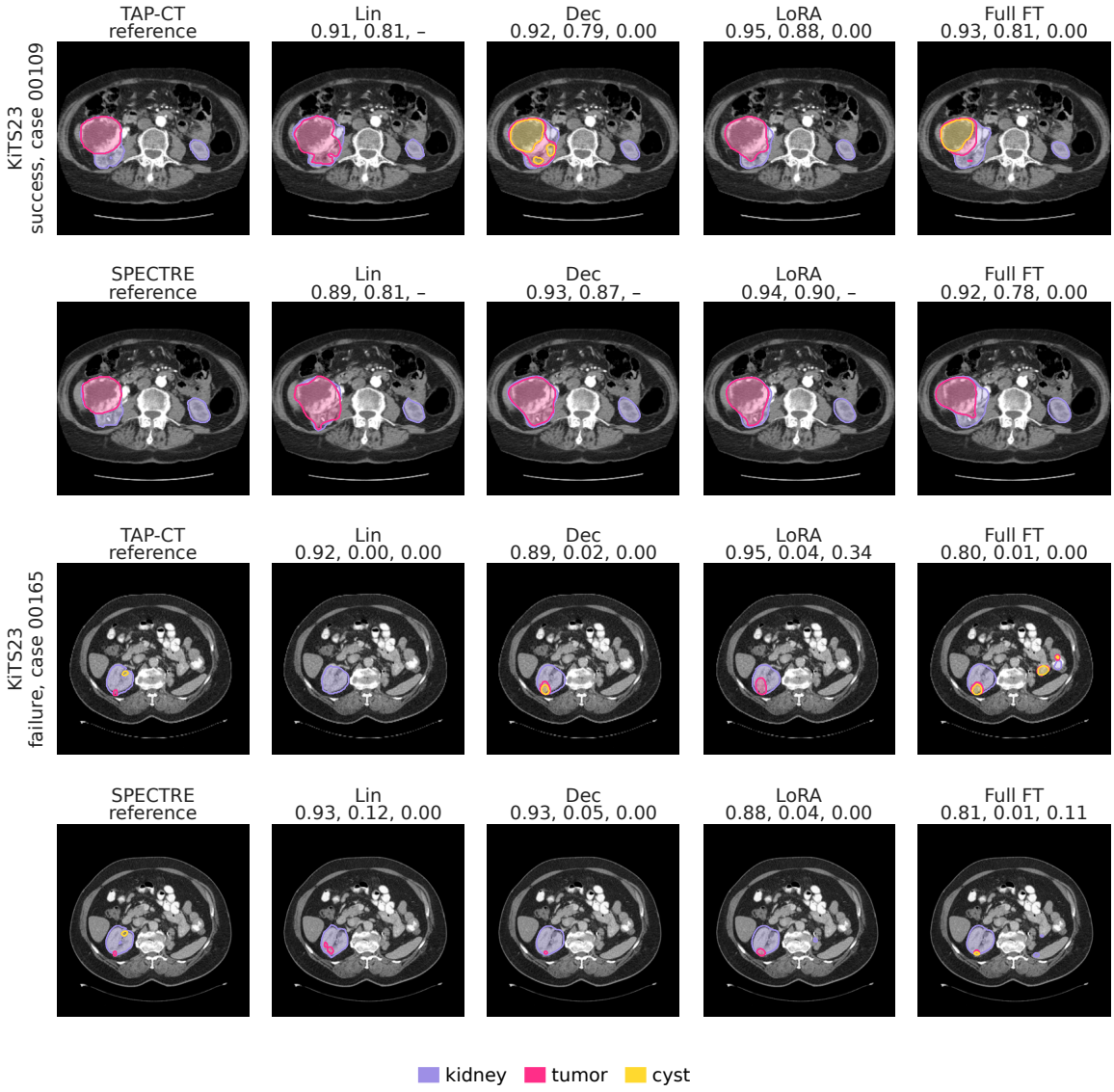


Figure 22: Segmentation success and failure cases on KiTS23, shown as in Fig. 20, except that panel titles give the Dice of each structure (kidney / tumor / cyst), where – marks a structure absent from both reference and prediction, and that a predicted tumor or cyst is drawn over the kidney wherever it is predicted, as in the reference; where both are predicted, the cyst is shown.

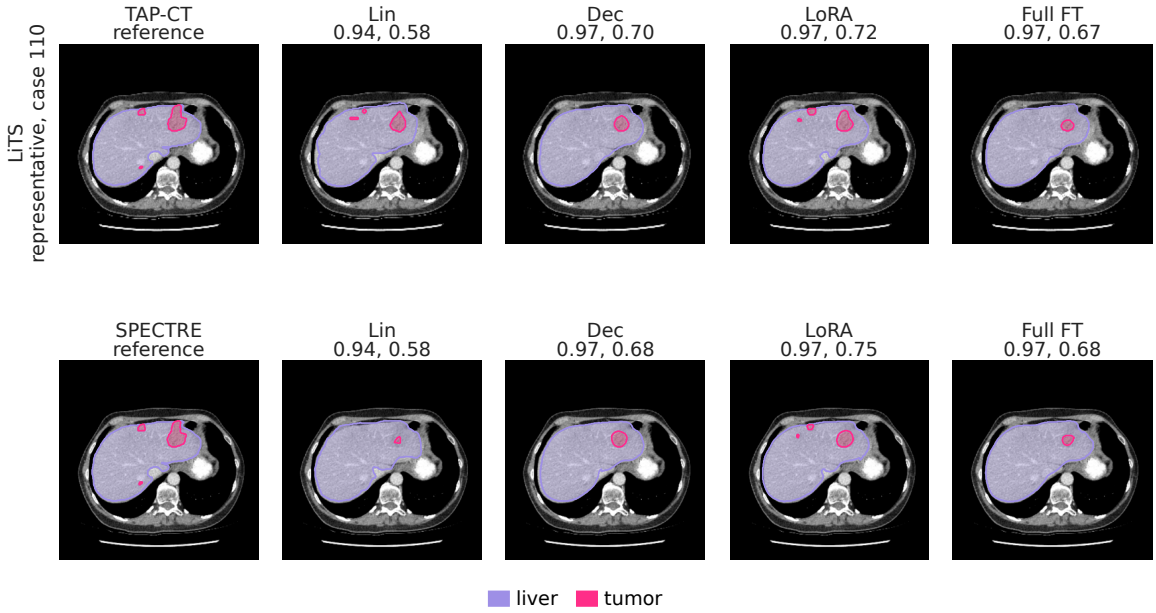


Figure 23: Representative segmentation case on LiTS for the joint models, shown as in Fig. 20, except that panel titles give the Dice of each structure (liver / tumor) and that the predicted tumor is drawn over the liver wherever it is predicted, as in the reference. Among the 17 held-out cases with a reference tumor (Fig. 18 counts the 18 cases with a valid tumor Dice in every run, one of which has no reference tumor), the case was chosen by the rule of Fig. 18: on both backbones, its best and worst methods by mean Dice over liver and tumor agree with those of the dataset means, and among the 5 cases that qualify, its pairwise Dice differences are closest to the mean differences. As on average, the frozen decoder improves on the linear probe, LoRA has the highest Dice, and full fine-tuning does not improve on LoRA.

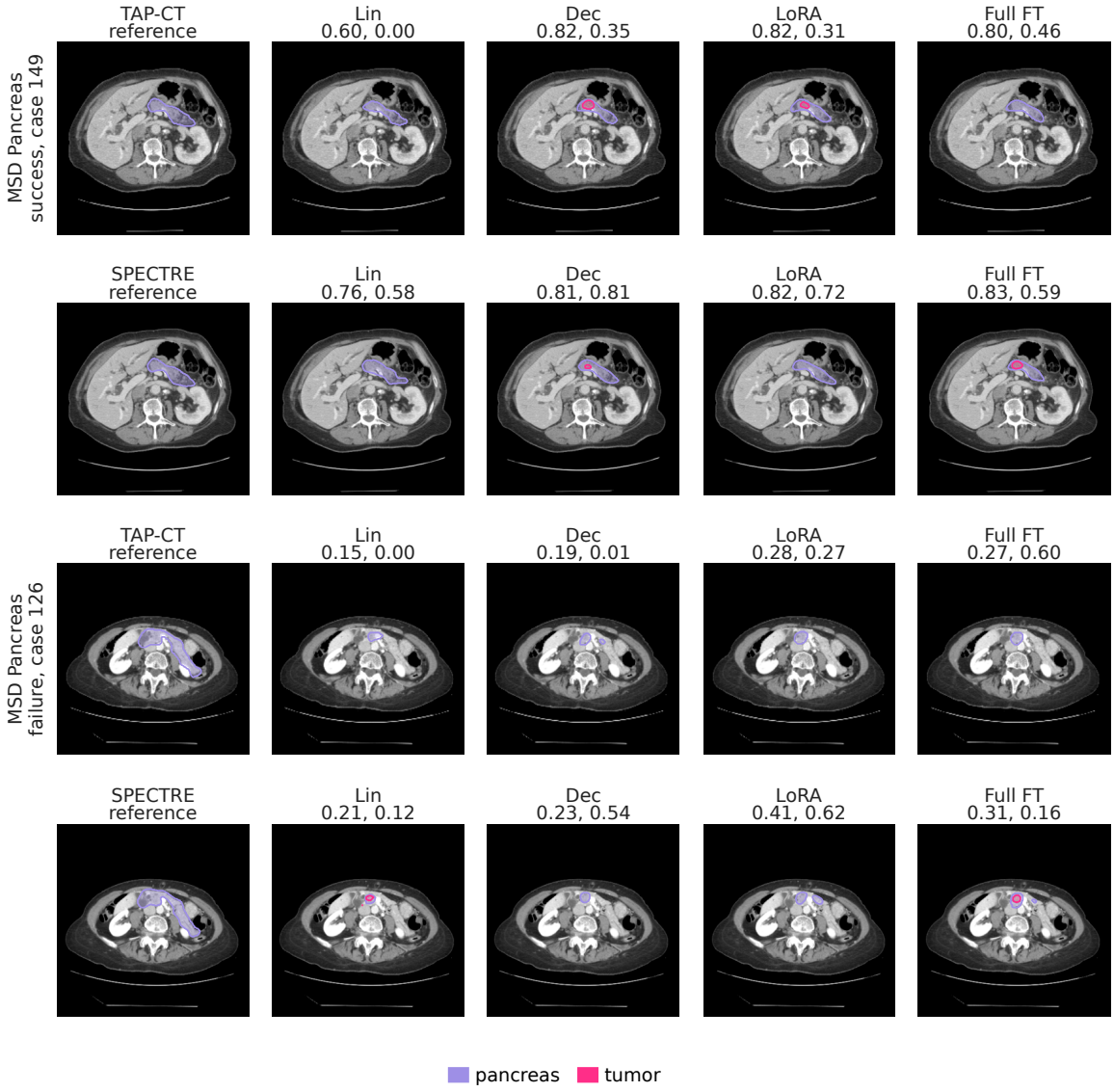


Figure 24: Segmentation success and failure cases on MSD Pancreas, shown as in Fig. 20, except that panel titles give the Dice of each structure (pancreas / tumor), where – marks a structure absent from both reference and prediction, and that the predicted tumor is drawn over the pancreas wherever it is predicted, as in the reference.

Appendix H. Extended discussion

This appendix extends Section 5.

Attention-only fine-tuning and LoRA cost about the same per training step because both back-propagate through every encoder block to reach the adapted weights of the first

block (Tab. 4). Our rankings also depend on the training budget (Sections 4.1 and 4.2), and the joint segmentation models were limited in part by their budget per structure.

Our study has several limitations. Each method was trained with a single default setting, with all adapters of rank 16 on the query, key and value projections, and only full fine-tuning was re-tuned after it failed; consequently, the ranking among the stronger methods may change with method-specific tuning, and seeds were repeated only for joint models. Future work could tune each method on selection data under a matched budget. Per-label tests also have limited power: LoRA was significantly ahead of the linear probe on none of the labels with fewer than 100 held-out cases and on 88% of those with 400 or more, although all of the latter come from RSNA19 ICH. Segmentation models had not converged at our training budgets and were scored on each backbone's working grid, so our results rank adaptation methods rather than measure their distance to nnU-Net (Isensee et al., 2021) or to models pretrained with segmentation labels (Li et al., 2024; He et al., 2025). We also report overlap (Dice) only: Dice does not measure boundary accuracy, and on a grid coarser than the image, such as the 224 in-plane grid of TAP-CT, it penalizes small structures most. Future work could compare adaptation methods at convergence with boundary metrics and evaluate structure-balanced crop sampling and a Dice computed over the batch for joint segmentation models. Lastly, two caveats concern the data: SPECTRE saw AMOS, LUNA25, LiTS, MSD Pancreas and most held-out KiTS23 cases, without labels, during self-supervised pretraining (Tab. 26), which may favor it on these datasets, and the AMOS release gives no patient identifiers, so patient-level separation of the AMOS part of the CVPR26 cohort cannot be confirmed (Tab. 24).